Noise Covariance Defines Reconstruction Geometry: Optimal Filtering, Weighted Subspaces, and Tied Linear Autoencoders

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Draft of 11 September 2026, revision after the progress review (arXiv v1 not yet posted)

Abstract

Under a linear Gaussian measurement model the inverse noise covariance Σ^{-1} is the metric in which a detector reconstruction is learned; the architecture fixes only the admissible class and the training data the excited directions. Optimal filtering, covariance-weighted PCA, tied noise-aware linear autoencoders and their Kronecker-separable restriction are one Σ^{-1}-orthogonal projection differing only in class; the weighted residual predicts whitened, not Euclidean, clean-signal error. Equal-norm covariance errors need not produce equal representation errors: to first order the learned subspace moves by the retained–discarded coupling block of the relative error divided by the pairwise gaps, and diagonal error is harmless until it reorders the spectrum; equal-norm perturbations differed a hundredfold in effect. The separable class's contrast with the unrestricted one splits exactly into a provably positive approximation cost and two estimation excesses; on fresh seeds it won at $N = 800$ and lost at $N = 1600$. Excitation in training did not make a physical direction recoverable; the induced population spectrum and loadings did. On seven public CRESST channels covariance weighting lowered every held-out weighted residual (mean 21.6%). On TIDMAD data with an injected reference, no learned reconstruction beat predicting zero on fourteen predeclared held-out files at dimension-to-sample ratio 187. Restricting the same rank-2 fit to a nine-bin band at the same window, a second predeclared hold-out, moved it from the trivial-predictor level to below a no-denoising bin read, but its dependence assessment did

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not stabilise, so all three registered verdicts are withheld; the metric's contribution vanished with the band's dynamic range.

1 Introduction

Scientific detectors often record a structured response superposed with electronic, thermal, or environmental fluctuations. These fluctuations are rarely independent and identically distributed. They may be colored in frequency, heteroskedastic across samples, correlated between readout channels, or slowly varying in time. Reconstruction must therefore do more than smooth a waveform: it must estimate signal amplitudes or shapes without spending model capacity on directions dominated by detector noise.

Three communities describe closely related solutions in different language. Signal processing uses matched or optimal filters to project a trace onto a known template with inverse-noise weighting [1, 2]. In cryogenic calorimetry, this construction is the standard optimal-filter (OF) amplitude estimator [3–5]. Statistics replaces the fixed template by a learned low-dimensional signal model. Probabilistic PCA assumes isotropic noise [6], factor analysis allows diagonal noise [7], and weighted or expectation–maximization PCA (EMPCA) accommodates nonuniform weights [8, 9]. Machine learning expresses a similar projection as an encoder, a latent representation, and a decoder; under Euclidean mean-squared error, tied linear autoencoders recover the PCA subspace [10–12].

The common object is the observation-space metric. For a trace $x = s + n$ with $n \sim \mathcal{N}(0, \Sigma)$, the negative log-likelihood of a candidate reconstruction $\hat{s}$ contains $(x - \hat{s})^\dagger \Sigma^{-1} (x - \hat{s})$. Thus Σ^{-1} determines which residual directions are costly. OF, EMPCA, and the tied noise-aware linear autoencoder can all be viewed as projections in this metric. They differ in their admissible signal model: a known one-dimensional template for OF and a learned k-dimensional subspace for EMPCA and the tied linear autoencoder. This viewpoint separates three choices that are often conflated: the likelihood determines residual geometry, the signal model determines admissible reconstructions, and the algorithm or architecture parameterizes the projection.

This distinction matters because isotropic mean-squared error (MSE) is not a neutral default. It is the Gaussian likelihood only when $\Sigma = \sigma^2 I$. Under colored or correlated noise, MSE ranks residuals in a different geometry and can favor variance-rich but low-information directions. A method may therefore reduce raw MSE while worsening the covariance-weighted residual that follows from the assumed detector model. The effect is a likelihood misspecification, not merely a change of units. Which ordering is scientifically right is not settled by the likelihood alone: the weighted residual predicts the population whitened clean-signal error of a fixed linear reconstruction exactly, up to a trace correction (Proposition 7.1), and it need not predict the Euclidean clean-signal error or a physical-

parameter error, as a two-dimensional example shows (Example 7.2). The paper therefore names the target of every reconstruction claim it makes.

Our theoretical contribution is a compact hierarchy of constrained Gaussian likelihood projections. We identify classical OF as the fixed rank-1 case, state the exact finite-sample degeneracy of a rank-1 weighted subspace projection to OF when its basis spans the template, and give the corresponding population result for a learned rank-1 EMPCA direction. We then establish a bridge from weighted subspace estimation to representation learning: a tied linear autoencoder whose encoder is the Σ^{-1}-adjoint of its decoder and whose loss is covariance-weighted has the same optimal subspace as EMPCA. The bridge is a deterministic identity between two optimization problems — it holds for any data and any positive-definite weight, with the noise model entering only through the choice of weight — and we state, rather than assume, what untying the encoder changes: not the optimal reconstruction, but the parameterization and the landscape.

Two further statements fix the division of labor between the metric and the architecture. The symmetry argument that selects the weighted loss acts on the residual vector alone; stated as the axiom it is — for quadratic residual losses, invariance under the isometries of the measured noise is *equivalent* to $M \propto \Sigma^{-1}$ — it makes the metric architecture-neutral, so that a reconstruction method is the pair $(\Sigma^{-1}, \mathcal{F})$ of a residual geometry fixed by the likelihood and an admissible class $\mathcal{F}$ fixed by the architecture. Holding the metric fixed, we add a fourth class that *restricts* the learned subspace rather than enlarging it: a Kronecker-separable channel–time subspace, the noise-factored projection autoencoder (NFPA), for which we prove the restricted bridge — it solves the same objective over a product Grassmannian, with equality to the unrestricted optimum exactly when that optimum is separable — and its degeneracy to the optimal filter at rank one. The separable class matters because it is the first at which the architecture, not the metric, decides what can be represented, and because a planted channel–time structure can be compared with the learned factors axis by axis.

Contributions. The question the paper asks is what determines which representation a detector model learns, and when covariance weighting and a structural restriction of the learned class improve recovery of a physical signal. For a calibrated readout three determinants are visible in the theory: the metric the measurement fixes, the class the architecture fixes, and the support the training data excite. The paper establishes the first two, states the third as an exact population identity, and then asks what each does at finite sample, where the covariance is estimated and the data are limited. Its results are the following. (i) *One projection, four classes.* OF, covariance-weighted PCA, the tied noise-aware linear autoencoder and its Kronecker-separable restriction are one Σ^{-1}-orthogonal projection whose classes differ only in the admissible set, with the tied encoder as an

adjoint condition, the finite-sample degeneracies and rank-deficient cases made explicit, and untying stated as a change of parameterization rather than of optimum. (ii) *Which risk the residual predicts.* The weighted residual predicts the population whitened clean-signal error of a fixed linear map exactly up to a trace correction, and a two-dimensional example shows that it need not predict Euclidean clean-signal error (Section 7.1). (iii) *Covariance error acts locally.* The displacement of the learned subspace under a relative covariance error E is, to first order, the coupling block of E between retained and discarded eigenspaces divided by the pairwise gaps (Proposition 5.2); a diagonal error in the fitted eigenbasis is exactly harmless until it reorders the spectrum, and a near-singular estimate is a separate regime. The global bound in relative covariance error (Remark 5.2) is valid but uninformative where it applies, and the controlled study of Section 9.2.2 found perturbations of equal norm differing by two orders of magnitude in effect, as the local statement predicts and the bound cannot. (iv) *A structural prior trades approximation for estimation.* The finite-sample contrast between the separable and the unrestricted class decomposes exactly into a population approximation cost, for which a computable lower bound is given, and two estimation excesses (Proposition 6.3); on fresh seeds the restricted learner won at $N = 800$ and lost at $N = 1600$, so both regimes have been observed and the crossover is bracketed (Section 9.2.3). (v) *Excited is not detectable.* Scaling the excitation of one physical latent in the training draw did not set its recovery; the induced population spectrum and the direction's eigenvector loadings did, on an exploratory reading that is recorded for an independent test (Section 9.2.4). (vi) *The two determinants trade off on real data.* Restricting the same rank-2 fit to a nine-bin band at the same TIDMAD window — a second predeclared hold-out, executed once — moved it from the trivial-predictor level to below a no-denoising bin read, confirming the dimensional diagnosis in direction; but its dependence assessment did not stabilise, so all three registered verdicts are withheld, and the metric's contribution vanished with the band's dynamic range (Section 9.5). (vii) *An evidence programme with named boundaries*, reporting for each public dataset which of three questions — reconstruction in the assumed geometry, calibration of the assumed covariance, and agreement with recorded truth in a named metric — it can and cannot answer, including the failures.

What is recalled. The identities that organize (i) are largely classical: that the generalized least-squares amplitude is the optimal filter, that covariance-weighted low-rank approximation reduces to an SVD of whitened data [13, 14], that Euclidean linear autoencoders recover PCA [10], and that alternating per-axis projections are multilinear PCA [15]; Section 8 marks each statement of Sections 2–7 as established, specialized, a corollary, a synthesis, or derived here, and claims no statement as an independently substantiated novelty. Likewise (ii) specializes classical unbiased-risk accounting to a fixed

linear reconstruction in whitened coordinates, (iii) is first-order invariant- subspace perturbation theory applied to the operator the estimator actually diagonalizes, and (iv) is the ordinary approximation–estimation decomposition of a constrained estimator with a Ky Fan bound. What is added is their statement in one geometry — the pullback form that carries Σ^{-1} to any representation (Section 7.2) is stated for the same reason — and what the controlled programme found when it tested them. One block of controlled planted-physics evidence for the separable class — the July programme (P1-E2, E6, E9, E11, E12): whether the wrong metric learns the wrong physical structure, and how much of the excited span a finite sample recovers under two binary support choices; together with the September zero-cost check on the same family (T23) — is complete and is held from this version until the double-blind review of its companion workshop submission concludes on 29 September 2026. Nothing else is held. The September programme on the same planted family is reported in full: the finite-sample class contrast over sample size and nonseparability (T14), its population risk accounting with the fresh-seed crossover (T20), and the continuous-excitation study (T21) (Section 9.2). This paper is the theory and controlled-evidence half of a two-part programme; a companion study is designed to take the same determinants to frozen models on real detectors (Section 10). Full derivations, algorithmic updates, conditioning details, and gauge arguments are placed in the supplement so that the main text remains focused on the likelihood geometry.

The empirical contribution follows a deliberately asymmetric evidence arc. Controlled simulations first test consequences that are known under the data- generating model. Public CRESST-II/III pulse-shape data [16] then provide a positive but narrow real-detector test: within each of seven channels, we compare fixed-template and learned linear reconstructions on held-out pulses. Pulse and noise streams were each split chronologically, separately; no common cross-stream timestamp cutoff is implied. A post-freeze channel-then-event bootstrap tests whether the pooled result survives equal channel weighting. The OF–EMPCA contrast tests the benefit of enlarging the admissible signal model, whereas the same-rank PCA–EMPCA contrast and covariance controls test the contribution of the noise geometry. We keep these questions separate because a gain over a fixed template is not, by itself, evidence that inverse-covariance weighting caused the gain. Public H1/L1 data around GW150914 from the Gravitational Wave Open Science Center (GWOSC) [17] then provide a negative calibration stress test. Neither of these releases records the signal that was actually present, so both compare residuals rather than distance to truth. We therefore add public TIDMAD data from the ABRACADABRA axion search [18], whose validation files carry an injected reference channel alongside the detector readout. It supplies the real-data ground truth the other two lack, and it delivers two positive results and two negative ones: its noise model passes second-moment calibration checks where GWOSC's does not, and on the primary validation file the reconstruction

preferred by the covariance-weighted residual is also the one with the lower injected-amplitude error against the recorded reference, while raw MSE prefers the other — an ordering result, because at the frozen operating point both reconstructions recover the injected amplitude less accurately than predicting zero, and a predeclared held-out test on fourteen further files at a second operating point finds the same, with both reconstructions 2.7× worse than a direct read of the injection bin, at a dimension-to-sample ratio of 187; a narrowband restriction of the same fit at the same window then moved it below the bin read on a second predeclared hold-out, with every verdict withheld by its own dependence rule; against that, the rank-2 quadrature prediction is not confirmed on any real file, and the linear estimator does not compete on the release's own detector-facing denoising score. This order distinguishes reconstruction performance from detection sensitivity and implementation identity from statistical calibration.

The scope is linear and Gaussian. The exact statements do not extend to nonlinear autoencoders, non-Gaussian transients, or a covariance model that is incorrect for the evaluation data. CRESST is used only to assess within-channel reconstruction of public held-out pulses, not cross-channel transfer or detector sensitivity. GWOSC is used to test whether specific global and local PSD models calibrate a matched statistic on one public interval, not to estimate GW150914 significance or general sensitivity. TIDMAD is used at two frozen operating points on named validation files, the second a predeclared held-out test executed once; it is not a survey of the release's parameter space, its metric-reversal confirmation rests on a single file rather than a replicated effect, no useful recovery is established at either operating point, it does not confirm the rank prediction on real data, and the linear estimator is not competitive on the release's own detector-facing denoising score.

The remainder of the paper introduces the common objective and whitening map, treats OF and EMPCA as rank-constrained cases, states the tied-linear- autoencoder bridge and its Kronecker-separable restriction, derives the consequence for MSE, and then presents synthetic, CRESST, GWOSC, and TIDMAD evidence in that order. The discussion separates supported conclusions from limitations. Supplementary material contains the extended OF derivations, full proofs, EM implementation details, symmetry and gauge analysis, evidence provenance, and the status of deferred supplementary diagnostics.

2 The Unified Noise-Aware Objective

This section formulates the reconstruction problem as a single Gaussian maximum-likelihood (ML) objective and shows that optimal filtering (OF) and expectation–maximization PCA (EMPCA) are constrained solutions of it. We distinguish three levels of equivalence used throughout the paper: *estimator equivalence* (identical scalar

amplitude estimates under matched preprocessing and weighting), *subspace equivalence* (identical signal subspace up to gauge freedom), and *amplitude equivalence* (identical coefficients once projected onto the same subspace with consistent normalization).

Notation and conventions. Vectors are columns. We work in $\mathbb{C}^d$ to cover frequency-domain representations; $\dagger$ denotes the conjugate transpose, and all statements specialize to real-valued data by replacing $\dagger$ with the transpose T. Throughout, $x \in \mathbb{C}^d$ is one observed trace, s its unknown noiseless signal, $\hat{s}$ a candidate reconstruction, s_0 a fixed signal template, $\Sigma \succ 0$ the noise covariance with inverse Σ^{-1} and whitening map $\Sigma^{-1/2}$, $U \in \mathbb{C}^{d\times k}$ a learned rank-k basis with coefficient vector $c \in \mathbb{C}^k$, and $P \in \mathbb{C}^{d\times k}$ the decoder basis of the linear autoencoder. Whitened quantities carry tildes, e.g. $\tilde{x} = \Sigma^{-1/2}x$. Events are indexed by $i = 1, \ldots, N$, frequency bins by k (the rank is also written k; the meaning is always clear from context), and channels by $c = 1, \ldots, C$ where needed.

2.1 Generative model

For each event i we observe $x_i \in \mathbb{C}^d$, a time-domain trace or its frequency-domain coefficients, with the generative model

$$x_i = s_i + n_i, \qquad n_i \sim \mathcal{N}_{\mathbb{C}}(0, \Sigma), \qquad \Sigma \succ 0, \tag{1}$$

where s_i is the unknown noiseless signal for event i and the covariance Σ is known or estimated from dedicated noise data that are independent of the fitted trace. The reconstruction $\hat{s}_i$ is restricted to a specified signal model. In the fixed-template special case, $s_i = a_i s_0$ and $\hat{s}_i = \hat{a}_i s_0$ for a scalar amplitude a_i.

Define the whitening transform

$$\tilde{x}_i = \Sigma^{-1/2}x_i, \qquad \tilde{s} = \Sigma^{-1/2}s_0. \tag{2}$$

In whitened coordinates the noise is isotropic,

$$\tilde{x}_i = \Sigma^{-1/2}s_i + \tilde{n}_i, \qquad \tilde{n}_i \sim \mathcal{N}_{\mathbb{C}}(0, I), \tag{3}$$

so every Σ^{-1}-weighted projection becomes an ordinary orthogonal projection. Stacking events row-wise gives the whitened data matrix $\tilde{X} \in \mathbb{C}^{N\times d}$ with rows $\tilde{x}_i^\dagger$.

2.2 The Gaussian ML objective

For a single trace x under model (1), the Gaussian negative log-likelihood is, up to additive constants,

$$\chi^2(\hat{s}) = (x - \hat{s})^\dagger \Sigma^{-1}(x - \hat{s}). \tag{4}$$

Every estimator in this paper minimizes objective (4); the methods differ only in the constraint placed on the admissible reconstruction $\hat{s}$.

Optimal filtering: fixed rank-1 constraint. For a known template s_0 and unknown scalar amplitude A, the constraint is $\hat{s} = As_0$. Minimizing (4) over A gives the generalized least-squares estimator

$$\hat{A}_{\mathrm{OF}} = \frac{s_0^\dagger \Sigma^{-1} x}{s_0^\dagger \Sigma^{-1} s_0}, \tag{5}$$

which is exactly the optimal filter in covariance form. For stationary noise in Fourier space, Σ is diagonal and (5) reduces to the familiar PSD-weighted OF formula (Section 3).

EMPCA: learned rank-k constraint. For a learned basis $U = [u_1, \ldots, u_k] \in \mathbb{C}^{d\times k}$, the constraint is $\hat{s} = Uc$, and

$$\chi^2(c) = (x - Uc)^\dagger \Sigma^{-1} (x - Uc) \tag{6}$$

is minimized by the weighted normal equations

$$\hat{c} = (U^\dagger \Sigma^{-1} U)^{-1} U^\dagger \Sigma^{-1} x, \qquad \hat{s}_{\mathrm{EMPCA}} = U\hat{c}. \tag{7}$$

Equation (7) makes the connection to OF explicit: OF is the special case $k = 1$ with $U = s_0$ fixed.

2.3 Whitening as the geometric normal form

Whitening converts the Σ^{-1}-weighted inner product into the standard Euclidean one: for any $u, v \in \mathbb{C}^d$,

$$u^\dagger \Sigma^{-1} v = (\Sigma^{-1/2} u)^\dagger (\Sigma^{-1/2} v) = \tilde{u}^\dagger \tilde{v}. \tag{8}$$

After whitening, the noise covariance is the identity and all estimators in this paper become ordinary orthogonal projections; this is the simplification that makes the geometric statements of Sections 3–5 exact.

In the most common detector setting the noise is wide-sense stationary. Its finite-window covariance is then Toeplitz, and it is circulant — exactly diagonalized by the DFT — under the periodic finite-grid noise model of Appendix G, which is the model the synthetic experiments use; for a general stationary process the diagonal form is an approximation in the asymptotic-equivalence sense of Section 3.4, not a uniform one. Under the periodic model,

$$\Sigma = F^\dagger \mathrm{diag}(J_1, \ldots, J_d) F, \tag{9}$$

where $J_k = \mathbb{E}[|N_k|^2]$ is the noise power spectral density (PSD) at frequency bin k. Whitening is then element-wise division in the frequency domain,

$$\tilde{X}_k = X_k / \sqrt{J_k}. \tag{10}$$

For non-circulant Σ (short traces, non-stationary noise, correlated channels), one uses any whitening map W with $W\Sigma W^\dagger = I$ — the Hermitian $\Sigma^{-1/2}$ from an eigendecomposition, or L^{-1} from a Cholesky factor $\Sigma = LL^\dagger$; the two differ by a unitary factor that leaves every Σ^{-1} inner product, projector and principal angle unchanged (Appendix C.8); numerically small PSD bins require a regularization floor or frequency masking. These implementation details, which do not affect the theory, are collected in Appendix C.

2.4 The estimator hierarchy

Figure 1 and Table 1 summarize the estimator family studied in this paper. Each row is a constrained minimizer of the same Gaussian ML objective (4); the rows differ only in how the admissible reconstruction is constrained.

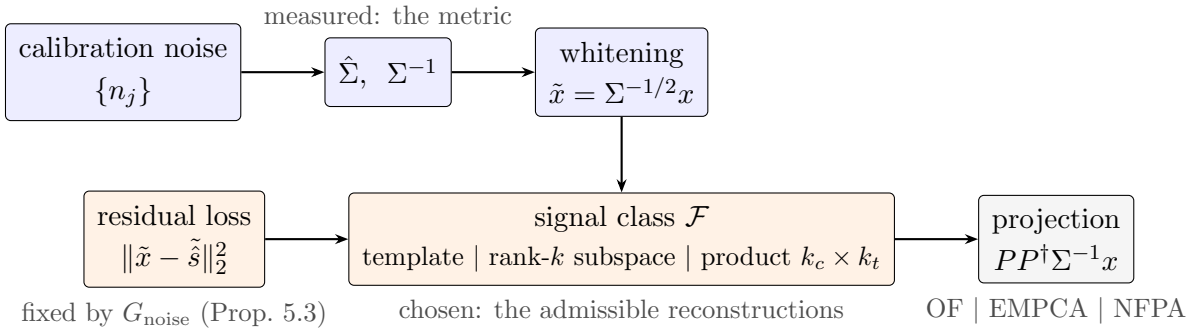


Figure 1: Every estimator in this paper is the pair $(\Sigma^{-1}, \mathcal{F})$. The metric is measured from calibration noise and is not a modelling choice; the loss is the unique residual functional invariant under the noise isometries; the only free decision is the admissible signal class $\mathcal{F}$, and the four conceptual classes of this paper — a fixed template (OF), a learned rank-k subspace (weighted PCA/EMPCA), its tied encoder–decoder form (the noise-aware linear AE), and its separable restriction (NFPA) — are four choices of $\mathcal{F}$ under one metric. Sections 3–6 take them in order.

Table 1: Hierarchy of constrained ML estimators under the unified noise-aware objective. All methods minimize $(x - \hat{s})^\dagger\Sigma^{-1}(x - \hat{s})$. "Degeneracy condition" states when a row coincides with the basic OF row.

Estimator	Constraint on $\hat{s}$	Subspace dim.	Solution form	Degeneracy condition
Basic OF	$\hat{s} = As_0,\ s_0$ fixed	$k = 1$ (fixed)	$\hat{A} = (s_0^\dagger\Sigma^{-1}x)/(s_0^\dagger\Sigma^{-1}s_0)$	Signal is exactly $\propto s_0$
Time-shift OF	$\hat{s} = Ae^{-i\omega t_0}s_0,\ t_0$ unknown	$k = 1$ (shift manifold)	$\hat{t}_0 = \arg\max_{t_0} \lvert\hat{A}(t_0)\rvert$	No timing jitter; collapses to basic OF
Joint-channel OF	$\hat{s} = As,\ s \in \mathbb{C}^{Cd}$ joint template	$k = 1$ (joint)	$\hat{A} = (s^\dagger\Sigma^{-1}x)/(s^\dagger\Sigma^{-1}s)$	Channels uncorrelated; factorizes per channel
Rank-k EMPCA	$\hat{s} = Uc,\ U \in \mathbb{C}^{d\times k}$ learned	k (learned)	$\hat{c} = (U^\dagger\Sigma^{-1}U)^{-1}U^\dagger\Sigma^{-1}x$	$k = 1$, $\text{span}(U) = \text{span}(s_0)$: equals basic OF
Noise-aware linear AE	$\hat{s} = PP^\dagger\Sigma^{-1}x,\ P^\dagger\Sigma^{-1}P = I$	k (learned)	P^* = top-k eigenvectors of $\hat{R}_x\Sigma^{-1}$	Equivalent to EMPCA at convergence (Theorem 5.1)
Separable (NFPA)	$\text{vec}(\hat{\tilde{Z}}) = (U_tU_t^T \otimes U_cU_c^T)\text{vec}(\tilde{Z})$	k_ck_t (learned, product)	Alternating Rayleigh–Ritz on U_c, U_t	$(1,1)$ and separable template: equals OF (Cor. 6.2)

Table 2 states the same comparison from an evaluation perspective. Ordinary PCA is included because it is the Euclidean baseline used in the loss-geometry experiments, even though it does not minimize the covariance-weighted objective unless the data have first been correctly whitened.

Table 2: Methods compared under a shared reconstruction task. "Uses covariance" refers to the fitted objective, not merely to post-hoc evaluation.

Method	**Model and output**	**Uses Σ**	**Known template**	**Colored or correlated noise**	**Likelihood interpretation**
OF / matched filter	Fixed rank-1; amplitude and template reconstruction	Yes	Yes	Yes, with valid Σ or PSD	Gaussian ML for the template amplitude
Ordinary PCA	Learned rank-k Euclidean subspace and scores	No	No	Only after exact whitening	Gaussian ML only for $\Sigma = \sigma^2 I$
Weighted PCA / EMPCA	Learned rank-k weighted subspace and coefficients	Yes	No	Yes	Gaussian ML for the specified covariance or weights
Noise-aware tied linear AE	Rank-k subspace, latent code, and reconstruction	Yes	No	Yes, in this paper	Same linear-Gaussian objective as weighted PCA / EMPCA

Three structural observations follow. First, constraint complexity increases monotonically down the first five rows: basic OF is the most constrained (one free parameter, fixed geometry) and the noise-aware linear AE the least (a full learned k-dimensional subspace); the separable row then *restricts* the fifth, learning a product of two smaller subspaces inside the same rank. Second, the noise model enters identically in every row through the same quadratic form — the methods differ in their signal model, not in how they treat noise. Third, degeneracy propagates upward: each row collapses to the row above when its extra freedom is unused. Sections 3–5 prove these degeneracies; exact improvement over OF occurs only when the true signal family has dimensions a fixed template cannot capture.

2.5 Multi-channel extension

Detectors with C simultaneously read-out channels are covered by stacking the channel traces into a joint vector $x \in \mathbb{C}^{Cd}$ with block-structured covariance $\Sigma_{\text{joint}} \in \mathbb{C}^{Cd \times Cd}$,

whose diagonal blocks are intra-channel covariances and whose off-diagonal blocks carry cross-channel noise correlations. The Gaussian ML objective (4) is unchanged in form; the rank-1 constraint gives joint-channel OF (Section 3.3) and the rank-k constraint gives multi-channel EMPCA. When cross-channel noise correlations vanish, the joint objective is a sum of per-channel quadratic forms; it decouples into C independent single-channel *problems* only when the signal model is itself separable across channels (a per-channel subspace with independent coefficients), which is a restriction of the class and not a consequence of the noise: a shared latent signal couples the channels in the optimal joint subspace even under independent noise (Appendix B.2). The full block-covariance construction and whitening details are given in Appendix B.2.

3 Optimal Filtering as Rank-1 Constrained ML

3.1 The rank-1 constraint and OF estimator

Specializing the unified objective (4) to the fixed-template constraint $\hat{s} = As_0$ in the stationary single-channel case, where Σ is diagonal in frequency space with PSD entries J_k, gives

$$\chi^2(A) = \sum_k \frac{|X_k - AS_k|^2}{J_k}, \tag{11}$$

the discrete Gaussian log-likelihood for uncorrelated frequency bins [2]; this frequency-domain form for microcalorimeters was first stated by Szymkowiak et al. [3]. For a real trace the sum runs over all d bins of the unitary DFT (the two-sided form), so that (11) equals $(x - As_0)^T\Sigma^{-1}(x - As_0)$ exactly; an implementation on the one-sided rFFT grid counts each interior bin twice and the DC and Nyquist bins once (Appendix A). Because the negative-frequency bins are conjugates of the positive ones, the estimator below is real for real A and the d bins are not d independent complex observations. The stationarity condition $\partial\chi^2/\partial A^* = 0$ yields

$$\hat{A} = \frac{\displaystyle\sum_k \frac{S_k^* X_k}{J_k}}{\displaystyle\sum_k \frac{|S_k|^2}{J_k}}, \tag{12}$$

which is the covariance-form estimator (5) in frequency coordinates. Defining the filter kernel and normalization

$$\Phi_k = \frac{S_k^*}{J_k}, \qquad N_\Phi = \sum_k \frac{|S_k|^2}{J_k}, \qquad \Phi_k^{(0)} = \frac{\Phi_k}{N_\Phi}, \tag{13}$$

the estimate $\hat{A} = \sum_k \Phi_k^{(0)} X_k$ is the filtered inner product of the data with the normalized kernel — the discrete matched filter [1]. Geometrically, $\hat{s}_{\mathrm{OF}} = \hat{A} s_0$ is the Σ^{-1}-orthogonal projection of x onto the fixed one-dimensional template subspace.

Assumption boundary. The covariance-form estimator requires a known template direction and a known or independently estimated positive-definite noise covariance. Stationarity is needed only for the convenient diagonal PSD representation in Eq. (11); the generalized least-squares expression remains valid for non-stationary or cross-channel covariance when the full matrix is used. Gaussianity is required for the likelihood and efficiency claims, while the best-linear-unbiased statement needs only the corresponding second-moment assumptions. OF is therefore optimal for this fixed-template linear problem. It is not a solution to general nonlinear waveform reconstruction, unknown signal morphology, or covariance mismatch.

3.2 Estimator variance and energy resolution

The OF estimator is unbiased, and its variance

$$\mathrm{Var}(\hat{A}) = \frac{1}{N_\Phi}, \qquad \sigma_A = \frac{1}{\sqrt{N_\Phi}}, \tag{14}$$

attains the Cramér–Rao bound (CRB) for the Gaussian model, making OF the minimum-variance unbiased estimator for a fixed template; the full Fisher information calculation is given in Appendix A. When the amplitude calibrates to deposited energy, $A = E/E_0$, the baseline energy resolution is

$$\sigma_E = \frac{E_0}{\sqrt{N_\Phi}} = E_0 \left(\sum_k \frac{|S_k|^2}{J_k} \right)^{-1/2}. \tag{15}$$

Equation (15) carries the design intuition: resolution improves where signal power is large and noise PSD small, and N_Φ is simultaneously the filter normalization and the total signal-to-noise budget available to any rank-1 estimator. The CRB normalization is verified empirically by the controlled test S1 of Section 9.

3.3 Extensions: the same projection principle

Two standard OF extensions are structured applications of the same rank-1 weighted projection principle, and both are absorbed naturally by the hierarchy of Table 1.

Time-shift OF. For an unknown arrival time, the admissible signals form the one-parameter manifold $\mathcal{M} = \{e^{-i\omega t_0} s_0 : t_0 \in \mathbb{R}\}$. Profiling the amplitude at each candidate shift gives $\hat{A}(t_0) = \sum_k e^{i\omega_k t_0} \Phi_k^{(0)} X_k$ — the inverse DFT of the filtered spectrum — and the ML delay maximizes the filtered amplitude, $\hat{t}_0 = \arg\max_{t_0} |\hat{A}(t_0)|$ [5]. For small shifts, a Taylor expansion shows $\mathcal{M}$ is approximately contained in $\mathrm{span}\{s_0, \partial_t s_0, \partial_t^2 s_0, \dots\}$, so a low-rank learned subspace can absorb small timing jitter to the order of the truncation; this is a local approximation (Appendix B.1), not a containment. The controlled test S6 of Section 9 shows the selected rank rising above one under timing variability; no quantitative timing-bias result is claimed in this version (Appendix F). The full profiling derivation and the formal degeneracy statement (Proposition B.1) are given in Appendix B.1.

Joint-channel correlated OF. For C channels stacked into $x, s \in \mathbb{C}^{Cd}$ with full joint covariance Σ, the GLS estimator $\hat{A}_{\mathrm{JOF}} = (s^\dagger \Sigma^{-1} x)/(s^\dagger \Sigma^{-1} s)$ is algebraically the rank-1 weighted subspace estimator (7) with $U = s$. Cross-channel noise correlations are handled identically in OF and EMPCA, through the same quadratic form; the formal statement (Proposition B.1) and the block-covariance details are in Appendix B.2.

3.4 The stationary basis as a positional encoding: the token-axis principle

The frequency-domain form (11) contains a statement about representation that is worth isolating, because it is the point at which the fixed-template estimator meets architectures that tokenize a measurement rather than project it.

Proposition 3.1 (Stationary noise selects the Fourier basis as the precision eigenbasis)**.** *Let the noise be wide-sense stationary and periodic with the record length d (equivalently, generated by independent DFT coefficients, the sampling contract of Appendix G), so that its covariance is circulant, $\Sigma = F^\dagger \mathrm{diag}(J_k) F$ with F the unitary DFT. Then*

$$\Sigma^{-1} = F^\dagger \,\mathrm{diag}(1/J_k)\, F. \tag{16}$$

The eigenvectors of the precision operator are the discrete Fourier modes and its eigenvalues are the inverse-PSD weights $1/J_k$ that appear in the OF kernel (13)*. Consequently an encoding of the time axis by the leading eigenvectors of Σ^{-1}, weighted by their eigenvalues, is the set of Fourier features weighted by inverse noise power — the matched-filter weighting itself.*

The proof is Eq. (16). For a stationary process that is not periodic the finite-window covariance T_d is Toeplitz rather than circulant, and the sense in which the DFT diagonalizes it must be stated with its norm. For an absolutely summable autocovariance, T_d

and its circulant counterpart C_d are asymptotically equivalent [19]: both have uniformly bounded operator norms and the normalized Hilbert–Schmidt distance $\|T_d - C_d\|_F/\sqrt{d}$ tends to zero, so eigenvalue distributions and normalized traces of functions of the two matrices agree in the limit. The operator-norm distance need not vanish: a covariance with diagonal 3 and nearest-neighbour entries 1 differs from its periodic wrap-around by two corner entries, and $\|T_d - C_d\|_{\mathrm{op}} = 1$ at every d. Consequently a statement about an inverse-covariance-weighted statistic on a finite window — the optimal-filter weights, a whitened residual — is not uniformly approximated by its circulant version, and the exact statements of this paper need the periodic model; what carries over to a general stationary process is the weak (normalized-trace) sense. The statement is elementary, but it identifies the detector-physics content of a construction that geometric deep learning arrives at from the other side: Bronstein et al. [20, §5.4] note that positional encodings for attention on a graph may be taken to be Laplacian eigenvectors of the assumed graph [21], and that the sinusoidal encodings of Vaswani et al. [22] are this construction for the cyclic grid. Here the operator whose eigenvectors encode position is not an assumed graph Laplacian but the measured noise precision: for stationary noise the two constructions coincide up to the weights $1/J_k$, which is exactly where the detector enters.

Two qualifications keep the proposition from saying more than it does. The eigenvalue weights $1/J_k$ of the precision operator are not the whitening scalings $1/\sqrt{J_k}$: the two constructions share the Fourier eigenbasis but are different operators, and the proposition identifies the basis and the weights, not a preferred encoding. And nothing here concerns a learned model: a learned query–key projection changes the dot-product geometry that attention computes, so no statement about attention scores follows from prewhitening the inputs. The design suggestions this observation invites — prewhitening before tokenization, and tokenizing along the axis in which Σ^{-1} is diagonal — are recorded as untested suggestions in Section 10; whitening supplies no locality, no graph structure and no deformation stability, which are properties of the detector domain rather than of the noise model (Appendix E.5).

3.5 Limitation: the fixed-template assumption

The OF derivation assumes every signal is a scalar multiple of the *same* known template. Realistic pulse families violate this in two generic ways: timing jitter spreads the signal over the shift manifold, and physical effects (position-dependent response, geometry inhomogeneity, channel cross-talk, saturation) make the pulse shape itself vary event-to-event. When the template direction s_0 is misaligned with the true signal s, the OF estimate is

biased,

$$\mathbb{E}[\hat{A}_{\mathrm{OF}}] = A \cdot \frac{s_0^\dagger \Sigma^{-1} s}{s_0^\dagger \Sigma^{-1} s_0} \neq A \quad (\text{if } s_0 \not\propto s), \tag{17}$$

with effective SNR loss $\cos^2\theta_w$, the squared cosine of the whitened-space angle between s_0 and s. The bias factor is a ratio of whitened inner products, not a shrinkage toward zero: a direction mismatch scales the expected amplitude by $\|\tilde{s}\|\cos\theta_w/\|\tilde{s}_0\|$ (with the modulus of the complex ratio in the complex case), which can exceed one, and a proportional template $s = \gamma s_0$ changes the amplitude calibration by γ with no direction mismatch at all (Appendix A). The $\cos^2\theta_w$ law is checked on real SQUID noise by the TIDMAD estimator-identity gates of Section 9.5. Replacing the fixed template by a *learned* k-dimensional subspace removes this assumption: when the true signal family lies in the learned subspace, the estimator is unbiased and attains the CRB of the subspace-constrained problem. This motivates the rank-k extension developed next.

4 EMPCA as Rank-k Constrained ML

4.1 The rank-k constraint

The optimal filter assumes a one-dimensional signal family. Realistic families have more structure: pulse shapes vary with interaction position, timing jitter generates a continuum of shifted templates, and multi-mode excitations populate several directions simultaneously. The natural relaxation constrains the signal to a *learned* k-dimensional subspace. For a single observation $x \in \mathbb{C}^d$ the rank-k constrained ML problem is

$$\min_{U,\,c} \ \chi^2(c;\,U) = (x - Uc)^\dagger \Sigma^{-1}(x - Uc), \qquad \text{subject to} \quad U^\dagger \Sigma^{-1} U = I_k, \tag{18}$$

and over N events with coefficient matrix $\Gamma = [c_1, \ldots, c_N] \in \mathbb{C}^{k\times N}$,

$$\min_{U,\,\Gamma} \ \mathcal{L}(U,\Gamma) = \frac{1}{N}\sum_{i=1}^{N}(x_i - Uc_i)^\dagger \Sigma^{-1}(x_i - Uc_i), \qquad U^\dagger \Sigma^{-1} U = I_k. \tag{19}$$

In whitened coordinates ($\tilde{x}_i = \Sigma^{-1/2}x_i$, $\tilde{U} = \Sigma^{-1/2}U$, $\tilde{U}^\dagger\tilde{U} = I_k$) the objective reduces to a standard low-rank approximation problem,

$$\mathcal{L}(U,\Gamma) = \frac{1}{N}\sum_{i=1}^{N}\|\tilde{x}_i - \tilde{U}c_i\|_2^2, \tag{20}$$

whose minimizer over subspaces is the span of the top-k eigenvectors of the whitened sample covariance $\Sigma^{-1/2}\hat{R}_x\Sigma^{-1/2}$, with $\hat{R}_x = N^{-1}\sum_i x_i x_i^\dagger$. Standard PCA is the ML estimator of this subspace only when $\Sigma = \sigma^2 I$; EMPCA restores ML optimality for general Σ by carrying the weight Σ^{-1} through the estimation.

Algorithm. EMPCA solves (19) by alternating minimization [8]: the E-step solves the weighted normal equations for each c_i with U fixed (closed form, Eq. (7)), and the M-step solves an unconstrained regression for U with Γ fixed, followed by re-orthonormalization in the Σ^{-1} inner product with the compensating change of coefficients (Appendix C.2). The objective (19) is convex in each block separately (the orthonormal feasible set is not convex, so the problem is not biconvex in the usual sense), each step is an exact block minimization, and the iteration decreases the objective monotonically. Its stationary points are *all* k-dimensional invariant subspaces of the whitened second moment, not only the top-k one; with a common covariance the exact-step iteration is orthogonal iteration on that matrix, which converges to the top-k eigenspace from almost every initialization when $\lambda_k > \lambda_{k+1}$ (Theorem C.1 and Remark C.1, Appendix C.5). The archived fits of Section 9 compute the optimum directly by an eigendecomposition of the whitened second moment rather than by the iteration. Implementation details — the frequency-domain variant of Yu [9], initialization, model-order selection, and conditioning safeguards — are collected in Appendix C.

Ordinary PCA versus weighted PCA / EMPCA. Ordinary PCA minimizes Euclidean reconstruction error in the original coordinates. The weighted problem above minimizes that error after the noise geometry has been applied. When every event shares the same positive-definite covariance Σ, exact whitening reduces the weighted problem to ordinary PCA on $\tilde{X} = X\Sigma^{-1/2}$; this is an equivalence between weighted PCA in detector coordinates and ordinary PCA in whitened coordinates. It is not, in general, an equivalence to ordinary PCA on the unwhitened traces. The latter coincides with the weighted solution when $\Sigma = \sigma^2 I$, or accidentally when the weighting leaves the relevant eigenspace and its ordering unchanged. With event-dependent variances, missing samples, or weights Σ_i^{-1}, no single global whitening need exist and the alternating EMPCA updates are required. Consequently, PCA and EMPCA should agree in the white-noise control but should not be expected to agree under genuinely colored weighting. Numerically identical outputs in that regime would require an explicit audit of whether the weights entered the fit, whether both methods received already whitened data, and whether a degenerate eigenspace obscured the comparison.

Gauge freedom. The basis U is identified only up to a unitary rotation $U \mapsto UQ$, $Q \in \mathcal{U}(k)$: the reconstruction $\hat{s} = Uc$ and the subspace are gauge-invariant, the individual coefficients are not. Physical amplitude estimates therefore use gauge-invariant functionals, either the whitened reconstruction norm $\|\hat{s}\|_{\Sigma^{-1}}$ or the projection onto a

calibration template,

$$\hat{A} = \frac{s_0^\dagger \Sigma^{-1} \hat{s}}{s_0^\dagger \Sigma^{-1} s_0}, \tag{21}$$

which reduces to the OF estimator when $k = 1$ and $\mathrm{span}(U) = \mathrm{span}(s_0)$. The geometric structure behind this gauge freedom (the Grassmannian picture of subspace optimization) is developed in Appendix E.

4.2 Equivalence theorems: rank-1 EMPCA and basic OF

We can now state the first main result. Recall the three levels of equivalence from Section 2: subspace equivalence implies amplitude equivalence implies estimator equivalence, provided weighting and normalization conventions are matched.

Theorem 4.1 (Rank-1 OF–EMPCA equivalence, population level)**.** *Assume (i) the generative model $x_i = a_i s + n_i$ with a fixed template $s \neq 0$ and scalar amplitudes with positive second moment $\nu = \mathbb{E}|a_i|^2 > 0$; (ii) Gaussian noise $n_i \sim \mathcal{N}(0, \Sigma)$ with known covariance and zero signal–noise cross moment, $\mathbb{E}[a_i n_i^\dagger] = 0$ (e.g. a_i independent of n_i); and (iii) identical whitening, baseline subtraction, and preprocessing conventions. Then, at the population level (sample second moment equal to its expectation):*

1. *the rank-1 EMPCA subspace equals* $\mathrm{span}(\tilde{s})$ *in whitened space;*
2. *the EMPCA coefficient estimator equals the generalized least-squares / optimal-filter estimator;*
3. *hence rank-1 EMPCA and OF are estimator-equivalent.*

Proof sketch. In whitened space the population second moment is the spiked identity $\mathbb{E}[\tilde{x}\tilde{x}^\dagger] = \nu\, \tilde{s}\tilde{s}^\dagger + I$, whose leading eigenvector is $\tilde{s}/\|\tilde{s}\|$ with eigengap $\nu\|\tilde{s}\|^2 > 0$. Rank-1 EMPCA recovers this eigenvector, and the rescaled projection coefficient is $\hat{a}_i = (s^\dagger \Sigma^{-1} x_i)/(s^\dagger \Sigma^{-1} s)$, which is exactly (5). The full proof, including the eigenvalue gap and the normalization bookkeeping, is given in Appendix D.1. □

Theorem 4.1 is a population-level statement; finite-sample deviations are controlled by the Davis–Kahan bound discussed in Section 4.4. The next result is exact at any sample size, for a fixed basis.

Theorem 4.2 (Exact degeneracy to basic OF)**.** *Assume $k = 1$; the EMPCA basis spans the template direction, $u = \gamma s_0$ with $\gamma \neq 0$; OF and EMPCA use the same weighting Σ^{-1}; and preprocessing and alignment conventions are matched. Then rank-1 EMPCA and basic OF yield identical reconstructed signals $\hat{s}$ and identical weighted residuals.*

Proof sketch. Substituting $u = \gamma s_0$ into the rank-1 coefficient $\hat{c} = (u^\dagger \Sigma^{-1} x)/(u^\dagger \Sigma^{-1} u)$ gives $\hat{c} = \hat{A}_{\mathrm{OF}}/\gamma$, hence $\hat{s}_{\mathrm{EMPCA}} = \hat{c}\, u = \hat{A}_{\mathrm{OF}} s_0 = \hat{s}_{\mathrm{OF}}$. The scalar γ is the rank-1 gauge freedom: it changes the coefficient convention but not the reconstruction. Details in Appendix D.1. □

4.3 Why $k > 1$ helps: the χ^2 improvement

In whitened coordinates, OF projects onto a one-dimensional subspace $\mathcal{S}_1$ and rank-k EMPCA onto a k-dimensional subspace $\mathcal{U}_k$. If $\mathcal{S}_1 \subseteq \mathcal{U}_k$, projection theory gives

$$\chi^2_{\mathrm{EMPCA},k} \leq \chi^2_{\mathrm{OF}} : \tag{22}$$

a larger admissible subspace can only reduce the weighted residual. The physical content is that pulse-shape variation, timing jitter, and channel-coupling modes are additional signal directions that a fixed template conflates into a single amplitude.

Caveat: containment is not automatic. The bound (22) requires the template direction to lie in the learned subspace. At the population level this is automatic for the rank-1 model: in whitened coordinates every noise eigenvalue equals one and the signal direction carries $1+\nu\|\tilde{s}\|^2$, so $\mathcal{S}_1$ is the leading direction whenever $\nu\|\tilde{s}\|^2 > 0$. The condition that matters is the finite-sample one. The nonzero eigenvalues of the sample covariance of N whitened noise vectors in d dimensions spread over the Marchenko–Pastur bulk, whose upper edge is $(1+\sqrt{d/N})^2$ (with $d-N$ further eigenvalues at zero when $N < d$) [23], and for an isolated signal direction in isotropic noise the leading sample eigenvector carries a nonvanishing overlap with $\tilde{s}$ only above the spiked-covariance transition $\nu\|\tilde{s}\|^2 > \sqrt{d/N}$ [24, 25]; below it, or with unfavorable initialization, containment can fail. These are asymptotic statements for one spike in isotropic noise and are used here as the relevant scale, not as a threshold theorem for the structured estimators of this paper. Only an explicit containment constraint — fixing the template direction and learning the remaining $k-1$ directions in its whitened orthogonal complement — restores the guarantee; a template-*seeded* initialization does not, because the iteration moves the seed toward the sample eigenvectors (Appendix C.2). Without the constraint, containment is an empirical property of the fit: the controlled test S2 of Section 9 finds the learned rank-1 direction within four degrees of the template direction in its regime.

4.4 Geometric interpretation and finite-sample stability

After whitening, both estimators are orthogonal projectors:

$$\hat{\tilde{x}}_{\mathrm{OF}} = P_{\tilde{s}_0}\tilde{x}, \qquad \hat{\tilde{x}}_{\mathrm{EMPCA}} = P_{\tilde{U}}\tilde{x}, \qquad P_{\tilde{U}} = \tilde{U}(\tilde{U}^\dagger\tilde{U})^{-1}\tilde{U}^\dagger. \tag{23}$$

OF projects onto a fixed rank-1 direction; EMPCA onto a learned rank-k subspace; when $k = 1$ and the learned direction equals the template direction, the projectors coincide. Because the objective depends on $\tilde{U}$ only through the projector, EMPCA optimizes over the Grassmannian of k-dimensional subspaces; that geometric picture, and its connection to the latent gauge group, is developed in Appendix E.

In finite samples, with the true Σ used for whitening, the estimated subspace is the top-k eigenspace of the whitened sample second moment $\hat{A} = \Sigma^{-1/2}\hat{R}_x\Sigma^{-1/2}$ and the population one is that of $A = \Sigma^{-1/2}R_x\Sigma^{-1/2}$; both live in the same whitened coordinates, and the Davis–Kahan theorem in the form of Yu, Wang and Samworth [26, 27] gives

$$\sin\Theta\big(\hat{\mathcal{S}}, \mathcal{S}\big) \le \frac{2\,\|\hat{A} - A\|_{\mathrm{op}}}{\Delta_\lambda}, \qquad \Delta_\lambda = \lambda_k(A) - \lambda_{k+1}(A), \tag{24}$$

with numerator and gap in the same (whitened) coordinates. For the rank-1 model the gap is $\Delta_\lambda = \nu\|\tilde{s}\|^2 = \nu N_\Phi$ (Theorem 4.1) and grows with SNR. The numerator is a covariance-estimation error: when the whitened observations $\tilde{x}_i$ are sub-Gaussian — Gaussian noise and amplitudes with a bounded sub-Gaussian norm, which excludes heavy-tailed amplitude distributions — one has $\|\hat{A} - A\|_{\mathrm{op}} \le C\|A\|_{\mathrm{op}}\max(\delta, \delta^2)$ with $\delta = \sqrt{d/N} + t/\sqrt{N}$, with probability at least $1 - 2e^{-ct^2}$, for constants C, c depending only on the sub-Gaussian norm [28, Thm. 5.39 and Cor. 5.50]; the $\delta^2 = d/N$ term matters outside the $N \gg d$ regime, and $\|A\|_{\mathrm{op}} = 1 + \nu N_\Phi$ carries the signal scale. Gaussianity of the noise alone does not supply such a bound for arbitrary amplitude distributions. At high SNR and $N \gg d$ the bound is of order $\sqrt{d/N}$, which is the regime of the controlled test S2 of Section 9, in which the learned rank-1 direction lies within four degrees of the OF direction in whitened space. The extended stability discussion, and the corresponding bound for an *estimated* covariance, are given in Appendix E.7 and Remark 5.2.

From algorithm to objective. As defined above, EMPCA is an algorithmic procedure — alternating E- and M-steps. For comparing estimators at the level of the loss, and for replacing the linear encoder by a nonlinear one while preserving the noise-aware structure, an objective-first formulation is preferable. The next section provides it: the same estimator is the minimizer of a noise-aware tied linear autoencoder loss.

5 The Noise-Aware Linear Autoencoder and the Bridge Theorem

5.1 The encoder–decoder language

We now translate the weighted-projection estimators of Sections 3 and 4 into encoder–decoder language. A linear autoencoder consists of an encoder $E : \mathbb{C}^d \to \mathbb{C}^k$ producing a latent code $z = Ex$ and a decoder $D : \mathbb{C}^k \to \mathbb{C}^d$ producing the reconstruction $\hat{x} = Dz$. This is the division of labor already present in OF and EMPCA: OF computes one weighted coefficient along a fixed template and decodes it into a rank-1 waveform; EMPCA computes k weighted coefficients in a learned basis. The autoencoder formulation introduces no new estimator by itself — the question is which loss and which encoder–decoder relationship recover the Gaussian ML projection.

For ordinary Euclidean linear AEs, "tied weights" means the encoder is the Hermitian transpose of the decoder. In detector coordinates this is the wrong adjoint, because the data-space inner product is set by Σ^{-1}, not the Euclidean metric. The noise-aware tied form is

$$D(z) = Pz, \qquad E(x) = P^\dagger \Sigma^{-1} x, \qquad P^\dagger \Sigma^{-1} P = I_k, \tag{25}$$

i.e. the encoder is the Σ^{-1}-adjoint of the decoder. This section shows that this architecture, trained with the Mahalanobis reconstruction loss, is exactly EMPCA.

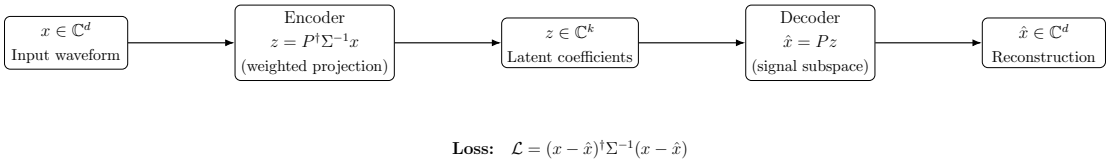


Figure 2: The noise-aware tied linear autoencoder. The encoder is the Σ^{-1}-adjoint of the decoder, and the training loss is the Gaussian χ^2 objective (4).

5.2 The noise-aware loss and its minimizer

With the tied form (25), the reconstruction is $\hat{x}_i = PP^\dagger \Sigma^{-1} x_i$ and the noise-aware reconstruction loss over N events is

$$\mathcal{L}(P) = \frac{1}{N} \sum_{i=1}^{N} (x_i - \hat{x}_i)^\dagger \, \Sigma^{-1} \, (x_i - \hat{x}_i). \tag{26}$$

Expanding with $P^\dagger \Sigma^{-1} P = I_k$,

$$\mathcal{L}(P) = \frac{1}{N} \sum_{i=1}^{N} x_i^\dagger \Sigma^{-1} x_i \; - \; \mathrm{Tr}\big(P^\dagger \Sigma^{-1} \hat{R}_x \Sigma^{-1} P\big), \qquad \hat{R}_x = \frac{1}{N} \sum_i x_i x_i^\dagger, \tag{27}$$

so minimizing $\mathcal{L}(P)$ is a Rayleigh–Ritz problem whose optimum spans the top-k eigenvectors of the Σ^{-1}-weighted covariance operator — precisely the EMPCA subspace. Equivalently, in whitened coordinates ($W = \Sigma^{-1/2}P$, $W^\dagger W = I_k$) the loss is the classical tied linear AE loss $\|\tilde{X} - \tilde{X}WW^\dagger\|_F^2$, minimized by the top-$k$ right singular vectors of $\tilde{X}$.

5.3 The Bridge Theorem

Theorem 5.1 (Bridge Theorem: noise-aware linear AE ≡ EMPCA)**.** *Let $x_1, \dots, x_N \in \mathbb{C}^d$ be any observations, let $\Sigma \succ 0$ be a fixed Hermitian positive-definite weighting matrix, and let $1 \le k \le d$. Consider the noise-aware tied linear autoencoder loss*

$$\mathcal{L}(P) = \frac{1}{N}\sum_{i=1}^{N}(x_i - PP^\dagger\Sigma^{-1}x_i)^\dagger\,\Sigma^{-1}\,(x_i - PP^\dagger\Sigma^{-1}x_i), \qquad P^\dagger\Sigma^{-1}P = I_k, \tag{28}$$

and write $\lambda_1 \ge \cdots \ge \lambda_d \ge 0$ for the eigenvalues of the whitened second moment $\Sigma^{-1/2}\hat{R}_x\Sigma^{-1/2}$, $\hat{R}_x = N^{-1}\sum_i x_i x_i^\dagger$, with the convention $\lambda_{d+1} := 0$ (at $k = d$ the only feasible subspace is $\mathbb{C}^d$ and every statement below is trivial). Then:

1. *P^* minimizes $\mathcal{L}$ if and only if $\Sigma^{-1/2}P^*$ spans a k-dimensional invariant subspace of $\Sigma^{-1/2}\hat{R}_x\Sigma^{-1/2}$ associated with $\lambda_1, \dots, \lambda_k$; equivalently, P^* spans a k-dimensional eigenspace of the generalized eigenproblem $\hat{R}_x\Sigma^{-1}v = \lambda v$ for the k largest eigenvalues.*

2. *If $\lambda_k > \lambda_{k+1}$, this subspace is unique, and $\mathrm{range}(P^*) = \mathrm{range}(\hat{U})$ for the rank-k EMPCA solution $\hat{U}$ with weight Σ^{-1}, up to a unitary rotation within the subspace. If $\lambda_k = \lambda_{k+1}$, the minimizers form a family: every minimizer contains the eigenspaces of all eigenvalues strictly greater than λ_k and is completed by an arbitrary subspace of the eigenspace of λ_k. In particular, when the data are rank-deficient ($\mathrm{rank}\,\hat{R}_x < k$, e.g. $N < k$), every direction in the whitened null space $\ker(\Sigma^{-1/2}\hat{R}_x\Sigma^{-1/2}) = \Sigma^{1/2}\ker\hat{R}_x$ is an equally good completion of $\Sigma^{-1/2}P^*$; in detector coordinates these are the directions $p \in \Sigma\ker\hat{R}_x$, the null space of $\hat{R}_x\Sigma^{-1}$, not $\ker\hat{R}_x$ itself.*

3. *The minimum value is $\mathcal{L}^* = \mathrm{Tr}(\Sigma^{-1}\hat{R}_x) - \sum_{j=1}^{k}\lambda_j$.*

Proof sketch. Whitening maps the constraint to $W^\dagger W = I_k$ and the loss to $\frac{1}{N}\|\tilde{X}(I - WW^\dagger)\|_F^2$. By Rayleigh–Ritz the minimum is attained exactly on the top-k invariant subspaces of $\Sigma^{-1/2}\hat{R}_x\Sigma^{-1/2}$, which map back to the generalized eigenspace of $\hat{R}_x\Sigma^{-1}$; these are the fixed points of the EMPCA iteration that attain the global minimum (Section 4, Appendix C.5). The full proof, including the minimum value and uniqueness up to gauge, is given in Appendix D.3. □

Remark 5.1 (What the theorem uses)**.** *The Bridge Theorem is a deterministic identity between two optimization problems: it holds for any data matrix, any $\Sigma \succ 0$, and any $N \geq 1$, and its proof does not use a noise model. The measurement model $x_i = s_i + n_i$, $n_i \sim \mathcal{N}_{\mathbb{C}}(0, \Sigma)$, enters only in motivating the choice of weight — Σ^{-1} is the Gaussian likelihood metric (Section 5.6) — and in the population statements of Section 4. Two further distinctions matter for reading the evidence. The theorem characterizes the* global *minimizers; the alternating EMPCA algorithm is guaranteed to reach a stationary point (Theorem C.1), and every k-dimensional invariant subspace is stationary, so agreement with the global optimum is a property of the initialization and of the spectrum, not of the identity (Appendix C.5). And the minimizer is unique only up to the latent gauge rotation $P^* \mapsto P^*Q$, $Q \in \mathcal{U}(k)$; physical quantities depend only on the reconstruction $P^*P^{*\dagger}\Sigma^{-1}x$ and are gauge-invariant. The controlled test S3 of Section 9 verifies the identity numerically with an independently trained tied-AE implementation (mean relative objective gap 1.95×10^{-16}). The tied-AE row of the public CRESST run is a different kind of check: the archived fitting code evaluates the tied AE through its closed form, i.e. through the same weighted-PCA routine that fits EMPCA, so its maximum principal angle of 1.91×10^{-6} degrees (Section 9.3) is a parameterization-consistency check of the two code paths on real data, not an independent optimization.*

Remark 5.2 (Sensitivity to covariance estimation error)**.** *The Bridge Theorem is conditional on Σ. In practice one substitutes an estimate $\hat{\Sigma} = \Sigma + \Delta\Sigma$ obtained from calibration data, and the fitted subspace is then the top-k eigenspace of the perturbed whitened operator $A(\hat{\Sigma}) = \hat{\Sigma}^{-1/2}\hat{R}_x\hat{\Sigma}^{-1/2}$ rather than of $A = A(\Sigma)$. The relevant perturbation is that of A, not of Σ: scaling Σ, $\Delta\Sigma$ and $\hat{R}_x$ together by a common factor leaves A, its eigengap and the fitted subspace unchanged while $\|\Delta\Sigma\|$ changes arbitrarily, so no bound of the form $c\|\Delta\Sigma\|/\Delta_\lambda$ with a universal c can hold. The dimensionless quantity that controls the effect is the* relative *covariance error*

$$\varepsilon = \left\|\Sigma^{-1/2}\,\Delta\Sigma\,\Sigma^{-1/2}\right\|_{\mathrm{op}}, \tag{29}$$

which is invariant under the common scaling. Compare the two fitted subspaces in one coordinate system, the whitened coordinates of the true Σ: $\mathcal{S} = \mathrm{range}(\Sigma^{-1/2}P^*)$ and $\hat{\mathcal{S}} = \mathrm{range}(\Sigma^{-1/2}\hat{P})$, where $\hat{P}$ is the detector-coordinate basis fitted with $\hat{\Sigma}$. (The unwhitening map changes with the covariance, so a comparison in detector coordinates is a comparison in the Σ^{-1} metric of the same two subspaces; principal angles are taken in that metric.) If $\varepsilon \leq \frac{1}{2}$ and $\Delta_\lambda = \lambda_k(A) - \lambda_{k+1}(A) > 0$, then

$$\sin\Theta\big(\hat{\mathcal{S}}, \mathcal{S}\big) \;\leq\; 6\,\varepsilon\,\frac{\lambda_1(A)}{\Delta_\lambda}. \tag{30}$$

The proof is in Appendix E.7: with $E = \Sigma^{-1/2}\Delta\Sigma\Sigma^{-1/2}$ the perturbed problem is, in true whitened coordinates, the eigenproblem of $A(I + E)^{-1}$, whose invariant subspaces

are $(I+E)^{1/2}$ applied to those of the symmetric operator $\tilde{A} = (I+E)^{-1/2}A(I+E)^{-1/2}$; the Davis–Kahan theorem [26, 27] bounds the rotation between the top-k eigenspaces of $\tilde{A}$ and A by $2\|\tilde{A}-A\|/\Delta_\lambda$ with $\|\tilde{A}-A\| \le (2\varepsilon+\varepsilon^2)\lambda_1(A)$, and the factor $(I+E)^{1/2}$ adds at most ε. Three consequences matter for the experiments. First, the fitted subspace moves continuously with the relative covariance error away from the oracle one — a mildly misestimated Σ yields a mildly rotated subspace — at a rate set by the conditioning ratio λ_1/Δ_λ of the retained whitened spectrum and not by the scale of Σ or of the data; the algebraic equivalence between the tied AE and weighted PCA (Theorem 5.1) is untouched, since it holds exactly for any common positive weight, $\hat{\Sigma}$ included. Second, the bound is vacuous when the spectral gap is small, which is precisely the regime where the retained rank is not well determined by the data; rank selection and covariance accuracy are therefore not independent choices (Appendix C.7). Third, ε is the diagnostic to report for a covariance-calibration experiment: Experiment S7 (Section 9) varies the amount of calibration data and is the empirical counterpart of this remark, and the covariance-error study of Section 9.2 (T15) reports ε, Δ_λ and the measured subspace error together. That study found the inequality satisfied on every fit, the hypothesis $\varepsilon \le \frac{1}{2}$ met only at calibration sizes $m \ge 2d$, and the bound uninformative (above 1) wherever it applied; it also found ε an incomplete diagnostic — a structured misspecification and a calibration-sample error of comparable ε differed by two orders of magnitude in effect — which an upper bound does not contradict but does not explain. The first-order statement that explains it is Proposition 5.2 below: only the off-diagonal block of E between the retained and the discarded eigenspaces rotates the subspace, each entry divided by its own pairwise gap, and a diagonal error acts only when it reorders the spectrum. It was checked as a conditional statement at fixed training data (T19, Section 9.2.2): within a factor of two on every fit at $\varepsilon \le 0.2$, no rotation from ordering-preserving diagonal errors, and an exchange of the retained subspace exactly at the sample-spectrum reversal. A finite-perturbation bound of that form is left open, and the present bound is uninformative on the planted two-axis configuration of Section 9.2.3 at any ε of practical size ($\lambda_1/\Delta_\lambda \approx 2.7\times10^3$ at rank six). A covariance model that is wrong in structure rather than merely noisy need not have a large ε — $\hat{\Sigma} = \Sigma^{1/2}(I+\delta vv^T)\Sigma^{1/2}$ with $\|v\|=1$ and $\delta > -1$ is a covariance with $\varepsilon = |\delta|$ — but its ε does not vanish as the calibration sample grows: the estimate converges to the misspecified model, and ε converges to that model's nonzero relative distance from Σ rather than to zero. That case is the subject of the GWOSC stage.

Proposition 5.2 (First-order displacement under covariance error)**.** *Let A be the whitened operator whose top-k eigenspace the estimator returns, with eigenpairs (λ_i, q_i), retained set $\mathcal{R} = \{1,\dots,k\}$ and $\lambda_k > \lambda_{k+1}$; let the covariance supplied to the estimator be $\hat{\Sigma} = \Sigma^{1/2}(I+E)\Sigma^{1/2}$ with E symmetric and $\varepsilon = \|E\|_{\mathrm{op}} < 1$; write $E_{ji} = q_j^T E q_i$; and let P and $\hat{P}$ be the orthogonal projectors, in the whitened coordinates of the true Σ, onto the*

subspaces fitted with Σ and with $\hat{\Sigma}$. Then $\hat{P} = P + \delta P + O(\varepsilon^2)$ with, in the eigenbasis of A,

$$q_j^T\, \delta P\, q_i = q_i^T\, \delta P\, q_j = -\frac{\lambda_j\, E_{ji}}{\lambda_i - \lambda_j} \quad (i \in \mathcal{R},\ j \notin \mathcal{R}), \tag{31}$$

and every other entry of δP zero. Consequently:

1. *only the coupling block $E_{\mathcal{R}^\perp\mathcal{R}}$ moves the subspace to first order; the blocks $E_{\mathcal{R}\mathcal{R}}$ and $E_{\mathcal{R}^\perp\mathcal{R}^\perp}$ deform the coordinates within the retained and within the discarded subspace and rotate neither;*

2. *each coupling entry is weighted by the discarded eigenvalue and divided by its own pairwise gap $\lambda_i - \lambda_j$, not by the global gap Δ_λ, so that $\|\delta P\|_F^2 = 2\sum_{i\in\mathcal{R},\, j\notin\mathcal{R}} \lambda_j^2 E_{ji}^2/(\lambda_i - \lambda_j)^2$ and $\sin\Theta(\hat{\mathcal{S}}, \mathcal{S}) = \|\delta P_{\mathcal{R}^\perp\mathcal{R}}\|_{\mathrm{op}} + O(\varepsilon^2)$;*

3. *if E commutes with A (diagonal in its eigenbasis, entries $e_i > -1$) the statement is exact at every order: the fitted eigenvectors are unchanged, the eigenvalues become $\lambda_i/(1+e_i)$, and the fitted subspace equals the oracle one if and only if $\min_{i\in\mathcal{R}} \lambda_i/(1+e_i) > \max_{j\notin\mathcal{R}} \lambda_j/(1+e_j)$, i.e. until the perturbed spectrum reorders, at which point the retained set is exchanged;*

4. *if the estimator diagonalizes a sample operator $\hat{A} = A + D$ rather than A, the same expansion in the eigenbasis of A gives the joint first-order displacement from the population subspace as $(D_{ji} - \lambda_j E_{ji})/(\lambda_i - \lambda_j)$: the training-sample term and the covariance term add and are separated by the same pairwise gaps. Expanding instead in the eigenbasis of $\hat{A}$, at fixed training data, removes D and gives (31) with $(\hat{\lambda}_i, \hat{q}_i)$ in place of (λ_i, q_i); that is the form the controlled test of Section 9.2.2 checks.*

The expansion requires $I + E \succ 0$, and its remainder is of order $\varepsilon^2\lambda_1(A)^2/\delta^2$ with $\delta = \min_{i\in\mathcal{R},\, j\notin\mathcal{R}} |\lambda_i - \lambda_j|$: the second-order term of the invariant-subspace expansion is $O(\|M\|^2/\delta^2)$ with $\|M\| \le \varepsilon\lambda_1(A)$, and since $\delta \le \lambda_k \le \lambda_1$ that term dominates the $O(\varepsilon^2)$ truncation of $(I + E)^{-1}$, which contributes only $\varepsilon^2\lambda_1/\delta$. The first-order statement is therefore informative in the regime $\varepsilon\lambda_1/\delta \ll 1$, not merely $\varepsilon \ll 1$. A near-singular $\hat{\Sigma}$, $\lambda_{\min}(I + E) \to 0$, is outside it: nothing here, and nothing in the global bound (30), describes that regime, which Section 9.2.2 reports separately. The proof is in Appendix E.7.

The Bridge Theorem closes the equivalence chain of the paper. The noise-aware tied linear AE is not an unrelated neural architecture: in the linear tied case it is the encoder–decoder *parameterization* of the same weighted low-rank reconstruction problem solved by EMPCA, which in turn degenerates to OF at rank 1. It is therefore the correct linear baseline against which nonlinear autoencoders should be compared.

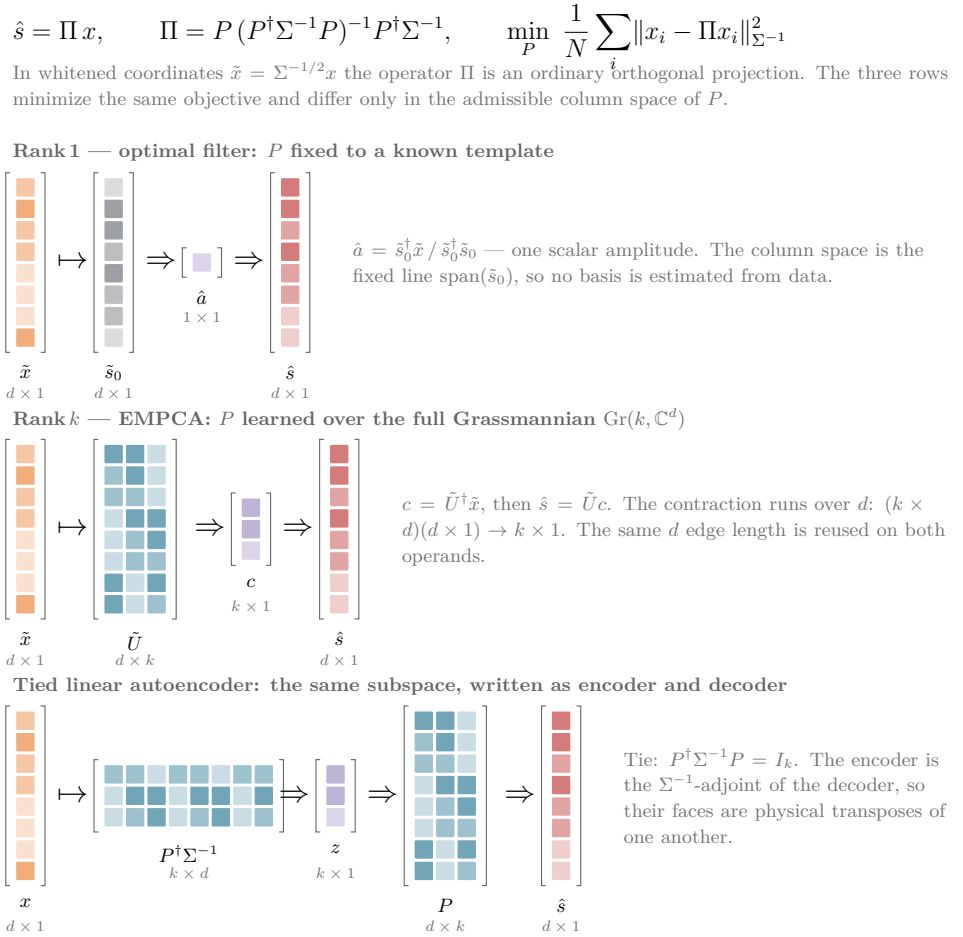


Figure 3: The three estimators of this paper as one Σ^{-1}-orthogonal projection, differing only in the admissible column space of P. Rank-1 optimal filtering fixes that column space to a known template; EMPCA learns a k-dimensional subspace over the full Grassmannian; the tied linear autoencoder writes the same subspace as an encoder and decoder related by $P^\dagger \Sigma^{-1} P = I_k$, so the two faces are transposes. Block widths and heights encode matrix shapes: equal shapes are drawn at equal edge lengths and the contracted dimension d is drawn at one length throughout. Cell counts illustrate axis structure and are not literal dimensions.

5.4 Why this architecture: a symmetry summary

There is a structural reason the tied Σ^{-1}-adjoint architecture and the χ^2 loss appear together. Two symmetry groups act on the problem: the *noise isometry group* of linear maps preserving the Σ^{-1} inner product, and the *latent gauge group* $\mathcal{U}(k)$ rotating the basis within the learned subspace. Requiring the quadratic reconstruction loss to be invariant under all noise isometries forces the loss weight to be proportional to Σ^{-1} (the Mahalanobis χ^2 loss, which is also the Fisher information metric of the Gaussian family [29]); requiring the reconstruction to be the metric-orthogonal projection onto the learned subspace forces the encoder to be the Σ^{-1}-adjoint of the decoder. Tied weights are thus not a regularization heuristic but an adjointness condition. What untying changes is stated precisely in Appendix E.2: under the same Σ^{-1} loss an unrestricted linear encoder–decoder pair has the same optimal reconstruction map — the Σ^{-1}-orthogonal projector onto the EMPCA subspace — whenever the whitened data have full rank and the spectral gap $\lambda_k > \lambda_{k+1}$ holds, so the tie does not enlarge or shrink the set of optimal reconstructions. It removes a redundant scale gauge $(D, E) \mapsto (DS, S^{-1}E)$ from the parameterization, and away from the optimum an untied product DE is in general neither idempotent nor Σ^{-1}-self-adjoint, so intermediate iterates are not projections at all. A controlled ablation of what this costs a first-order optimizer in practice belongs to the held evidence (Section 1).The formal statement (Theorem E.1) and its proof, together with the Grassmannian geometry of the subspace family, are given in Appendix E.

5.5 Scope of the equivalence claim

The equivalences of this section are rigorous for linear subspace models, Gaussian noise likelihood, matched whitening/weighting and preprocessing conventions, and identical rank k. They do not automatically extend to nonlinear autoencoders or mismatched objectives; what does transfer is the loss geometry, which the next subsection isolates and Section 7 develops.

5.6 The metric is architecture-neutral: the pair $(\Sigma^{-1}, \mathcal{F})$

The symmetry argument that selects the weighted loss for the linear architecture (Theorem E.1) acts on the residual vector alone and does not use linearity of the reconstruction map. We state it separately because it is an *axiom*, not a theorem about learning: it says what the residual metric must be *if* one requires invariance under the isometries of the measured noise, and that requirement is the modelling commitment this paper makes.

Proposition 5.3 (Architecture-neutral loss selection)**.** *Let $\hat{x} = D_\theta(x)$ be any reconstruction map, linear or not, and let $\mathcal{L}_M(x, \hat{x}) = (x - \hat{x})^\dagger M(x - \hat{x})$, $M \succ 0$, be a quadratic*

residual loss. Then $\mathcal{L}_M$ is invariant under the noise isometry group G_{noise} of Definition E.1 if and only if $M = \kappa\Sigma^{-1}$ for some $\kappa > 0$.

Proof. Invariance gives $Q^\dagger M Q = M$ for all $Q \in G_{\text{noise}}$. In whitened coordinates G_{noise} is the full unitary group, and the only positive-definite form invariant under it is a scalar multiple of the identity (Schur's lemma); transforming back gives $\kappa\Sigma^{-1}$. Conversely $\kappa\Sigma^{-1}$ is invariant by the definition of G_{noise}. The argument constrains the quadratic form on residuals and is silent about D_θ. □

The proposition is an equivalence between a symmetry requirement and a metric, so it does not by itself show that every learning objective ought to be Σ^{-1}-weighted; what it shows is that the single assumption "the loss treats all noise isometries alike" already contains the Gaussian likelihood metric, and that this assumption has nothing to do with the reconstruction map. Under that assumption the residual metric is fixed by the likelihood, independently of how the reconstruction is parameterized, and what an architecture decides is only the admissible reconstruction class $\mathcal{F}$. A reconstruction method is therefore the pair $(\Sigma^{-1}, \mathcal{F})$, and the classes of Sections 3–5 and Section 6 — a fixed template, a learned subspace, its tied encoder–decoder form, and its separable restriction — differ in $\mathcal{F}$ alone. Whether the metric choice *matters* for a given class is then a measurable property of the instrument and the class, which is what the experiments test.

6 Separable Subspaces: the Kronecker-Restricted Class

The three estimators assembled so far differ only in the admissible signal model: a fixed template, a learned rank-k subspace, and the encoder–decoder parameterization of that subspace. This section adds a fourth member that *restricts* the learned class rather than enlarging it. For a multi-channel measurement the natural representation is a matrix, one axis indexing the readout channel and the other the time sample, and the simplest structural prior one can impose on the learned subspace is that it factor across the two axes. The resulting estimator — a noise-factored projection autoencoder (NFPA) — keeps the likelihood of Section 2 unchanged and alters only the subspace family. It is the covariance-aware member of the multilinear-PCA family [15], related to generalized least-squares matrix decompositions [14] and matrix-normal PCA [30], and we claim no new factorization. Its role here is that it is the first class in the hierarchy for which the architecture, not the metric, decides what can be represented, and the class for which

recovery of a planted physical structure can be measured directly. Statements in this section are given for real data; the complex case follows with $T \to \dagger$.

6.1 Two-axis measurements and per-axis whitening

Let each event be a matrix $Z \in \mathbb{R}^{C\times d}$, with rows indexed by channel and columns by time sample, and let $z = \mathrm{vec}(Z) \in \mathbb{R}^{Cd}$ be its column-stacked vectorization. The joint-channel objective of Section 3.3 applies to z with a $Cd\times Cd$ covariance. When the noise covariance is itself separable,

$$\Sigma = \Sigma_t \otimes \Sigma_c, \qquad \Sigma_c \in \mathbb{R}^{C\times C},\ \Sigma_t \in \mathbb{R}^{d\times d}, \tag{32}$$

whitening factors across the axes,

$$\tilde{Z} = \Sigma_c^{-1/2}\, Z\, \Sigma_t^{-1/2}, \qquad \mathrm{vec}(\tilde{Z}) = \left(\Sigma_t^{-1/2} \otimes \Sigma_c^{-1/2}\right) \mathrm{vec}(Z) = \Sigma^{-1/2} z, \tag{33}$$

by the identity $\mathrm{vec}(AZB) = (B^T \otimes A)\,\mathrm{vec}(Z)$. The separable-covariance assumption is what makes the two-axis whitening well defined; for a non-separable Σ one whitens z with the full $\Sigma^{-1/2}$ and the factor structure below is imposed on the whitened vector directly, at the cost of the per-axis interpretation of the factors.

6.2 The NFPA class

Definition 6.1 (Noise-factored projection autoencoder)**.** *Fix ranks $k_c \le C$ and $k_t \le d$. An NFPA consists of orthonormal factor bases $U_c \in \mathbb{R}^{C\times k_c}$ and $U_t \in \mathbb{R}^{d\times k_t}$, $U_c^T U_c = I_{k_c}$, $U_t^T U_t = I_{k_t}$, with code and reconstruction*

$$\Gamma_i = U_c^T \tilde{Z}_i U_t \in \mathbb{R}^{k_c\times k_t}, \qquad \hat{\tilde{Z}}_i = U_c U_c^T\, \tilde{Z}_i\, U_t U_t^T, \qquad \hat{Z}_i = \Sigma_c^{1/2} \hat{\tilde{Z}}_i \Sigma_t^{1/2}, \tag{34}$$

trained by minimizing the whitened objective $N^{-1}\sum_i \|\tilde{Z}_i - \hat{\tilde{Z}}_i\|_F^2$, which equals the unified objective (4) *evaluated on z_i.*

The encoder is again the Σ^{-1}-adjoint of the decoder — the code is the whitened inner product with the factor bases — so the tied structure of the Bridge Theorem is inherited, not re-imposed. What is new is the admissible family.

Proposition 6.1 (Kronecker-restricted bridge)**.** *In vectorized form the NFPA reconstruction is*

$$\mathrm{vec}(\hat{\tilde{Z}}_i) = \left(U_t U_t^T \otimes U_c U_c^T\right) \mathrm{vec}(\tilde{Z}_i), \tag{35}$$

the orthogonal projector of rank $k_c k_t$ onto $\mathrm{range}(U_t) \otimes \mathrm{range}(U_c) \subset \mathbb{R}^{Cd}$*. NFPA therefore minimizes the rank-$k_c k_t$ EMPCA/tied-autoencoder objective of Theorem 5.1 restricted to the embedded product*

$$\mathrm{Gr}(k_c, C) \times \mathrm{Gr}(k_t, d) \ \hookrightarrow\ \mathrm{Gr}(k_c k_t, Cd), \tag{36}$$

and its optimum equals the unrestricted rank-$k_c k_t$ optimum if and only if at least one unrestricted minimizer has a (k_c, k_t)-separable subspace.

Proof. Equation (35) is the vectorization identity applied to (34). The Kronecker product of two symmetric idempotents is a symmetric idempotent of rank $k_c k_t$, hence the orthogonal projector onto the product subspace. Substituting into the whitened rank-k projection loss of Theorem 5.1 with $k = k_c k_t$ shows that NFPA is that objective with the feasible set reduced from $\mathrm{Gr}(k_c k_t, Cd)$ to the image of (36). Restricted and unrestricted minima coincide exactly when the feasible set contains an unrestricted minimizing projector. □

Two consequences deserve emphasis because they are easy to get wrong. First, the restriction stays active even when one factor is full rank: with $k_c = C$ the family is $\mathrm{range}(U_t) \otimes \mathbb{R}^C$, which is still a strict subset of $\mathrm{Gr}(Ck_t, Cd)$ unless $k_t = d$ as well, so "NFPA reduces to EMPCA when a factor is full rank" is false. Second, when the restriction binds, the gap between the two optima is the price of the architecture prior and not a failure of the metric; it is measured by fitting the restricted and unrestricted classes at matched rank $k_c k_t$ under the same Σ^{-1}.

Corollary 6.2 (Rank-one NFPA and the optimal filter)**.** *Assume isotropic whitened noise and a unique leading population signal eigendirection separated by a nonzero eigengap, as in Theorem 4.1. If that direction is generated by a separable whitened template $\tilde{s}_0 = v\, t^T$, then population NFPA with $k_c = k_t = 1$ learns $v/\|v\|$ and $t/\|t\|$ up to signs, and its scalar code is proportional to the joint-channel OF amplitude of Section 3.3.*

Proof. The leading vectorized direction of the population second moment is $(t \otimes v)/(\|t\|\|v\|)$, which is feasible for rank-one NFPA, so the restricted and unrestricted rank-one optima coincide (Proposition 6.1) and Theorem 4.1 applies. The code $U_c^T \tilde{Z}_i U_t = \langle \mathrm{vec}(\tilde{Z}_i),\, t \otimes v \rangle / (\|t\|\|v\|)$ is the whitened matched-filter coefficient, hence the OF amplitude up to normalization. □

The corollary places the separable class inside the same degeneracy structure as Table 1: at $(k_c, k_t) = (1, 1)$ and a separable template it is basic OF; at general ranks it is EMPCA restricted to a product subspace; between the two it is neither, and the restriction is a modeling choice whose cost is measurable.

6.3 Alternating least squares and gauge

The joint problem over (U_c, U_t) is nonconvex on the product manifold, but each factor subproblem is exact. With U_t fixed, the objective is maximized over U_c by the top-k_c eigenvectors of $\sum_i \tilde{Z}_i U_t U_t^T \tilde{Z}_i^T$; with U_c fixed, by the top-k_t eigenvectors of $\sum_i \tilde{Z}_i^T U_c U_c^T \tilde{Z}_i$.

Alternating these two Rayleigh–Ritz steps is monotone in the objective, so the objective values converge, and every accumulation point of the iterates at which both block eigenproblems have a spectral gap at the retained rank is a fixed point of the alternation and hence a stationary point of the product-manifold problem; convergence of the factor sequence itself to a single stationary point is not claimed, because a repeated block eigenvalue makes the block step nonunique and monotone objective values alone do not exclude oscillation between equivalent choices. It is the covariance-aware form of the alternating scheme used for multilinear PCA [15], with the Tucker-2 decomposition as the underlying algebra [31]. No statement is made about *which* stationary point is reached: unlike the unrestricted problem, whose non-global stationary points are saddles or, for the bottom-k subspace, a maximum (Remark C.1), the product-manifold problem has no such characterization here, and multiple restarts are the only check used. The learned factors carry the product gauge $U_c \mapsto U_c Q_c$, $U_t \mapsto U_t Q_t$ with $Q_c \in O(k_c)$, $Q_t \in O(k_t)$, a proper subgroup of the $O(k_c k_t)$ gauge of the unrestricted class; only the projectors $U_c U_c^T$ and $U_t U_t^T$ are physical, and every subspace comparison in this paper is gauge-invariant by construction (Appendix E.4).

Parameter count and cost. The three classes are also separated by how many numbers they learn. A rank-k subspace of $\mathbb{R}^{Cd}$ has $k(Cd-k)$ free parameters (the dimension of the Grassmannian), against $k_c(C-k_c)+k_t(d-k_t)$ for the product class and 0 for a fixed optimal-filter template; at the $C = 8$, $d = 256$, $(k_c, k_t) = (2, 3)$, $k = k_c k_t = 6$ configuration of the planted-subspace study these are 12252, 771, and 0. One ALS sweep forms two Gram matrices at cost $O(NCd(k_c + k_t) + N(C^2 k_t + d^2 k_c))$ and solves two eigenproblems of sizes C and d. The unrestricted estimator need not form the $Cd \times Cd$ covariance: a thin or truncated SVD of the $N \times Cd$ whitened data matrix costs $O(N(Cd)^2)$ for the thin factorization or $O(NCd\,k)$ per iteration for a truncated iterative one, and the dense $O(NC^2d^2) + O(C^3d^3)$ route is only the worst case; the experiment repository's weighted PCA uses the rectangular SVD. Any efficiency claim for the separable class is therefore to be measured against the thin/truncated-SVD comparator, with runtime and memory reported, and none is made in this paper. We report the parameter counts alongside every comparison between the classes so that a difference in recovery is not read as a difference in capacity.

6.4 Approximation error versus metric error

The discrepancy between a fitted NFPA subspace and the unrestricted EMPCA subspace has four sources that must not be conflated: a *metric* term, present only if the model is trained under a loss other than Σ^{-1} and not removable by more data; a *class-restriction*

(approximation) term, the population-level cost of the separability prior, which is nonnegative by construction because the product family is a subset of the unrestricted one; a *statistical* term from finite N; and an *optimization* term from the nonconvex alternation. Once the metric is correct the first vanishes identically, and the remaining three are the ordinary terms of any constrained estimator. The statement the evidence uses separates them exactly and bounds the approximation term from below by a quantity computable from the population covariance alone.

Proposition 6.3 (Exact risk decomposition and a lower bound on the restriction cost)**.** *Let $K \succeq 0$ be the whitened clean-signal second-moment operator on $\mathbb{R}^{Cd}$, with eigenvalues $\mu_1 \geq \mu_2 \geq \cdots$ and eigenvectors $v_1, v_2, \ldots$; for an orthogonal projector P of rank $k = k_c k_t$ let $R_K(P) = \operatorname{tr} K - \operatorname{tr}(PK)$ be its clean-signal projection risk; let $P_U^\star$ be the top-k eigenprojector of K, with gap $\Delta_k = \mu_k - \mu_{k+1}$; let $\mathcal{F}$ be the set of product projectors $P_t \otimes P_c$ of ranks (k_t, k_c); and let $P_{\mathcal{F}}^\star \in \arg\min_{\mathcal{F}} R_K$, which exists because $\mathcal{F}$ is compact. For any fitted $\hat{P}_{\mathcal{F}} \in \mathcal{F}$ and any fitted rank-k orthogonal projector $\hat{P}_U$,*

$$\begin{aligned} R_K(\hat{P}_{\mathcal{F}}) - R_K(\hat{P}_U) = \underbrace{R_K(P_{\mathcal{F}}^\star) - R_K(P_U^\star)}_{a \geq 0} + \underbrace{R_K(\hat{P}_{\mathcal{F}}) - R_K(P_{\mathcal{F}}^\star)}_{x_{\mathcal{F}} \geq 0} \\ - \underbrace{R_K(\hat{P}_U) - R_K(P_U^\star)}_{x_U \geq 0}, \end{aligned} \tag{37}$$

an identity. The approximation cost satisfies, for every $P_F \in \mathcal{F}$,

$$\begin{aligned} R_K(P_F) - R_K(P_U^\star) \;&\geq\; \Delta_k\big(k - \operatorname{tr}(P_F P_U^\star)\big), \\ \operatorname{tr}(P_F P_U^\star) \;&\leq\; \min\Big\{k,\; \sum_{i \leq k_c} \mu_i(M_c),\; \sum_{i \leq k_t} \mu_i(M_t)\Big\}, \end{aligned} \tag{38}$$

where $M_c = \operatorname{tr}_t P_U^\star \in \mathbb{R}^{C \times C}$ and $M_t = \operatorname{tr}_c P_U^\star \in \mathbb{R}^{d \times d}$ are the partial traces of the unrestricted projector over the temporal and the channel factor and $\mu_i(\cdot)$ are eigenvalues in decreasing order. Hence a is bounded below by a quantity computed from K alone, and when $\Delta_k > 0$ it vanishes if and only if $P_U^\star \in \mathcal{F}$. The proof is in Appendix E.4.

The identity is what a finite-sample class contrast measures; the bound is what makes the sign of its first term a fact about the generator rather than about the fits. On the planted family of Section 9.2.3 it gives a provably positive cost at both nonzero nonseparabilities (0.10 and 0.55, against best-found values 2.71 and 2.82), while at $\alpha = 0$ the gap Δ_k is numerically zero and $a = 0$ follows from the support argument instead (the excited signal lies in the product class). Two cautions apply to measuring the terms. An observed signed contrast between the held-out weighted residuals of NFPA and of the unrestricted class at matched rank is a *finite-sample class contrast*: it mixes the class-restriction term with the statistical and optimization terms, and it can be negative — the

restriction acting as a regularizer at finite N — without the population approximation error being negative, which it cannot be. The controlled evidence reports it under that name (the controlled separable-class evidence, held from this version (Section 1)), and a map of it over N, α and SNR is in Section 9.2 (T14 and T20): at $T = 256$ the contrast is negative or indistinguishable from zero in every cell and no crossover was observed; at $T = 32$ the population restriction cost at $\alpha > 0$ was established independently of the fits (best-found 2.7–2.8, provably at least 0.10 and 0.55 on the reference covariance by (38)), and on fresh seeds the contrast reverses sign between $N = 800$ and $N = 1600$, so the regularizing and the approximation-dominated regimes have both been observed on one configuration, with the crossover bracketed rather than located. And agreement of the objective across random restarts bounds the optimization term only for the restarts tried; it does not prove that the landscape is unimodal or that the global optimization error is zero. A subspace bound of the form $\sin\Theta(\mathcal{S}_{\mathrm{NFPA}}, \mathcal{S}_{\mathrm{EMPCA}}) \leq F(\epsilon_{\mathrm{kron}}, \Delta_\lambda)$, with ϵ_{kron} the best Tucker-2 approximation error to the EMPCA projector, would be a stronger statement in the spirit of Appendix E.7; we do not pursue it. The risk decomposition is the quantity the class contrast measures and the evidence reports, and a principal-angle bound would add nothing to that evidence.

What the separable class makes available, and what the unrestricted class cannot, is a truth-referenced test: when the generating signal is planted with a known channel factor and a known temporal factor, the learned $U_c U_c^T$ and $U_t U_t^T$ can be compared with the planted projectors axis by axis, so the question "did the metric change what physical structure was represented?" has a direct answer rather than one mediated by a residual. That test is the subject of the controlled evidence for this class, which is complete and is held from this version pending an external double-blind review (Section 1).

7 Loss Geometry: Why MSE Can Be Wrong

The Bridge Theorem has an immediate consequence for machine-learning loss design. Under the measurement model $x_i = s_i + n_i$, $n_i \sim \mathcal{N}_{\mathbb{C}}(0, \Sigma)$, the Gaussian negative log-likelihood of a reconstruction $\hat{x}_i$ is, up to constants,

$$\mathcal{L}_{\Sigma^{-1}} = \frac{1}{N} \sum_{i=1}^{N} (x_i - \hat{x}_i)^\dagger \Sigma^{-1} (x_i - \hat{x}_i), \tag{39}$$

whereas the standard isotropic loss used by ordinary PCA and most autoencoders is

$$\mathcal{L}_{\mathrm{MSE}} = \frac{1}{N} \sum_{i=1}^{N} \|x_i - \hat{x}_i\|_2^2. \tag{40}$$

The two coincide (up to scale) exactly when $\Sigma = \sigma^2 I$. Isotropic MSE is therefore not a neutral default: it is the Gaussian likelihood for white noise, and under colored, heteroskedastic, or correlated detector noise it is a *misspecified* likelihood.

The residual metric is therefore fixed by the likelihood whatever the reconstruction map: Proposition 5.3 (Section 5.6) shows that invariance under the measured noise isometries is equivalent to $M \propto \Sigma^{-1}$, so a reconstruction method is the pair $(\Sigma^{-1}, \mathcal{F})$ and the choice of loss geometry is separable from the choice of class. This section first states which risk that geometry predicts, and then asks what follows when it is replaced by the Euclidean one.

7.1 Which risk the weighted residual predicts

The likelihood fixes a residual geometry. It does not by itself say which scientific error a reconstruction chosen in that geometry makes small, and the claims of this paper have to name their target. Four risks are in play for a reconstruction $\hat{x} = \hat{x}(x)$ of $x = s + n$, written here for real data with $\mathbb{E}[n] = 0$ and $\mathbb{E}[nn^T] = \Sigma$ (the complex circular case is identical with $T \to \dagger$):

$$\mathcal{R}_{\text{obs}}(\hat{x}) = \mathbb{E}\,(x - \hat{x})^T \Sigma^{-1} (x - \hat{x}) \qquad \text{(observation residual)}, \tag{41}$$

$$\mathcal{R}_{\text{clean},\Sigma}(\hat{x}) = \mathbb{E}\,(s - \hat{x})^T \Sigma^{-1} (s - \hat{x}) \qquad \text{(whitened clean-signal risk)}, \tag{42}$$

$$\mathcal{R}_{\text{clean},I}(\hat{x}) = \mathbb{E}\,\|s - \hat{x}\|_2^2 \qquad \text{(Euclidean clean-signal risk)}, \tag{43}$$

$$\mathcal{R}_{\vartheta}(\hat{\vartheta}) = \mathbb{E}\,\|\hat{\vartheta} - \vartheta\|^2 \qquad \text{(physical-parameter risk)}. \tag{44}$$

Each fit in this paper minimizes the sample version of the residual it is defined by: the covariance-weighted classes minimize (41), the raw-PCA and isotropic comparison arms minimize the Euclidean residual, and every residual *comparison* in Section 9 reports the held-out sample version of (41) for both. The TIDMAD stage scores (44) for the injected amplitude, and the synthetic tests with a recorded clean signal score (43) in their archived feature representation. The following identity says exactly when the observation residual predicts the whitened clean-signal risk. It is elementary unbiased-risk accounting of the kind introduced by Mallows and Stein [32, 33], specialized to a fixed linear map in whitened coordinates.

Proposition 7.1 (Fixed affine-estimator risk identity)**.** *Work in real whitened coordinates, $y = \Sigma^{-1/2} x = t + w$ with $t = \Sigma^{-1/2} s$ and $w = \Sigma^{-1/2} n$. Let $\hat{y} = By + b$ be an affine reconstruction with $B \in \mathbb{R}^{d \times d}$ and $b \in \mathbb{R}^d$ fixed — fitted on other data, or population quantities — and let the evaluation noise satisfy, conditionally on (B, b), $\mathbb{E}[w] = 0$, $\mathbb{E}[ww^T] = I$ and $\mathbb{E}[tw^T] = 0$. Then, with expectations conditional on (B, b),*

$$\mathcal{R}_{\text{clean},\Sigma} = \mathbb{E}\|(I-B)t - b\|^2 + \operatorname{tr}(BB^T), \qquad \mathcal{R}_{\text{obs}} = \mathbb{E}\|(I-B)t - b\|^2 + \operatorname{tr}\big((I-B)(I-B)^T\big), \tag{45}$$

so that

$$\mathcal{R}_{\text{clean},\Sigma} = \mathcal{R}_{\text{obs}} + 2\,\mathrm{tr}(B) - d. \tag{46}$$

Every idempotent B of rank k — an orthogonal projector or an oblique one — has $\mathrm{tr}\,B = k$ *and correction $2k - d$; the zero map is the case $k = 0$; a shrinkage estimator $B = \sum_j \beta_j u_j u_j^T$ has correction $2\sum_j \beta_j - d$. For an orthogonal projector Π with $b = 0$ the two risks separate as $\mathbb{E}\|(I-\Pi)t\|^2 + k$ and $\mathbb{E}\|(I-\Pi)t\|^2 + (d-k)$; that separate form uses orthogonality, the trace correction* (46) *does not. In the complex circular convention $T \to \dagger$ and* $\mathrm{tr}\,B \to \mathrm{Re}\,\mathrm{tr}\,B$.

Proof. $\hat{y} - t = (B-I)t + b + Bw$ and $y - \hat{y} = (I-B)t - b + (I-B)w$. Conditionally on (B, b) the cross terms vanish, $\mathbb{E}\big[((B-I)t+b)^T Bw\big] = \mathrm{tr}\big(B\,\mathbb{E}[wt^T](B-I)^T\big) + b^T B\,\mathbb{E}[w] = 0$ and likewise for the second, while $\mathbb{E}\|Bw\|^2 = \mathrm{tr}(BB^T)$ and $\mathbb{E}\|(I-B)w\|^2 = \mathrm{tr}((I-B)(I-B)^T) = d - 2\,\mathrm{tr}\,B + \mathrm{tr}(BB^T)$. Subtracting gives (46); for idempotent B, $\mathrm{tr}\,B = \mathrm{rank}\,B$. □

Five remarks fix what the identity does and does not say, and they bound what the rest of the paper may claim.

(a) *Population, conditional.* The identity is a statement about population risks conditional on the fitted (B, b). Its moment conditions must hold for the evaluation noise given the training and calibration data: a chronological split makes them plausible but does not by itself establish them under dependent noise, and a wrong Σ violates $\mathbb{E}[ww^T] = I$ and biases the correction.

(b) *Ordering.* Within any set of fixed estimators of equal trace — in particular all rank-k idempotent maps, oblique ones included, so that the raw-PCA projector expressed in the true whitened coordinates is covered as well — the population ordering by the observation residual is the population ordering by the whitened clean-signal risk. This is the precise sense in which the covariance-weighted subspace of Theorem 5.1 is the better reconstruction, and the sense in which a held-out $\mathcal{R}_{\text{obs}}$ estimates clean-signal error. Two realized held-out averages differ by the constant only in expectation: on a finite evaluation sample the orderings can differ, and every comparison in Section 9 is reported with its own interval.

(c) *Roles.* Training minimizes a sample objective; a held-out residual estimates the population risk of the fitted map; validation selects among fitted maps by such estimates; a final test compares the selected maps once. The identity connects the second of these to the clean-signal risk and says nothing about the first: a training-sample optimum need not minimize the population risk.

(d) *Silent about other targets.* Maps of different trace need the correction (Appendix C.7); a nonlinear or data-dependent map is outside the identity, and for

a nonlinear same-input reconstruction the likelihood geometry alone does not even exclude the identity map, whose observation residual is zero; and neither the Euclidean clean-signal risk nor a physical-parameter risk is implied.

(e) *Supervised MSE.* Training on a clean target with isotropic MSE is therefore not an assertion that the measurement noise is white: it is the choice of $\mathcal{R}_{\mathrm{clean},I}$ as the prediction risk, a legitimate scientific target that the weighted residual need not predict.

The following example makes the last point exact.

Example 7.2 (The weighted residual and the Euclidean clean-signal error can order oppositely)**.** *Let $d = 2$, $k = 1$, with independent zero-mean Gaussian signal and noise, $\mathbb{E}[ss^T] = \mathrm{diag}(100, 30)$ and $\Sigma = \mathrm{diag}(4, 1)$. Raw PCA retains the first coordinate, because the observed variances are $(104, 31)$; weighted PCA retains the second, because the whitened observed variances are $(26, 31)$. With the exact covariance, infinite data and exact optimization,*

Population risk	*raw PCA (e_1)*	*weighted PCA (e_2)*	*preferred*
$\mathcal{R}_{\mathrm{obs}}$ *(weighted residual)*	31	26	*weighted PCA*
$\mathcal{R}_{\mathrm{clean},\Sigma}$ *(whitened clean error)*	31	26	*weighted PCA*
$\mathcal{R}_{\mathrm{clean},I}$ *(Euclidean clean error)*	34	101	*raw PCA*

Both arms are rank-one idempotent maps ($\mathrm{tr}\, B = 1$; with a diagonal Σ the raw-PCA projector is orthogonal in whitened coordinates as well), so Proposition 7.1 applies to both with $2k - d = 0$, and the first two rows agree as it requires. The third does not: the Euclidean clean-signal error of the projector onto e_j is the signal variance it discards plus the noise variance it keeps, $30 + 4$ against $100 + 1$. The example does not contradict Theorem 5.1; it separates the theorem's objective from another legitimate risk, and it shows that a disagreement between the two orderings is not by itself evidence for the likelihood-aligned one.

7.2 From observation geometry to representation geometry

Two consequences of the same metric are stated here because they are needed to say what the controlled evidence for the separable class (the controlled separable-class evidence, held from this version (Section 1)) does and does not measure, and because the companion study of failure diagnostics uses them as its instruments; the proofs are in Appendix E.6. Both concern a forward model $x = g(\vartheta) + n$ with $\vartheta \in \mathbb{R}^m$ the physical latent variables (amplitude, rise and decay time, position, energy, ...), an observation $x \in \mathbb{R}^d$, real

Gaussian noise $n \sim \mathcal{N}(0, \Sigma)$, and Jacobian $J_g(\vartheta) = \partial g/\partial \vartheta \in \mathbb{R}^{d\times m}$. Statements are given for real observations and latents; for the complex circular convention used elsewhere in the paper the same quadratic forms appear with the overall factor recorded in Appendix A. The linear classes of Sections 3–6 are the case $g(\vartheta) = A\vartheta$ for a fixed $A \in \mathbb{R}^{d\times m}$.

Proposition 7.3 (Pullback of the noise metric)**.** *For fixed Σ the Fisher information of ϑ is*

$$I(\vartheta) = J_g(\vartheta)^T \, \Sigma^{-1} \, J_g(\vartheta), \tag{47}$$

a positive-semidefinite quadratic form on latent displacements, positive definite exactly where $J_g(\vartheta)$ has full column rank. A latent displacement $\delta\vartheta$ has squared length $\delta\vartheta^T I(\vartheta)\, \delta\vartheta$: the Gaussian likelihood pulls the Σ^{-1} inner product back onto latent space, and the length of $\delta\vartheta$ is the distance it moves the observation in units of what the noise can resolve. More generally, let h be any representation of the data and let $y = g_{\text{out}}(h) \in \mathbb{R}^d$ be a prediction in observation space*, compared with x under the same covariance Σ, with Jacobian $J_h = \partial y/\partial h$; then the same construction gives the pulled-back form $J_h^T \Sigma^{-1} J_h$ on the representation, degenerate on* $\ker J_h$*. Where $I(\vartheta)$ is singular, the latent coordinates are not locally identifiable to first order: there is a direction v with $J_g(\vartheta)v = 0$, so $g(\vartheta + \epsilon v) = g(\vartheta) + o(\epsilon)$ (with $O(\epsilon^2)$ when g is twice continuously differentiable).*

The pulled-back form is a metric only where it is positive definite; elsewhere it is a degenerate quadratic form, which is exactly the information that the companion study uses. A Euclidean displacement $\|\delta h\|_2^2$ is the wrong statistic in the same sense that isotropic MSE is (Proposition 5.3), with the same qualification: the two coincide whenever $J_h^T \Sigma^{-1} J_h$ is a multiple of the identity, as it is in whitened orthonormal coordinates, so Euclidean latent distance is not wrong in general but is correct only in such coordinates. Singularity of $I(\vartheta)$ is a first-order degeneracy; it does not by itself imply that g fails to be locally injective (a map can be injective with a vanishing Jacobian), only that no estimator can resolve the direction v at rate ϵ.

Proposition 7.4 (Excited support, linear class)**.** *Let $g(\vartheta) = A\vartheta$ and let the training distribution vary only the latent coordinates $S \subseteq \{1, \dots, m\}$, so that $\vartheta = \vartheta_0 + \delta$ with δ supported on S. Write $t = \Sigma^{-1/2} A\delta$ for the whitened signal displacement, $T_S = \operatorname{range}(\Sigma^{-1/2} A_{\cdot S})$ for the excited span in whitened coordinates, and $R_t = \mathbb{E}[tt^T]$ for its second moment, and assume (i) the whitened noise $w = \Sigma^{-1/2} n$ satisfies $\mathbb{E}[w] = 0$, $\mathbb{E}[ww^T] = I$; (ii) $\mathbb{E}[tw^T] = 0$; (iii) the fixed mean $\Sigma^{-1/2} A\vartheta_0$ is either subtracted before reconstruction or included in the model, so that the reconstruction acts on $t + w$; and (iv) $\mathbb{E}[t] = 0$. Consider the population objective over the class of rank-k orthogonal projectors Π in whitened coordinates, $J(\Pi) = \mathbb{E}\|(I - \Pi)(t + w)\|^2$. Then:*

1. *$J(\Pi) = \operatorname{tr} R_t - \operatorname{tr}(\Pi R_t) + (d-k)$, so J depends on Π only through $\operatorname{tr}(\Pi R_t)$ and is* exactly *invariant, not merely flat to second order, under every rotation $Q \in O(d)$ that fixes T_S pointwise: $J(Q\Pi Q^T) = J(\Pi)$.*

2. *Let $r = \operatorname{rank} R_t \le \dim T_S$. If $k \ge r$, Π minimizes J if and only if $\operatorname{range}\Pi \supseteq \operatorname{range} R_t$; the remaining $k-r$ dimensions may lie anywhere in $(\operatorname{range} R_t)^\perp$ and are not determined by the population objective. If $k < r$, the minimizers are the top-k eigenspaces of R_t, contained in T_S, with the same gap condition as Theorem 5.1.*

3. *Write $B = \Sigma^{-1/2} A_{\cdot S}$ and $K = \mathbb{E}[\delta_S \delta_S^T] \succeq 0$, so that $R_t = BKB^T$. Then $\operatorname{range} R_t = B(\operatorname{range} K) \subseteq T_S$, with equality — the excited span is fully excited — if and only if K is positive definite on the observable latent subspace $\operatorname{range}(B^T) = (\ker B)^\perp$, i.e. $\ker K \cap \operatorname{range}(B^T) = \{0\}$. In particular $K \succ 0$ is sufficient whatever the rank of B; it is also necessary only when B is injective. Latent coordinates that B does not see may be unexcited without loss, and an excited direction of K lying in $\ker B$ contributes nothing to T_S.*

Three qualifications keep the proposition from saying more than it does. First, it is a statement about the population objective over *unrestricted* projectors; for the separable class of Section 6 the objective restricted to product projectors is still a function of $\operatorname{tr}(\Pi R_t)$ alone, but a rotation fixing T_S does not in general map a product projector to a product projector, so the unrestricted invariance statement is not inherited by NFPA and no claim about its complement directions follows from it. Second, nothing here is a finite-sample statement. With N training events the sample second moment of $t + w$ replaces $R_t + I$, its noise part is no longer isotropic, and which of the $k - r$ complement directions an estimator selects is decided by the sampling fluctuations, the initialization and the optimizer together; the population identity does not predict that those directions are free, only that the population objective is indifferent to them. (In the finite-sample regime of the controlled evidence, $N < d$, the sample noise covariance also has $d - N$ zero eigenvalues, so the sampling geometry is further from the isotropic population one than the bulk edge alone suggests.) Conversely, a finite-sample observation that the complement is reproducible across initializations on a fixed dataset does not show that it was determined by the physical signal. The controlled experiment P1-E11 (the controlled separable-class evidence, held from this version (Section 1)) measures exactly this finite-sample behaviour and is reported separately from the identity above. Third, the nonlinear case is a remark, not a proposition: for g continuously differentiable at ϑ_0 the same computation applies to the linearization there with $A = J_g(\vartheta_0)$ and $T_S(\vartheta_0) = \operatorname{range}(\Sigma^{-1/2} J_g(\vartheta_0)_{\cdot S})$, up to $o(\|\delta\|)$ terms in the signal ($O(\|\delta\|^2)$ when J_g is locally Lipschitz, in particular when g is twice continuously differentiable), and says nothing about which parameterization of a nonlinear reconstruction map the invariance would refer to.

Together with Proposition 5.3 these identify three determinants of what a calibrated readout can learn: the measurement fixes the metric Σ^{-1}, the architecture fixes the admissible class $\mathcal{F}$, and the training data fix the excited support S. The first two are the subject of this paper; the controlled evidence for the separable class (the controlled separable-class evidence, held from this version (Section 1)) tests whether the wrong metric learns the wrong physical structure when $\mathcal{F}$ and S are held fixed. The deployment consequences of the third — what a frozen model does when it meets data outside its excited support, or noise whose covariance differs from the training one — are stated in Section 10 as predictions for the companion study, not as results of this paper.

The geometric mechanism. The effect of the misspecification is not a rescaling of the objective; the two losses define different inner products and hence different notions of which directions in waveform space matter. Isotropic MSE weights every time or frequency bin equally, so the learned subspace is driven by raw waveform variance — which, under colored noise, is typically dominated by noise-rich, low-information directions. The Σ^{-1}-weighted loss weights residuals by inverse noise power, so the learned subspace is driven by signal-to-noise structure: noisy directions are down-weighted and informative directions emphasized. The same model class, rank, data, and optimizer can therefore converge to substantially different subspaces under the two losses.

Metric reversal. A concrete failure mode follows. Because each objective is optimal in its own geometry, a method can simultaneously achieve *lower raw MSE* and a *worse weighted residual* (equivalently, worse held-out Gaussian likelihood), while learning a subspace nearly orthogonal — in the whitened metric — to the one required for efficient estimation. We refer to this as *metric reversal*: the ranking of methods reverses depending on whether the evaluation metric respects the noise geometry. Selecting models by raw MSE under colored noise can therefore systematically prefer a reconstruction misaligned with the assumed Gaussian noise model. Whether that preference is a scientific error depends on the target risk: it is one when the target is the whitened clean-signal risk or the likelihood (Proposition 7.1), and it need not be one for the Euclidean clean-signal risk (Example 7.2).

Testable predictions. The argument yields three predictions that the experiments of Section 9 test directly, keeping the model class, latent rank, training data, preprocessing, and train/test split fixed while changing only the loss geometry:

(P1) under white noise, isotropic PCA and noise-aware EMPCA should recover the same subspace and agree on all metrics;

(P2) under colored noise the objectives should diverge: the Σ^{-1}-weighted objective should win on the weighted residual and on held-out negative log-likelihood, while isotropic PCA may win on raw MSE — the metric it optimizes;

(P3) where a clean signal or an injected reference is recorded, the reconstruction preferred by the weighted residual should also have the lower clean-signal error in a named metric. For the *whitened* clean-signal risk (42) and fixed maps of equal trace this is the population statement of Proposition 7.1; on a finite held-out sample it is a statistical comparison with its own interval. For the *Euclidean* clean-signal risk (43) it is not implied (Example 7.2), and the synthetic test S5 records that metric as an archived diagnostic (Section 9). For a *physical parameter* scored against a reference (44) — the quantity the TIDMAD stage measures — it is an empirical hypothesis of this paper, not a consequence of the argument. (P3) is the prediction that distinguishes the claim from a tautology, because (P2) alone only says each objective wins in its own metric.

The white-noise case (P1) acts as an internal control: if the two objectives coincide when $\Sigma \propto I$ but separate under colored noise, the observed effect is attributable to loss geometry rather than to rank, optimization, or preprocessing differences. (P1) and (P2) are testable on any release; (P3) requires a dataset that records the injected signal, which is why the TIDMAD stage is included and why neither CRESST nor GWOSC can supply it.

We state deliberately what these predictions do *not* assert. A gain in the weighted residual is not by itself a gain in energy resolution or in detector sensitivity: those additionally require a frozen signal-injection and background protocol, which Section 9 reports as underpowered on the available amplitude grid. Nor is a disagreement between raw MSE and the weighted residual by itself a verdict for the latter: which ordering is scientifically right is decided by the target risk (Section 7.1), which is why (P3) is tested against a recorded reference rather than assumed. The predictions above are therefore about reconstruction in a stated geometry and about agreement with recorded truth in a named metric, not about detector performance.

8 Related Work

The equivalences assembled here each have an established lineage, and covariance-aware low-rank approximation itself is not new. This section places the paper against the nearest prior work, marks each statement of Sections 2–7 as established, specialized, a corollary, a synthesis, or derived here (Table 3), and states what the paper adds beyond the placement.

Inverse-covariance weighting and matched filtering. Generalized least squares dates to Aitken [34], and the matched or optimal filter is its signal-processing form [1, 2, 35]. In cryogenic calorimetry the same construction is the standard amplitude estimator [3–5]. This literature fixes the weighting but assumes a known one-dimensional template; it does not ask what happens when the signal model itself is learned.

Weighted and probabilistic low-rank models. Weighted low-rank approximation with a general quadratic weight goes back to Gabriel and Zamir [13], and its alternating solution and the distinction between element-wise weights (no closed form) and a full row or column metric (a closed form after transformation) are classical [36]. Probabilistic PCA assumes isotropic noise [6], factor analysis allows a diagonal covariance [7, 37], and weighted or expectation–maximization PCA accommodates nonuniform weights [8, 9], with the underlying EM machinery from Dempster et al. [38] and convergence results for block-coordinate schemes from Tseng [39]. The symmetric-definite generalized eigenproblem $\hat{R}_x v = \lambda \Sigma v$ and orthogonal iteration for it are textbook material [40, Ch. 8]. This line is usually presented in terms of weights rather than of a stated measurement model, and its rank-1 case is not identified there with the detector-physics optimal filter; those are matters of presentation, and nothing in Section 4 is new as an estimator.

Covariance-aware and multilinear matrix decompositions. Both learned classes of this paper sit in an established family, and the precise optimization problems should be compared. Allen et al. minimize $\|X - UDV^T\|_{Q,R}^2 = \operatorname{tr}\big(Q(X - UDV^T)R(X - UDV^T)^T\big)$ over factors orthonormal in the Q- and R-inner products, with fixed row and column precision operators, and reduce the positive-definite case to an ordinary SVD of $Q^{1/2}XR^{1/2}$ [14]. The Bridge Theorem is the one-sided case $Q = \Sigma^{-1}$, $R = I$ of that problem, written in encoder–decoder form; Theorem 5.1 is therefore a specialization of a known result, and what Section 5 adds is the tied-adjoint reading, the explicit gap and rank-deficient cases, and the untied statement of Appendix E.2. The separable class is different from the generalized-least-squares decomposition rather than a case of it: the GMD has an unrestricted rank-k factor pair under a two-sided norm, whereas NFPA has a *product* subspace $\operatorname{range}(U_t) \otimes \operatorname{range}(U_c)$ of rank $k_c k_t$ under a Kronecker-separable covariance — the Tucker-2 structure of multilinear PCA [15, 31] applied to per-axis whitened data, close in spirit to matrix-normal PCA [30]. The Kronecker-restricted bridge (Proposition 6.1) is not a new estimator in this family; what it adds is the placement — the same Σ^{-1} metric, the same tied adjoint encoder, and the same degeneracy to the optimal filter at rank one, with the restriction stated as a change of admissible class rather than of objective — so that the cost of the prior and the effect of the metric can be measured separately at matched rank.

Positional encodings and the noise precision. Laplacian-eigenvector positional encodings for attention on graphs [21], and their reading of the sinusoidal encodings of Vaswani et al. [22] as the cyclic-grid case, are summarized by Bronstein et al. [20, §5.4]. Proposition 3.1 is the observation that, for periodic stationary detector noise, the eigenbasis of the precision operator is the same Fourier basis with the inverse-PSD weights of the optimal filter attached, which supplies a measured rather than assumed operator for that construction; the design suggestions it invites are recorded as untested in Section 10, and no empirical claim about attention models is made.

Linear autoencoders and PCA. That linear autoencoders under Euclidean loss recover the PCA subspace is classical [10, 11], with the loss landscape and the role of weight tying clarified more recently [12, 41]. These autoencoder results are stated in the Euclidean metric; the covariance-weighted optimum itself is the weighted low-rank approximation of the previous paragraphs. The step taken here is to impose the constraint $P^\dagger \Sigma^{-1} P = I_k$ and a covariance-weighted loss in the encoder–decoder parameterization, so that the tied Σ^{-1}-adjoint encoder appears as an adjointness condition rather than an architectural convenience. The subspace geometry itself is standard [42, 43], as is the perturbation theory used in Remark 5.2 [25, 26, 44].

Noise-aware learned reconstruction in physics. Learned denoisers are increasingly applied to detector traces. The Xenon-TPC study of Parker et al. [45] trains its unsupervised model under a covariance-dependent Stein unbiased risk estimate, whose divergence term is weighted by the noise covariance; that is an estimate of a *clean-signal* risk, not a residual likelihood, and it is the nearest published instance of the distinction drawn in Section 7.1. Related strands train under structured noise [46, 47], predict heteroskedastic uncertainty [48], or embed physical constraints in the objective [49, 50]; geometric deep learning supplies the language for architecture priors [20], and latent-manifold treatments of event reconstruction are developing separately [51]. The present paper argues that, for a quadratic residual loss, the residual metric is not a free choice once a covariance model has been asserted — isotropic MSE on the residual asserts $\Sigma = \sigma^2 I$ — while a supervised clean-target loss is a different and legitimate choice of risk (Section 7.1).

Learned geometry versus measured geometry. A separate line of detector machine learning learns the geometry rather than measuring it. GravNet and GarNet map every sensor to coordinates in a learned space and weight messages between sensors by a distance potential in that space [52]; object condensation constructs a potential-based loss whose minimizer places one representative per physical object and attaches to each vertex a learned weight expressing whether it can represent its object [53]; and cross-detector

pre-training recipes such as Panda Diplomacy learn one sensor-level representation for several imaging detectors from the data alone [54]. These are the natural tools when the geometry of the measurement is irregular or unknown. The present paper treats the complementary case in which the geometry *is* known, because a calibration run measures it: the inner product between readout samples is Σ^{-1}, its eigenbasis under stationary noise is the Fourier basis with the inverse-PSD weights attached (Proposition 3.1), and the loss is then not a design choice but the unique *quadratic* residual functional invariant under the noise isometries (Proposition 5.3, Appendix E.1). Where object condensation constructs a loss and shows that its minimizer has the wanted structure, the argument here runs the other way: the noise model fixes the loss, and the question becomes which admissible signal classes—template, subspace, product subspace—the fixed metric admits. A learned coordinate space applied to a calibrated readout must rediscover Σ^{-1} or pay for the difference in the likelihood; Section 9 measures that difference at matched rank.

What this paper adds. Table 3 marks the status of each formal statement. Placing recalled identities in one metric is not by itself a contribution, and the paper does not rest on it. What is added is, first, the risk distinction of Section 7.1: the classical unbiased-risk identity, specialized to fixed linear reconstructions in whitened coordinates, that says exactly which clean-signal risk the weighted residual predicts, the counterexample showing that the Euclidean clean-signal risk can order the other way, and the Davis–Kahan corollary in relative covariance error (Remark 5.2); second, the excited-support observation and the pullback form (Propositions 7.3 and 7.4), which carry the metric to the learned representation and to the training distribution; third, the finite-sample and rank-deficient bookkeeping of the degeneracies (Theorems 4.2 and 5.1); and fourth, an empirical programme that separates the three questions these identities invite — whether a larger or restricted signal model reconstructs better in the assumed geometry, whether the assumed covariance calibrates on the target data, and whether the preferred reconstruction is closer to a recorded truth in a named metric — and reports its failures. The last question requires a release with an injected reference channel, which is why the TIDMAD stage exists.

Table 3: Status of the formal statements against the nearest prior work. "Established" means the result is standard and is restated for the hierarchy; "specialized" or "corollary" means a known result is written for the one-sided precision Σ^{-1}, the whitened data, or this geometry; "synthesis" means a known argument assembled as the axiom it is; "derived here" means an elementary observation proved in this paper for which no priority is claimed. No statement in the table is claimed as an independently substantiated novelty; the contribution of the paper lies in the unified interpretation, the explicit assumptions and limitations, and the evidence that separates numerical equivalence, calibration and scientific recovery.

Statement	Status	Nearest prior work
OF as GLS amplitude (§3)	established	Aitken; matched filtering [1, 3, 34]
Rank-1 population equivalence (Thm. 4.1)	established / specialized	spiked-covariance leading eigenvector [6]
Exact degeneracy to OF (Thm. 4.2)	derived here, elementary	—
Bridge Theorem (Thm. 5.1)	specialized	weighted low-rank approximation, GMD with $Q = \Sigma^{-1}$, $R = I$ [13, 14, 36]; linear AE = PCA [10]
Untied optimum (Prop. E.3)	specialized	Eckart–Young in whitened coordinates
Architecture-neutral metric (Prop. 5.3)	synthesis (axiom)	Fisher metric of the Gaussian family [29]
Kronecker-restricted bridge (Prop. 6.1)	specialized	MPCA / Tucker-2, matrix-normal PCA [15, 30, 31]
Stationary precision eigenbasis (Prop. 3.1)	established	circulant diagonalization [19]; Laplacian PE [21]
Fixed-map risk identity (Prop. 7.1)	established / specialized	unbiased risk estimation [32, 33]; covariance-weighted SURE in detector denoising [45]
Relative covariance-error bound (Rem. 5.2)	corollary	Davis–Kahan [26, 27]
First-order displacement under covariance error (Prop. 5.2)	specialized	invariant-subspace perturbation [55], applied to $A(I+E)^{-1}$
Risk decomposition and restriction-cost bound (Prop. 6.3)	derived here, elementary	approximation–estimation split; Ky Fan maximum principle
Pullback of the noise metric (Prop. 7.3)	established	Fisher information under reparameterization [29]
Excited support (Prop. 7.4)	derived here, elementary	—

9 Experimental Validation

The experiments test progressively less controlled consequences of the likelihood geometry. Synthetic data test estimator identities and model consequences when the generating covariance is known. Public CRESST pulse traces test the within-channel linear reconstruction hierarchy on held-out detector data. Public GWOSC strain then tests a boundary condition: whether correctly implemented PSD-weighted statistics are calibrated when a simple stationary or time-local Gaussian noise model is applied to real interferometer noise. Because neither release records the true signal, a fourth stage uses public TIDMAD data, whose validation files carry an injected reference channel alongside the detector readout and therefore permit scoring a real-data reconstruction against known truth. The stages are not interchangeable. A synthetic identity does not establish real-data calibration, and reconstruction residuals do not establish trigger sensitivity.

9.1 Evaluation conventions

All reconstruction comparisons use the likelihood-aligned weighted residual $(x-\hat{s})^\dagger\Sigma^{-1}(x-\hat{s})$ defined in Eq. (4). Ordinary PCA is also evaluated by raw MSE to expose disagreement between the Euclidean and covariance-weighted geometries. Subspace agreement is measured by principal angles after applying the same whitening convention. Unless otherwise stated, confidence intervals are percentile bootstrap intervals over the paired unit named in the result. Covariance estimates use calibration data disjoint from evaluation traces. For stationary single-channel data, the covariance is represented by its PSD; the discrete Fourier and normalization conventions are held fixed across methods within each comparison. Table 4 lists the operating point of each stage as read from the frozen configuration files and run manifests, and Appendix I carries the full mechanically extracted inventory.

Predeclaration and multiplicity. All acceptance gates, operating points, and interval types reported below were frozen before the corresponding evaluation data were scored, and every frozen gate is reported here whether it passed or failed; no gate was added, dropped, or reweighted after seeing an outcome. Intervals are nominal and are not corrected for the number of gates, so the count of passed checks should not be read as a joint confidence statement. The paper's conclusions rest on the two primary predeclared contrasts named below and on the reported failures, not on the number of gates that passed. Analyses performed after the freeze are labeled post-freeze wherever they appear and never replace a primary gate.

The comparisons answer different questions. OF versus rank-6 EMPCA primarily tests signal-model capacity: a fixed template is compared with a learned six-dimensional subspace. Ordinary rank-6 PCA versus rank-6 EMPCA holds the nominal subspace dimen-

Table 4: Protocol summary: the values used by the runs reported in this paper, per stage. Values marked as overrides are taken from the run manifests (`transfer_paper/data/paper1/*/manifest.json`) where a run departed from the frozen default; all other values are read from the frozen configuration files (full inventory in the separately built `supplement_configuration.pdf`, Appendix I). A question mark marks a value that the inventory does not carry under the expected name and that the author should confirm.

Setting	Value
CRESST stage	
rank k / covariance model / shrinkage	6 / psd_stationary / 0.1
release split / per-channel calibration fraction	test / 0.7
event-bootstrap resamples / seed	300 / 22
GWOSC stage	
analysis window (s) / PSD estimator	4 / hann, median
PSD calibration fraction / off-source split seed	0.75 / 150914
local PSD radii (s)	[32, 64, 96]
TIDMAD stage	
window samples / rank	4096 / 2
windows / calibration windows / fit fraction / guard	512 / 256 / 0.7 / 4
bootstrap resamples / block size / seed	300 / 16 / 20260727

sion fixed and is therefore the more direct descriptive contrast for covariance weighting. The tied linear-AE comparison is a subspace implementation check. GWOSC PSD and shared-FIR agreements are also implementation checks, whereas held-out score dispersion is a calibration test. We do not use success on one axis as evidence for another.

9.2 Controlled synthetic validation

The controlled evidence has four parts. A suite of nine frozen regression gates verifies the estimator identities and model consequences of Sections 2–7 (Section 9.2.1, Fig. 4); it is verification, not the paper's scientific evidence. The three studies that follow are: the anatomy of covariance-perturbation error, which tests Remark 5.2 and Proposition 5.2 (Section 9.2.2, Fig. 5); the structural prior at finite sample, which measures the terms of Proposition 6.3 (Section 9.2.3, Fig. 6); and excitation against detectability, which probes the regime between the two cases of Proposition 7.4 (Section 9.2.4, Fig. 7). All three were predeclared (`plan/MLST_EXPERIMENT_PROGRAMME.md`, Parts I and II), executed once, and are reported with the registered predictions that were not met.

9.2.1 Regression gates (S1–S9)

The controlled suite contains nine frozen tests. S1–S4 and S6–S9 were run with ten seeds; S5 was subsequently repeated as a confirmatory 30-seed analysis. S1 checks OF normalization against the Cramér–Rao bound; S2 checks recovery of the OF direction by rank-1 EMPCA; S3 compares an independent tied linear-AE implementation with the EMPCA optimum; S4 is a white-noise PCA/EMPCA control; S5 tests colored-noise metric reversal; S6 introduces timing variability and varies rank; S7 varies the amount of covariance-calibration data; S8 checks the whitened residual χ^2; and S9 compares full and diagonal multichannel covariances. Frozen configurations and seed-level outputs are archived with the paper evidence.

All nine regression gates passed (Fig. 4). The OF standard deviation divided by its predicted bound was 0.994 (ten-seed 95% interval $[0.985, 1.002]$). The independent tied linear AE reached the EMPCA objective with mean relative gap 1.95×10^{-16}, and PCA and EMPCA in white noise differed by only 1.48×10^{-6} degrees in maximum principal angle. For S5, each seed generated 512 traces of $d = 256$ samples, split at the trace level into equal training and test sets with the same generated data supplied to the two rank-2 estimators; all S5 values below are the 30-seed confirmatory run. The held-out PCA and EMPCA weighted residuals were 48.419 and 47.869, respectively; the paired PCA-minus-EMPCA difference was 0.550 (30-seed bootstrap 95% interval $[0.530, 0.571]$), a fractional improvement of 1.14%. Held-out negative log-likelihood agreed: the PCA-minus-EMPCA difference was 11.67 with 95% interval $[11.24, 12.11]$ against a PCA mean

of 1026.92. Raw MSE preferred the opposite reconstruction (2330.27 for PCA and 7412.67 for EMPCA), while the mean maximum whitened principal angle was 86.25° (95% interval $[85.16, 87.21]$). Thus the ordering is metric-dependent under the stated synthetic model. The same run archives the Euclidean clean-signal error (43) of each reconstruction against the generated clean signal, in the stacked real/imaginary rFFT feature representation the fits use, as a diagnostic that was not a predeclared gate: 5110.9 for PCA (30-seed mean; 95% interval $[4979.0, 5236.7]$) against 58.2 for EMPCA ($[45.8, 73.6]$), with EMPCA lower in all thirty seeds. On this model, then, the Euclidean clean-signal error and the weighted residual order the two reconstructions the same way while raw MSE orders them oppositely; Example 7.2 shows that the first agreement is a property of the model, not a guarantee.

The last two numbers are worth reading together, because a near-orthogonal subspace and a one-percent-level residual difference (1.14%) sound inconsistent. They are not. With $d = 256$ and $k = 2$, the weighted residual is dominated by noise power in the $d - k$ directions that neither estimator models; only the component attributable to the retained two-dimensional subspace can differ between them. A large rotation of that subspace therefore moves the total residual very little, even though the two estimators are recovering almost unrelated directions. The principal angle measures *which* directions are retained and the residual measures *how much* total error remains; the two are not redundant, and the small fractional difference is not evidence that the choice of geometry is unimportant—on CRESST at rank 6 the corresponding same-rank contrast was an equal-channel reduction of 21.6%. With timing variability, the mean best rank by clean-signal MSE was 4.6, rather than one. At the largest covariance-calibration size the reconstructed uncertainty was 1.032 times the oracle value; the mean whitened residual per degree of freedom was 0.997; and diagonal rather than full multichannel covariance increased uncertainty by a factor 1.025. These results support the implemented consequences of the stated synthetic model, not real-detector adequacy of any covariance estimate.

9.2.2 Covariance-perturbation anatomy (T15, T19)

Two studies on one frozen generator ask how a wrong covariance moves the learned subspace. The first (T15) varies the source of the error — calibration-sample size or structured misspecification — and finds the global bound of Remark 5.2 valid but uninformative and ε an incomplete summary; the second (T19) tests the local statement of Proposition 5.2 that explains why, with engineered perturbations at fixed training data. Figure 5 collects the three regimes they found.

Covariance-error regime (T15). Remark 5.2 was tested on a generator frozen for the purpose (`reviews/2026-09-10-experiment-phase-report.md`, §5): $d = 256$; an

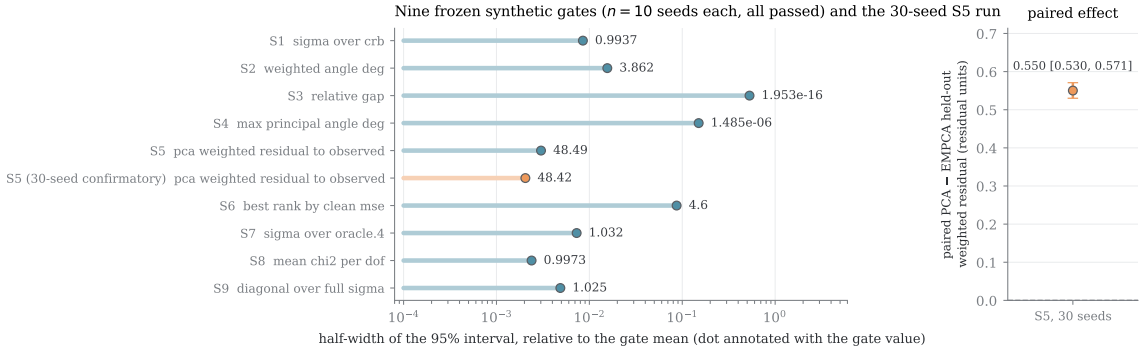


Figure 4: Controlled validation summary. The nine frozen experiments test OF normalization (S1), rank-1 direction recovery (S2), the independent tied-linear-AE/EMPCA bridge (S3), the white-noise control (S4), colored- noise metric reversal (S5), timing-dependent rank (S6), covariance estimation (S7), residual calibration (S8), and multichannel covariance (S9). Each bar is the half-width of the gate's 95% interval expressed as a fraction of its own mean, so gates on very different scales can be read on one logarithmic axis; the dot is annotated with the gate value. The nine teal rows are the archived ten-seed frozen gates, including the ten-seed S5 gate; the orange row is the separate 30-seed S5 confirmatory run whose values the text quotes (PCA held-out weighted residual 48.419, 95% interval $[48.320, 48.518]$), read from its archived summary — a level, not an effect. The right panel, in residual units on its own axis, shows the paired PCA−EMPCA held-out weighted-residual difference of that 30-seed run, 0.550 with 95% interval $[0.530, 0.571]$, which is the S5 contrast itself. Intervals and acceptance rules are archived with the seed-level evidence.

exponent -1 power-law PSD with knee $4/d$ and the exact circulant Σ it implies, normalized to unit average variance; a generic rank-2 signal basis with amplitudes at SNR 3 and 3/2; 512 training traces; and eight paired arms over ten seeds (80 fits): oracle Σ; $\hat{\Sigma}$ estimated from $m \in \{64, 128, 256, 512, 2048\}$ noise-only traces; and two structured misspecifications, the PSD exponent replaced by -1.5 or -0.5 and the correlation length doubled or halved, each renormalized to the same trace so that misspecification changes shape and never scale. The regime has $\lambda_1/\Delta_\lambda = 3.74$. A known-truth check corrected the comparison before the run: the bound governs the rotation caused by handing the estimator a perturbed covariance, so it is compared with the angle to the *paired oracle fit on the same training draw*, not to the planted subspace, which the oracle arm itself misses by $\sin\Theta = 0.32$ at $N = 512$ from training-sample error alone. The displayed inequality was satisfied on all 80 fits, but the remark's hypothesis $\varepsilon \le \frac{1}{2}$ holds only for the oracle arm and the two largest calibration sizes ($\varepsilon = 0.38$ at $m = 512$ and 0.19 at $m = 2048$); on the five other arms the inequality is an observation, not a validation of the theorem. Where the theorem applies it is valid and not informative: $6\varepsilon\lambda_1/\Delta_\lambda = 8.6$ and 4.3, above the value 1 that $\sin\Theta$ cannot exceed, and by the fitted scaling $\varepsilon \approx 0.59\sqrt{d/m}$ an informative value would need $m \approx 4.5 \times 10^4$ calibration traces at this d. The arms outside the domain are still instructive about ε as a diagnostic. The structured correlation-length misspecification has $\varepsilon = 0.557$ and is nearly harmless — $\sin\Theta$ to the paired oracle 0.012, held-out amplitude-probe R^2 0.921 against the oracle's 0.922 — while the covariance estimated from $m = 256 = d$ calibration traces has the comparable $\varepsilon = 0.575$ and is catastrophic ($\sin\Theta$ 0.94, R^2 0.11); recovery is well under way at $m = 2d$ (R^2 0.83) and near complete at $m = 8d$ (0.91). An upper bound in ε never implied that perturbations of equal ε act alike, so this does not contradict the remark; it shows that ε is an incomplete summary, and that the two arms are different regimes: the calibration-sample arms at $m \le d$ hand the estimator a near-singular $\hat{\Sigma}$ whose inversion is unstable, whereas the structured arm perturbs a well-conditioned whitening map mostly within the retained eigenspace. Note that these arms vary the number m of noise-only calibration traces at fixed signal-training size $N = 512$, so their error is covariance-estimation error, not the training-sample term. The statement that separates the cases is local (Proposition 5.2, item 4): in the eigenbasis of A, with D the training-sample perturbation of A and E the relative covariance error in the same coordinates, the first-order displacement of retained direction i toward discarded direction j is $(D_{ji} - \lambda_j E_{ji})/(\lambda_i - \lambda_j)$, which separates the two error sources and the pairwise gaps and requires the eigenvalue ordering to be preserved (a diagonal perturbation with zero off-diagonal block can reorder the spectrum and exchange the subspace). A finite-perturbation bound of that form with explicit assumptions is left open. The operator and Frobenius errors also differ materially (0.557 against 0.307 for the correlation-length arm), so the theorem's operator-norm quantity is the one reported.

Coupling, separation and conditioning (T19). Proposition 5.2 was tested on the same frozen generator in a second, locally executed phase (`reviews/2026-09-11-experiment-phase-report.md`, §4; 38 arms, ten seeds, 380 fits). Because the fit diagonalizes the *sample* whitened operator $\hat{A} = \hat{Q}\hat{\Lambda}\hat{Q}^T$, the perturbation families were built in $\hat{Q}$, $\hat{\Sigma} = \Sigma^{1/2}(I + E)\Sigma^{1/2}$, with the same training draw in every arm, so that the training-sample term D drops out of the comparison and the displacement of the fitted subspace from the paired oracle fit isolates the covariance term: retained direction i moves toward discarded direction j by $-\hat{\lambda}_j E_{ji}/(\hat{\lambda}_i - \hat{\lambda}_j)$ to first order. This is a conditional check at fixed data with engineered E, not a test of an independently estimated covariance. On the designed coupling and random families and on the stationary (Toeplitz) estimate at $\varepsilon \leq 0.2$ the measured Frobenius norm of the displacement was within a factor of two of the first-order value on 70 of 70 fits (medians 0.98–1.02), with the residual scaling as $\varepsilon^{2.05}$ for the random family and as $\varepsilon^{2.9}$ for the pure coupling family, whose second-order term vanishes identically. A diagonal E in $\hat{Q}$ that preserves the eigenvalue ordering left the fitted subspace unchanged ($\sin\Theta \leq 4 \times 10^{-8}$ on 40 of 40 fits up to $\varepsilon = 0.4$), and an order-reversing one exchanged the retained subspace exactly when the sample-spectrum check $\hat{\lambda}_2(1 + \varepsilon) < \hat{\lambda}_3(1 - \varepsilon)$ flipped (30 of 30; the sample threshold is 0.35–0.43 at $d/N = 1/2$ against the population value 0.72). This is the precise content of the T15 observation that ε alone does not predict the outcome: what moves the subspace is the off-diagonal block of E in the basis the fit diagonalizes, divided by the gap, and a large diagonal error is harmless until it reorders the spectrum. Conditioning is a third regime: an empirical covariance from $m \leq d$ noise-only traces has $\lambda_{\min}(I + E) < 0.05$ on all 20 fits and at $m = d$ destroys the subspace ($\sin\Theta$ 0.99), while a calibration-selected shrinkage toward the identity restores $\lambda_{\min}$ to 0.21–0.35 and leaves $\sin\Theta$ at 0.21–0.99: the regularizer repairs the spectrum floor, not the direction of the sampling error. (An encoder/decoder inconsistency in the first evaluation of the eigenvalue-floored $m = d/2$ arm was found by an independent audit and corrected before these numbers were read; with a consistent decoder that arm loses the subspace only partially, median $\sin\Theta$ 0.73, so how badly a near-singular estimate fails depends on the regularizer.) A finite-perturbation bound with explicit assumptions remains open, and the remark's own bound is uninformative on the planted two-axis configuration of the next paragraphs at any ε of practical size, since there $\lambda_1/\Delta_\lambda \approx 2.7 \times 10^3$ at rank six (T23, report §3).

9.2.3 Structural prior at finite sample (T14, T20)

The separable class restricts the learned subspace, and Proposition 6.3 says what the restriction costs and buys: a population approximation cost $a \geq 0$, paid at every N, and a lower estimation excess, paid by the unrestricted class and shrinking with N. Two

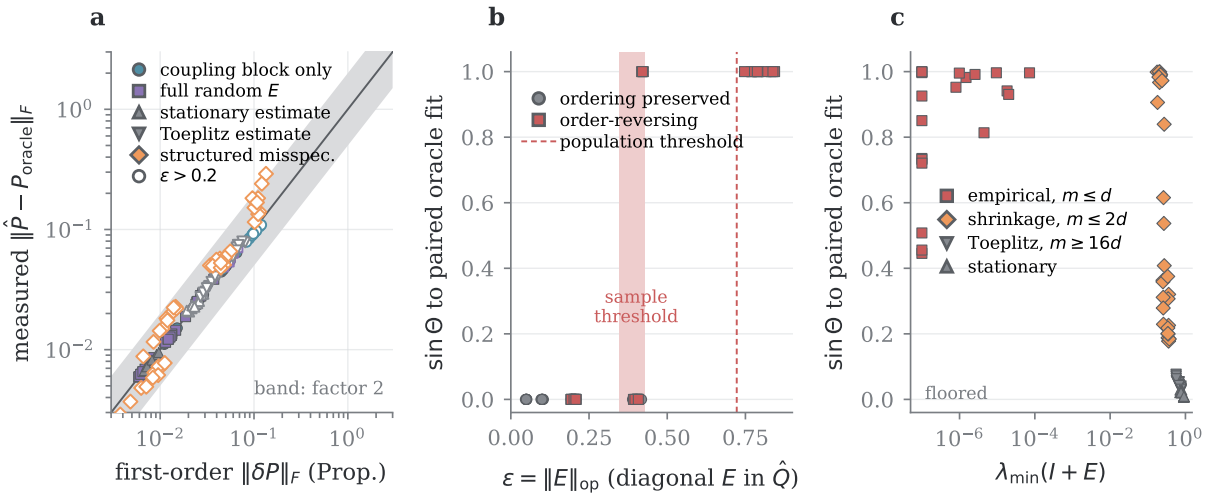


Figure 5: Covariance-perturbation anatomy (T19; ten seeds per arm, $d = 256$, $k = 2$, $N = 512$, displacement measured against the paired oracle fit on the same training draw). (a) Measured Frobenius displacement of the fitted projector against the first-order value of Proposition 5.2 for perturbations built in the sample eigenbasis: a pure coupling block (C), a full random symmetric E (F), and the estimated covariances of the stationary and Toeplitz arms and of the structured misspecifications (M); filled markers are the primary band $\varepsilon \leq 0.2$, open markers larger ε; the band is a factor of two about the identity. (b) Diagonal E in the fitted eigenbasis: ordering-preserving perturbations (P) leave the subspace unchanged at every ε tested, and order-reversing ones (R) exchange it exactly when the sample spectrum reorders (shaded: the per-seed sample thresholds, 0.35–0.43; dashed: the population value 0.72). (c) Conditioning: the subspace error of the empirical covariance from $m \leq d$ noise-only traces (consistent decoder, v2), of the calibration-selected shrinkage estimates, and of the well-conditioned Toeplitz and stationary estimates, against the smallest eigenvalue of $I + E$; the near-singular arms are outside the domain of both the proposition and the global bound, and shrinkage repairs the spectrum floor without repairing the direction. Registered inputs and field names: `generated/SOURCE_MAP_part2.md`.

studies on the planted two-axis family map the finite-sample contrast (T14) and then measure the three terms separately (T20); Fig. 6 shows the accounting.

Finite-sample map (T14). The finite-sample class contrast of Section 6.4 was mapped on the planted two-axis generator with oracle covariance at $(k_c, k_t) = (2, 3)$, $C = 8$, $T = 256$, over $N \in \{200, 400, 800, 1600, 3200\}$, nonseparability $\alpha \in \{0, 0.4, 0.8\}$ at SNR 10 and $\alpha \in \{0, 0.8\}$ at SNR 3, ten seeds and three ALS restarts per cell against the thin-SVD unrestricted fit at matched rank six (25 cells, 1000 fits; report §6). The contrast $d = \bar{\chi}^2_{\text{test}}(\text{NFPA}) - \bar{\chi}^2_{\text{test}}(\text{CW-PCA})$ was negative in every cell at SNR 10, with paired intervals excluding zero in fourteen of fifteen; negative at $\alpha = 0$ for every N at SNR 3; and indistinguishable from zero (all five estimates within 2.5×10^{-5}) at $\alpha = 0.8$, SNR 3. No crossover between the restriction's approximation cost and the unrestricted class's variance excess was found anywhere on the grid, so the predeclared confirmation stage was not triggered and its seeds were not spent. The retained statement is therefore that the separable restriction improved held-out risk across the tested SNR-10 grid and at $\alpha = 0$ at SNR 3, and that the crossover to approximation-dominated behaviour was not observed; the original gate, a transition within the grid, was not met. The null is not explained by the exploration itself. The unrestricted class has 12 252 subspace parameters against 771, and the largest N supplies 3200 observations of dimension 2048, but neither count locates the crossover, which depends on the whitened spectrum, the excitation and the noise; and the population approximation cost of the restriction at $\alpha > 0$ was never established independently of the finite-sample fits, so a strictly positive cost that consistent fits must eventually expose has not been ruled in or out (at $\alpha = 0$ it is identically zero and no crossover can exist). The follow-up is a population-level accounting against the generator's own covariance, the identity (37) of Proposition 6.3, which separates the approximation cost from the two estimation excesses exactly; it was carried out on a configuration in which the cost is first shown to be positive (T20, below). Where the effect is smallest the optimizer limits it: at $\alpha = 0.8$ the restart-to-restart spread of the held-out objective is 35% of $|d|$, and training-loss restart selection missed the held-out-best restart in 35% of seeds, so that line is optimization-limited and its small values should not be over-read.

Population risk accounting (T20). The accounting was carried out on the planted family with $T = 32$ ($d = 256$), $(k_c, k_t) = (2, 3)$, oracle covariance, SNR 10 and $\alpha \in \{0, 0.4, 0.8\}$ (report §5). The population reference is the whitened centred clean-signal covariance K estimated from 2×10^5 clean events per α; $R_K(P) = \operatorname{tr} K - \operatorname{tr}(PKP)$ is the clean-signal projection loss, and for equal-rank orthogonal projectors the retained-noise term cancels from every contrast. The unrestricted reference $P_U^\star$ is the top-six

eigenprojector of K; the restricted reference is the best of 51 alternating-least-squares solutions on K (HOSVD-initialized plus fifty random starts, all agreeing to 10^{-13}), so the cost $a = R_K(P^\star_{\mathcal{F}}) - R_K(P^\star_U)$ is the cost of the best solution found — an upper bound on the minimum restriction cost, not a certified optimum. It is -7×10^{-15} at $\alpha = 0$, where the product family contains the excited signal exactly (the planted rank-six class has only four excited directions there, because the coefficient map ties two of its entries), and 2.71 and 2.82 at $\alpha = 0.4$ and 0.8, about 3% of $\operatorname{tr} K$. The lower bound (38) on the minimum restriction cost, computed from the gap and the partial traces of $P^\star_U$ on the same K, gives 0.10 and 0.55, so the cost is provably positive at both nonzero α and the optimum lies between the bound and the value found. Over $N \in \{50, \dots, 1600\}$, ten seeds and three training-loss-selected restarts, the identity holds exactly by construction (180 paired comparisons), $x_U > x_F$ throughout (fitted constants c_U/c_F between 3 and 5), and the registered c/N form of the estimation excesses is refuted: the unrestricted excess decays with effective slope $N^{-0.49}$ and $N^{-0.38}$ over the grid, and the crossover the c/N constants forecast, near $N = 160$, is not there. On fresh seeds 10–19, chosen after the exploration and evaluated once, the contrast $R(\hat{P}_{\mathcal{F}}) - R(\hat{P}_U)$ is negative at $N = 800$ (-0.39 $[-0.49, -0.29]$ and -0.29 $[-0.40, -0.18]$) and positive at $N = 1600$ ($+0.90$ $[+0.83, +0.96]$ and $+0.87$ $[+0.80, +0.94]$) at $\alpha = 0.4$ and 0.8 respectively: at $C = 8$, $T = 32$ and SNR 10 the restricted learner wins at $N = 800$ and the unrestricted one at $N = 1600$ for both tested nonseparabilities, while at $\alpha = 0$ the contrast is negative at every N. An exploratory log-linear interpolation places the sign change near $N = 960$; the registered $1/N$ extrapolation did not predict it, and $N^\star$ is bracketed, not located. The T14 null at $T = 256$ stands as a null: the two configurations differ in dimension and in the pulse family relative to the fixed pulse widths, and the whitened spectra at the three α were not matched as the protocol asked, so cross-α differences are not a pure approximation-cost intervention and the within-α comparisons across N are the evidence. Restart selection by training loss coincided with the population-best restart in 50–90% of fits per cell, with restart spreads up to 0.18 — of the order of the contrast near the sign change — so the optimization term is visible exactly where the two classes are closest.

9.2.4 Excitation and detectability (T21)

Proposition 7.4 covers two cases, a direction the training data excite and one they do not; the study of this section scales the excitation of one physical direction continuously between them and asks what sets its recovery at finite N. The answer, Fig. 7, is not the excitation.

Excitation and recovery (T21). Proposition 7.4 concerns directions the training data never excite; the intermediate regime was probed on the planted family at $\alpha = 0$, $C = 8$,

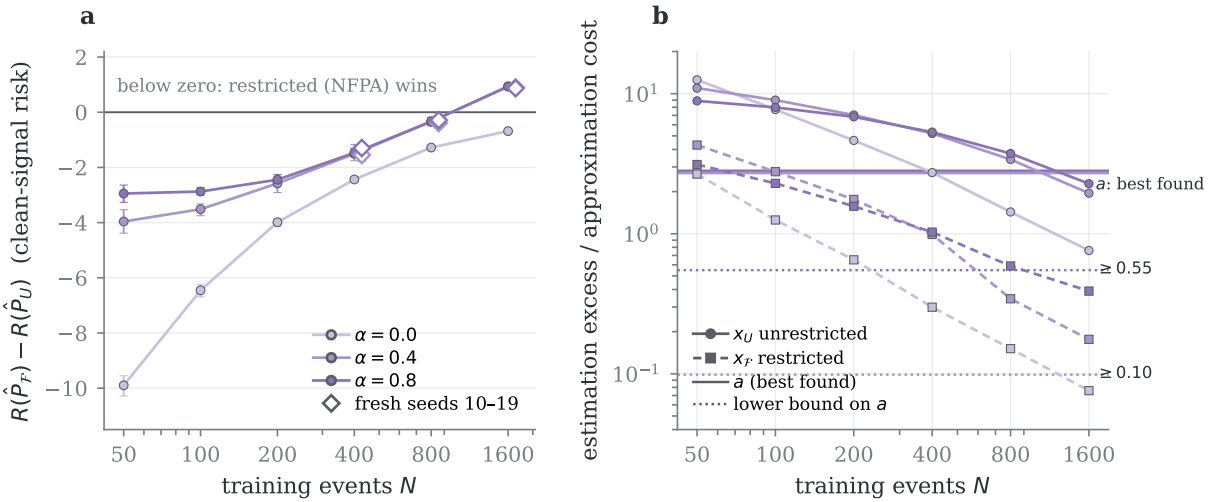


Figure 6: Structural prior at finite sample (T20; planted two-axis family, $C = 8$, $T = 32$, $(k_c, k_t) = (2, 3)$, SNR 10, oracle covariance). (a) Clean-signal risk contrast between the fitted separable (NFPA) and unrestricted (CW-PCA) subspaces against N: exploration seeds 0–9 with seed-paired 95% intervals (lines), and the fresh seeds 10–19 evaluated once at $N \in \{400, 800, 1600\}$ (open diamonds). At both nonzero nonseparabilities the contrast is negative at $N = 800$ and positive at $N = 1600$; at $\alpha = 0$, where the product class contains the excited signal and $a = 0$, it is negative throughout. (b) The terms of Proposition 6.3: the unrestricted excess x_U (solid) and the restricted excess $x_{\mathcal{F}}$ (dashed) against N, with the best-found approximation cost a (solid horizontal) and its provable lower bound (38) (dotted) for $\alpha = 0.4$ and 0.8. The contrast changes sign where $x_U - x_{\mathcal{F}}$ falls through a. Registered inputs: `generated/SOURCE_MAP_part2.md`.

$T = 256$, oracle covariance and SNR 10, with the standard deviation of one physical latent (amplitude, rise time, decay time, or the secondary channel-mixing mode) scaled in the training draw only by $s \in \{0, 0.03, 0.1, 0.3\}$ with the others at 1, plus three controls (19 cells, ten seeds, 190 closed-form rank-four fits; report §6). Recovery of direction j is $\|Pu_j\|^2/\|u_j\|^2$ for the whitened Jacobian column u_j at the reference point. None of the five registered predictions was met as registered: recovery is not a function of the excitation ratio λ_j/Δ_λ alone; at $s = 0$ recovery is 0.13–0.37 for three of the four directions against the geometric chance $k/d \approx 0.002$, because the planted directions are not orthogonal (Gram entries to -0.65); the weak-excitation variance signature and the 0.95 strong-excitation level were not reached at $N = 800$, and only two directions reached 0.95 at $N = 3200$; and the across-seed recovery variance is peaked at the detection edge rather than monotone. The exploratory reading, recorded for an independent test and not claimed here, is that recovery is set by the spike strengths of the population signal covariance in the training distribution relative to d/N and by each direction's loadings on its eigenvectors: the fourth population spike, 0.54 above the noise floor, is below $\sqrt{d/N}$ at both N, and the isolated-spike overlap formula combined with those loadings reproduces all eight observed full-excitation recoveries to within 0.01. Excited and detectable are therefore different properties of a planted direction, and no abstention threshold for a downstream diagnostic is derived from this study.

9.3 Public CRESST within-channel reconstruction

We used the public CRESST-II/III pulse-shape release [16, 56]. The frozen run at commit `9c9f696ff6bca6e6b9729cdb90e41ee92d576787` used all seven channels in the test release. Each run/channel pair was treated independently and sorted by the release time variable. Pulse and noise streams were each split chronologically, separately, using the first 70% for calibration and the remainder for evaluation. Calibration pulses fitted the signal models and calibration noise fitted the PSD; no evaluation pulse entered a fitted object. The separate stream splits do not impose a common cross-stream timestamp cutoff: six and three evaluation pulses in channels `35/57.0` and `35/58.0`, respectively, preceded the last calibration-noise timestamp. Within each stream, row and trace-hash overlap was zero. Depending on channel, calibration contained 3,195–12,692 pulses and 512–537 noise traces, while evaluation contained 1,369–5,440 pulses. The pooled held-out comparison comprised 19,335 pulses.

The frozen signal model was rank-6 EMPCA with a stationary PSD covariance and shrinkage 0.1. Baselines were single-template OF, a multi-template OF bank, ordinary PCA, and a tied linear AE evaluated through its closed form (a parameterization-consistency check, not an independently optimized model; see below). The learned models

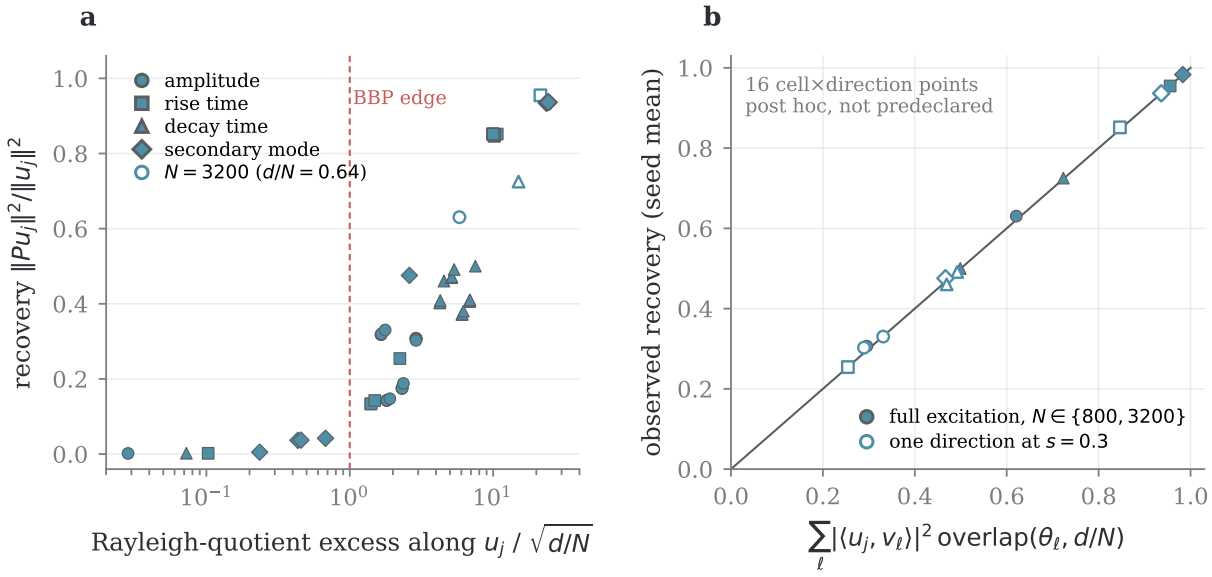


Figure 7: Excitation and detectability (T21; planted family at $\alpha = 0$, $C = 8$, $T = 256$, rank four, oracle covariance, ten seeds per cell). (a) Seed-mean recovery of each planted physical direction against the Rayleigh quotient of the training-distribution signal covariance along that direction, in excess of the noise floor and relative to the spiked-model threshold $\sqrt{d/N}$, for all nineteen cells (filled, $N = 800$) and the $N = 3200$ control (open). Recovery is not a function of this scalar: directions at equal excess differ by a factor of two or more, and the excitation scale s enters only through it. (b) Post-hoc reading, not predeclared: the recovery predicted by the isolated-spike overlap of the population eigenvalues at d/N, weighted by each direction's squared loadings on the population eigenvectors, against the observed seed mean, for the sixteen cell–direction pairs whose population loadings were archived (the eight full-excitation recoveries quoted in the text, filled, and the two $s = 0.3$ cells, open). Agreement is within 0.015 on every point; it is a description of these runs, and its test on fresh data is left open. Registered inputs: `generated/SOURCE_MAP_part2.md`.

were fitted with the calibration mean subtracted and restored, i.e. as affine reconstructions $\mu + P(x - \mu)$, whereas single-template OF is a line through zero; the same-rank PCA–EMPCA contrast shares the fitted mean and is unaffected, while the OF–EMPCA capacity contrast compares a one-parameter template with an affine rank-6 model and its difference includes the fitted intercept. The template bank used rise- and decay-time quantile cells and integer shifts of $-4, -2, 0, 2$, and 4 samples. The primary OF comparison was the paired difference $R_{\mathrm{OF}} - R_{\mathrm{EMPCA}}$. The predeclared competitiveness margin was $1.1R_{\mathrm{multiOF}} - R_{\mathrm{EMPCA}}$. Positive values favor EMPCA. The primary uncertainty calculation pooled paired event-level differences and resampled the 19,335 evaluation pulses 300 times. This event-bootstrap gate is retained as the primary predeclared analysis.

We also performed a post-freeze channel-aware sensitivity analysis from the archived event residuals. Its point estimand gives each of the seven run/channel clusters equal weight. For uncertainty, each of 10,000 bootstrap replicates (seed 20260713) sampled seven clusters with replacement and then sampled the observed number of events with replacement within each selected cluster occurrence. This tests robustness to unequal channel sizes; it does not replace the pooled primary gate.

Weighted residuals are unnormalized and their per-channel scales differ by orders of magnitude, so we report the fractional reduction $1-R_{\mathrm{EMPCA}}/R_{\mathrm{baseline}}$ as the primary effect measure and give the raw paired differences alongside it, since the predeclared acceptance gates were defined on the raw scale.

Same-rank contrast: the covariance-weighting evidence. Holding the nominal subspace dimension fixed at six, EMPCA had the lower channel-mean weighted residual in all seven channels. The fractional reduction $1 - R_{\mathrm{EMPCA}}/R_{\mathrm{PCA}}$ ranged from 1.6% to 62.0%, with an equal-channel mean of 21.6%; the corresponding unnormalized equal-channel difference was 3.78, with hierarchical 95% interval $[1.22, 6.69]$. The fractional estimand itself has hierarchical 95% interval $[10.3\%, 38.8\%]$ (channel-fixed $[17.4\%, 27.1\%]$; ratio-of-channel-means sensitivity $[12.1\%, 39.8\%]$; block lengths 16–256 move the interval by less than 0.35 percentage points), from a channel-then-event bootstrap with 10 000 replicates on the archived residuals (`reviews/2026-09-10-experiment-phase-report.md`, §3); all seven channel effects are individually positive with intervals excluding zero, and nearly all of the uncertainty is channel-level (hierarchical standard deviation 7.5 against channel-fixed 2.5 points). This is a sensitivity analysis of the archive, not an independent replication: seven channels from two runs are seven correlated clusters. Because rank is held fixed, this contrast isolates the contribution of the noise geometry, and it is the paper's direct real-data evidence for covariance weighting. The effect is directionally consistent and strongly heterogeneous rather than uniformly large.

Signal-model contrast: capacity, not geometry. Rank-6 EMPCA also improved the weighted residual relative to single-template OF in every channel (Fig. 8A), by an equal-channel fractional reduction of 96.2% (range 89.7%–98.8%). On the raw scale the pooled difference was 1012.27 with 95% event-bootstrap CI $[927.90, 1097.61]$, and the equal-channel mean was 731.87 with channel-then-event 95% interval $[110.69, 1537.48]$. EMPCA met the multi-template competitiveness gate in every channel: pooled margin 352.17 with CI $[272.40, 424.01]$, equal-channel mean 279.26 with interval $[36.52, 569.01]$. Both directions survived channel balancing, although seven clusters from two runs make these percentile intervals coarse and potentially unstable. This contrast is an order of magnitude larger than the same-rank one, and it primarily measures the benefit of replacing a fixed template with a rank-6 signal model. It should not be read as evidence about covariance weighting. The normalized ratios are descriptive functions of channel means rather than additional acceptance gates.

All five negative controls degraded at the pooled level (Fig. 8B). The weakest was wrong-PSD weighting, whose paired difference from EMPCA was 1.57 with CI $[0.97, 2.08]$. At the channel level two of the seven wrong-PSD intervals include zero (`33/37.0`, CI $[-0.64, 0.75]$, and `35/58.0`, CI $[-0.010, 0.0001]$); only `35/58.0` has a negative point estimate (-0.00496), so it is the sole sign reversal, but "did not degrade" applies to both channels. The other controls had larger pooled mean degradations: 489.40 for noise-only training, 4013.47 for a random weighted basis, 6911.82 for a mismatched OF template, and 59812.72 for permuted time bins. These scale-dependent values are reported to verify direction and falsification behavior, not to compare the scientific severity of different controls. All five control differences also had positive post-freeze hierarchical intervals, despite the two wrong-PSD channel exceptions.

The tied linear AE and EMPCA bases agreed to numerical precision on the public traces: the maximum principal angle across channels was 1.91×10^{-6} degrees. This row must be read for what the archived code computes. At the cited commit the CRESST fitting path evaluates the tied AE through its closed-form solution, which delegates to the same weighted-PCA routine that fits EMPCA; the angle therefore verifies that the tied parameterization $P^{\dagger}\Sigma^{-1}P = I_k$ and the weighted-PCA basis coincide on real data after the two code paths' whitening and unwhitening, and it is not the output of an independently initialized optimizer. In the default build the independently trained tied-AE evidence is the synthetic test S3. Trigger efficiency was inconclusive over the frozen amplitude grid for six of seven channels at the primary 5% false-positive rate, and no channel reached 90% efficiency. We therefore draw no detector-sensitivity conclusion from this stage.

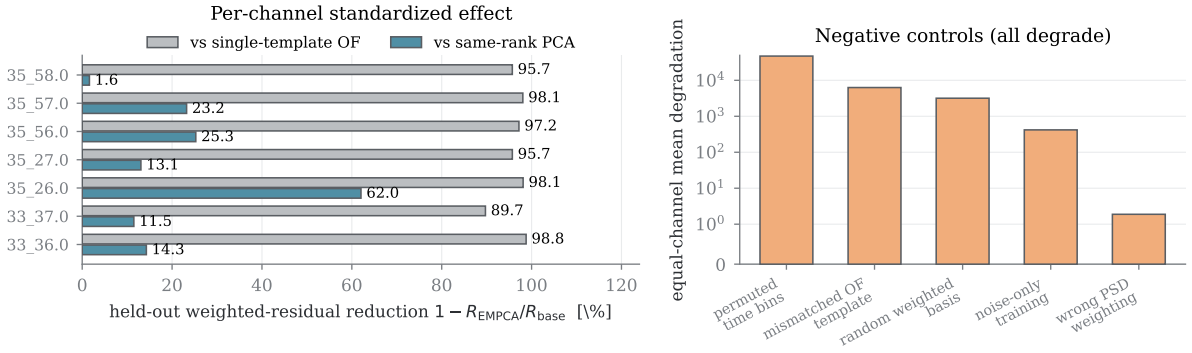


Figure 8: Public CRESST within-channel validation. (A) Per-channel *standardized* effects: the fractional reduction $1 - R_{\mathrm{EMPCA}}/R_{\mathrm{base}}$ against single-template OF (gray) and against same-rank PCA (teal). Fractions are used rather than raw paired differences because per-channel residual scales differ by orders of magnitude. The gray bars are large because they compare a rank-6 model with a fixed template; the teal bars isolate covariance weighting at fixed rank and are the direct evidence about noise geometry. (B) Equal-channel mean degradations for five negative controls on a symmetric-log axis; all five degrade. Wrong-PSD weighting is the weakest: two channels have intervals including zero (`33/37.0`, `35/58.0`) and its sole per-channel sign reversal was on `35/58.0` (mean -0.00496, CI $[-0.010, 0.0001]$). All values are generated from the archived run JSON; $n = 19{,}335$ held-out pulses across seven channels.

9.4 Public GWOSC calibration boundary

The GWOSC stage used public 4-kHz H1 and L1 strain around GW150914 from the O1 release [17, 57] as a stress test, not an event search. The selected 256-second interval supplied 62 valid off-source windows per detector. A Hann-windowed, bias-corrected median-Welch PSD was estimated from calibration windows and template-score dispersion was evaluated on held-out windows, over five random splits (46 calibration, 16 evaluation) and five chronological blocks (36 calibration, 12–13 evaluation); a null spread divided by its PSD prediction near one was the calibration target. Two implementation checks accompanied it: comparison of the repository PSD estimator with GWpy [58] on identical inputs, and a shared-FIR check forcing explicit FFT convolution and GWpy convolution onto the same FIR coefficients and normalization. A frozen follow-up then compared the global PSD with local radii of 32, 64, and 96 seconds, the 64-second model and acceptance range $[0.8, 1.2]$ being predeclared as primary.

The implementation checks passed and the calibration tests did not (Table 5). PSD ratios were one at recorded precision and the shared-FIR paths agreed far below their 10^{-10} tolerances, yet the global PSD failed the aggregate random and chronological gates for both detectors, with individual null-spread ratios reaching 10.480 for H1 and 7.619 for L1. The predeclared 64-second local model passed its stationary synthetic control but not real H1 or L1, and therefore did not repair calibration. Event scores and nominal injections remain archived diagnostics; we do not convert them to significance, false-alarm probability, or validated sensitivity.

This outcome is a boundary result rather than a verdict on the geometry, and it is the reason for the next stage. Because GWOSC carries no injected reference, a calibration failure there cannot be separated from the question the theory actually asks—whether the covariance-weighted reconstruction is closer to the true signal. Answering that on real data requires a detector release in which the truth is recorded alongside the readout.

The shared-FIR result has a deliberately narrow interpretation: two software paths agree when they implement the same finite-duration statistic. It does not establish mathematical equivalence between the original PSD-domain GLS statistic and a finite FIR statistic. Likewise, agreement with GWpy for the PSD estimator isolates one implementation question but cannot make a global or local Gaussian PSD model statistically adequate for nonstationary real noise.

9.5 Public TIDMAD ground-truth validation

Neither CRESST nor GWOSC records what the true signal was. Both therefore compare residuals rather than distances to truth, which is the quantity the metric-reversal claim is actually about. The public TIDMAD release from the ABRACADABRA axion search

Table 5: Compact GWOSC boundary results. PSD agreement and shared-FIR identity are implementation checks. Global and local score-spread results are statistical calibration checks on the selected interval.

Check	Synthetic/reference	H1	L1
Implementation checks			
PSD ratio, identical windows	reference = 1	1.000	1.000
Shared-FIR max relative L_2 difference	1.47×10^{-15}	1.70×10^{-13}	3.04×10^{-13}
Statistical calibration checks			
Global null-spread ratio, random median	—	1.986	1.263
Global null-spread ratio, chronological median	—	3.183	2.964
Local-64 score standard deviation	1.148 (pass)	9.424 (fail)	8.827 (fail)

[18] closes that gap: its validation files record an injected reference channel alongside the SQUID readout, at 10 MHz sampling, so a reconstruction can be scored against a real injected signal in real detector noise. We use the release's own channel convention (`channel0001` readout, `channel0002` reference). The frozen operating point was 4096-sample windows, rank 2, 512 windows split 70/30 chronologically into 358 fit and 150 evaluation windows with a four-window guard gap, and 300 bootstrap resamples; window-hash and window-index overlap between fit and evaluation were both zero.

As an implementation and data-integrity anchor, the release's own unmodified benchmark script was run on the full 20-file validation set before any analysis. The `none` baseline reproduced at coarse score 1.1026 against a published 1.10, fixing channel semantics, data integrity, and environment. The `savgol` baseline did not (-2.7708 against a published -2.35), which is a property of the published hyperparameters rather than of this reproduction: in the environment recorded with the run, at `polyorder` 11 the Savitzky–Golay coefficient sum collapses from approximately one to 1.7×10^{-10}, so the published score depends on numerical details of the coefficient computation and is not expected to reproduce bit-exactly across implementations or library versions. We treat the `none` match as the operative check and make no reproduction claim for `savgol`; the library versions used are in the archived environment record.

On real science-file noise with a synthetic injection, all six estimator- identity gates passed: fractional amplitude bias -3.0×10^{-4} ($z = -0.68$), empirical over PSD-predicted standard deviation 0.9986, a null consistent with zero ($z = -0.78$), 2.33° maximum principal angle between the learned rank-2 subspace and the analytic quadrature pair, the $\cos^2$ SNR-loss law to within 0.019, and a subspace-error scaling exponent of -0.9996 against the predicted -1. This validates the whitening and estimator machinery on real noise; it is not the metric-reversal test.

Table 6: Compact TIDMAD results, grouped by what each check can establish. Implementation identity, ground-truth reconstruction, statistical calibration, and the release's own detector-facing score are separate questions, and success on one is not evidence for another. Values are taken from the archived run JSON at commit `59db37151d45f18e67d2d60142a7140eaf02f47e`.

Check	Reference	Observed	Status
Implementation and data-integrity checks			
G0 `none` coarse score	published 1.10	1.1026	anchor matches
G0 `savgol` coarse score	published −2.35	−2.7708	ill-conditioned baseline
Phase 1 six estimator-identity gates	—	all passed	all passed
Ground-truth reconstruction comparison			
Phase 2 truth-error difference, file `0000`	> 0	$+0.0190$, CI $[0.0045, 0.0321]$	excludes zero
Phase 2 truth-error difference, file `0010`	> 0	$+0.0036$, CI $[-0.0040, 0.0133]$	consistent, not significant
Phase 2 truth-error difference, file `0005`	> 0	$+0.0402$, CI $[-0.0793, 0.1779]$	consistent, not significant
Amplitude RMSE, file `0000`: PCA / EMPCA / zero predictor	—	0.1209 / 0.0880 / 0.0747	both above zero predictor
Statistical calibration			
Phase 4 global-model χ^2/dof	band $[0.9, 1.1]$	1.0031	calibrates
Phase 4 score dispersion	band $[0.9, 1.1]$	1.0142	calibrates
Detector-facing benchmark			
Phase 5 coarse denoising score	no-denoising 1.10 (published, ref. [18])	−1.8025	below baseline

Reconstruction against real injected truth. The primary comparison scores rank-2 PCA and rank-2 EMPCA reconstructions of the readout channel against the file's own injected reference. The amplitude score projects each window's reconstructed quadrature coefficients onto the phase direction obtained from the evaluation reference itself; it is a truth-referenced diagnostic of the physical-parameter risk (44), usable only where the reference is recorded, and not an amplitude estimator available at deployment. The reference channel is treated as the injected signal in its own units; its gain, phase and frequency-dependent coupling into the readout are not modelled here, which is one reason the result is an ordering rather than a recovery in physical units. On `abra_validation_0000.h5` all three metrics ordered as the theory predicts, and the paired truth-error interval excluded zero. Raw MSE preferred PCA (182.40 against 212.44); the covariance-weighted residual preferred EMPCA (9.4324×10^{13} against 9.4994×10^{13}); and the injected reference also preferred EMPCA, with amplitude RMSE 0.0880 against 0.1209 (Fig. 9). The comparison must be read next to a trivial baseline: the root-mean-square of the injected amplitude itself over the same 150 evaluation windows — the error of predicting zero — is 0.0747 (recorded in the Phase 6 controls run on the same file at the same operating point, whose nominal EMPCA RMSE reproduces the 0.0880), so at this operating point both estimators recover the injected amplitude less accurately than a constant, and the result is an *ordering* of the two reconstructions, not a demonstration of useful recovery. The paired absolute-error difference $e_{\mathrm{PCA}} - e_{\mathrm{EMPCA}}$ was $+0.0190$ over 150 held-out windows, with 95% window-level interval $[0.0045, 0.0321]$ and block-level interval $[0.0027, 0.0344]$ at block size 16; both exclude zero. The leakage audit recorded zero window-hash and

zero window-index overlap between the 358 fit and 150 evaluation windows, at a realized four-window guard gap. The paired difference of per-window absolute errors (+0.0190) and the difference of the two aggregate RMSEs ($0.1209 - 0.0880 = 0.0329$) are different statistics and are both reported. This is the only real-data confirmation of metric reversal in the paper: raw MSE and the likelihood-aligned residual selected different reconstructions, and the recorded truth agreed with the likelihood-aligned one. It is a single file at a single frozen operating point below the zero-predictor level, and §11 states what it does not extend to. The archived run also records a paired clean-waveform MSE contrast in the reference channel's units (32.38 for PCA against 2.35 for EMPCA; paired difference 30.03, interval $[24.79, 37.07]$); it is not reported as a recovery result because the frozen protocol carried no zero or mean baseline for it and no reference-to-readout transfer model. The operating-point sweep originally specified as P1-E13 was redesigned so that the number of evaluation windows does not fall with window length, predeclared, and executed once on held-out files; its outcome is reported below.

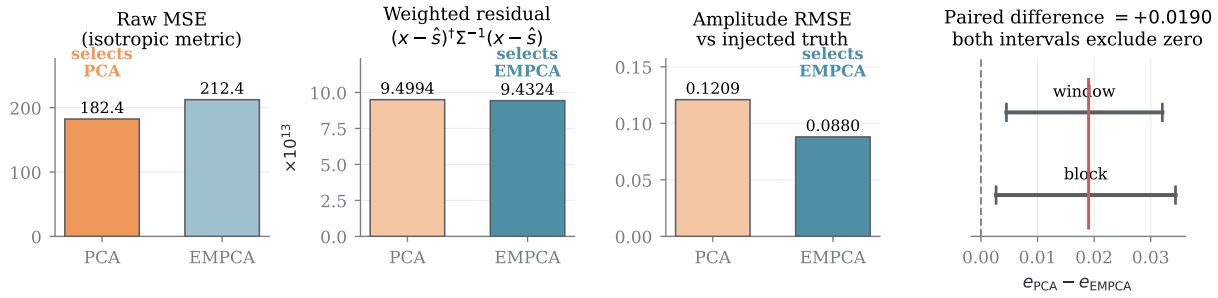


Figure 9: Metric reversal against recorded truth on TIDMAD validation file `abra_validation_0000.h5` (rank 2, 150 held-out windows). Raw MSE prefers the PCA reconstruction; the covariance-weighted residual prefers EMPCA; and the injected reference channel—the only panel that uses recorded truth rather than an assumed covariance—agrees with the weighted residual. The right panel shows the paired truth-error difference $e_{\mathrm{PCA}} - e_{\mathrm{EMPCA}}$ with window- and block-level 95% bootstrap intervals (300 resamples, block size 16); both exclude zero. Both amplitude RMSEs lie above the zero-predictor level of 0.0747 (Table 6). All values are generated from the archived run JSON.

Two further validation files are archived, and on both the predicted qualitative pattern held while the paired interval did not exclude zero. On `abra_validation_0010.h5`, raw MSE preferred PCA (171.15 against 199.36), the weighted residual preferred EMPCA (9.147×10^{13} against 9.197×10^{13}), and the injected reference also preferred EMPCA (amplitude RMSE 19.4799 against 19.4836); the paired absolute-error difference was +0.0036, with 95% window- and block-level intervals $[-0.0040, 0.0133]$ and $[-0.0089, 0.0127]$. The same three orderings held on `abra_validation_0005.h5` (raw MSE 131.13 against 148.93;

weighted residual 8.105×10^{13} against 8.058×10^{13}; amplitude RMSE 4.8103 against 4.7522), with paired difference +0.0402 and interval $[-0.0793, 0.1779]$. Both are consistent in direction with metric reversal and neither establishes it alone. Their limited resolving power is documented: the calibration tone couples into the reconstructed channel with ratio 3.22 on file `0000` but only 0.057 on file `0010`, the coupling falling off steeply with injection frequency across the release.

Held-out operating-range test, predeclared and executed once. The three files above were examined during development. A redesigned protocol (T10–T12 of `plan/MLST_EXPERIMENT_PROGRAMME_2026-09-10.md`; predeclaration `reviews/2026-09-10-PREDECLARED_T11.md`; report `reviews/2026-09-10-experiment-phase-report.md`, §4) then asked the recovery question on untouched data. Its endpoint is the physical injected amplitude in readout units, $a = 2q/\sqrt{w}$ with q the unitary DFT coefficient at the injection bin, so that window lengths are comparable; its truth label is the reference coefficient transferred to readout units through a complex gain fitted on 64 disjoint transfer-calibration windows per stretch, with a reference line SNR $\geq$ 100 required for eligibility; every stretch is characterized (main line, drift, coupling, multi-tone) on those calibration windows before any method is fitted; and the usefulness criterion is predeclared — an estimator is useful only if it beats both the zero predictor and the calibration-mean predictor with 95% intervals excluding zero. Three development files selected $w = 65536$ at rank 2 from $\{4096, 16384, 65536\}$; three validation files confirmed the direction; and the final test was run once on the remaining fourteen files, of which eight came out eligible-coupled, five uncoupled and one multi-tone. Baselines always run alongside PCA and EMPCA: zero, calibration mean, the no-denoising bin read $\hat{q}_i = c_i$, quadrature OF, a Wiener gain, and a calibration-shrinkage estimate.

Table 7 gives the outcome. Over the eligible stretches rank-2 PCA and EMPCA are indistinguishable from predicting zero (mean amplitude RMSE 0.661 and 0.655 against 0.654), and both are about 2.7× worse than reading the injection bin without any denoising (0.246). The combined paired contrasts of EMPCA against the two usefulness references include zero, and the contrast against the bin read excludes it in the bin read's favour; no "useful" verdict was issued for any file, both because the criterion fails on the combined estimate and because the frozen block-dependence assessment was inconclusive at nine evaluation blocks, which by the declared rule withholds the verdict. EMPCA does beat PCA consistently, in the direction the metric predicts, and the diagnostic shows why: EMPCA places 2.8×10^{-3} of its reconstruction energy at the injection bin against PCA's 2.1×10^{-4}, an order of magnitude more — but not enough to compete with the bin itself. The leading hypothesis for the cause is the dimension-to-sample ratio, and it is stated as a hypothesis. At $w = 65536$ the real dimension is $p = 65536$ against $n = 350$ fit

windows, $p/n = 187$. In the spiked-covariance model a population spike must exceed the noise level by about $\sqrt{p/n} \approx 14$ before the leading sample eigenvector overlaps it, and the injected tone's bin power is about 25 times the noise floor, so the operating point is near that threshold rather than clearly beyond it; the measured bin SNR is moreover not a population spike, since centering, phase variation between windows, leakage and dependence all enter. What is measured is the outcome — the injection bin carries 1.65×10^{-3} of the window energy while PCA's reconstruction places 2.1×10^{-4} there, an under-representation of about 8× — and whether estimation error in the whole-window subspace is its cause is the question a narrowband representation at the same window length is designed to decide (Section 10). Two further facts are recorded because they were predeclared expectations. High-frequency files were expected to be the uncoupled failure regime; they are the best coupled (readout line SNR 30–33 at 1.8 and 3.4 MHz), and the low-frequency and two-tone files fail instead — recorded before the method outcomes on those files were computed. And the precision-weighted combined estimate is effectively one file: `0019`, whose injected amplitudes are the smallest, carries 98.7% of the inverse-variance weight, so the weighted estimate is that file's value while the unweighted mean is $+7.8 \times 10^{-3}$; the protocol required the weighted estimand to be named as such and shown with every per-file value for exactly this reason (Fig. 10). Per file the picture is heterogeneous rather than a uniform null: three of the eight eligible test files and two of the three validation files would individually have met the criterion had the dependence assessment stabilized, one is significantly negative, and even on the files that would qualify EMPCA remains far from the bin read. Two sensitivities, computed after the outcome was seen and labelled as such, ask what the eight files say when no single one dominates. An equal-file mean of the per-file contrasts against the zero predictor is $+7.8 \times 10^{-3}$ with a file-level t-interval $[-4.2 \times 10^{-3}, +2.0 \times 10^{-2}]$ that includes zero. A DerSimonian–Laird random-effects combination, with per-file standard errors read from the archived intervals (which reproduces the archived inverse-variance weights to 0.1%), is $+6.5 \times 10^{-3}$ $[+2.0 \times 10^{-3}, +1.1 \times 10^{-2}]$: its between-file standard deviation, 5.9×10^{-3}, is the dominant variance term and its weights are 6–15% per file. The second interval excludes zero in EMPCA's favour and the first does not. With eight files, $I^2 \approx 0.98$ and a random-effects interval that under-covers at this size, neither replaces the predeclared primary, which stands as executed — no verdict issued, the combined usefulness contrasts including zero — and both say the same thing about scale: whatever EMPCA recovers here is of order 10^{-2} in squared readout amplitude against a bin read that is 0.40 better. The narrowband follow-up (`plan/MLST_EXPERIMENT_PROGRAMME.md`, II.T22) now predeclares the equal-file mean as its primary cross-file estimand, with the precision-weighted combination as a secondary sensitivity, so that the smallest-amplitude file cannot again define the conclusion. The quadrature-OF and bin-read estimators agree to 2.6×10^{-13}

on all eligible stretches, confirming the diagonal-Fourier-weighting identity on real data.

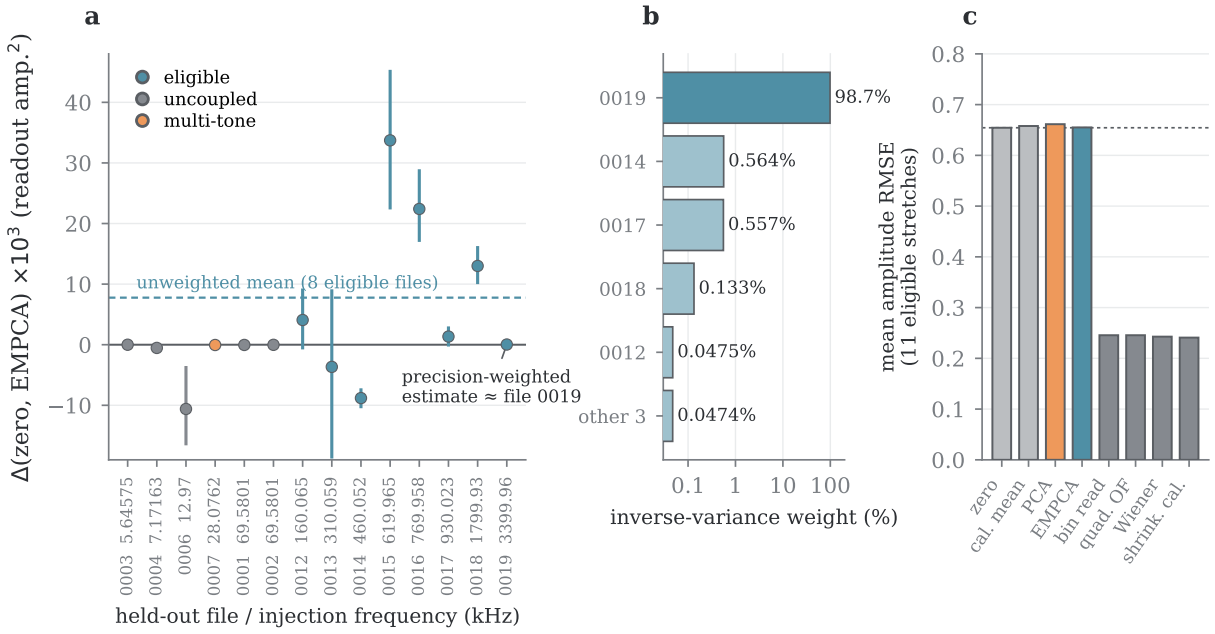


Figure 10: TIDMAD held-out operating-range test, per file ($w = 65536$, rank 2). (a) The paired contrast $\overline{\Delta}(\text{zero}, \text{EMPCA})$ on each of the fourteen final-test files, ordered by injection frequency, with its joint calibration/evaluation block-bootstrap 95% interval where the archive reports one; a positive value favours EMPCA over predicting zero; the eight eligible-coupled files enter the combined estimate. The dashed line is their unweighted mean; the precision-weighted estimate coincides with file `0019`, whose interval is the narrowest because its injected amplitudes are the smallest. (b) Inverse-variance weight shares of the eligible files, as archived. (c) Mean amplitude RMSE by method over the eleven eligible stretches: the learned rank-2 reconstructions sit at the trivial-predictor level (dotted), the frequency-domain estimators at 0.24–0.25. Values transcribed from the archived phase report, §4.5–4.6 (`evidence/data/paper1/mlst_2026-09-11/README.md`).

Narrowband representation at the same window, predeclared and executed once (T22). The whole-window stage left one hypothesis: that the failure was estimation error in a 65 536-dimensional subspace fitted from 350 windows. T22 tests it by removing that factor and keeping everything else — same window length, endpoint, truth labels, transfer calibration, rank, baselines, bootstrap and reporting — and fitting the rank-2 subspace instead on the $2B$ real coordinates of the B unitary-DFT bins centred on the injection bin. Two freezes are recorded: a stage-0 protocol (`plan/T22_STAGE0_PROTOCOL_2026-09-11.md` and its JSON) written before any code,

Table 7: TIDMAD held-out operating-range test ($w = 65536$, rank 2; eleven eligible-coupled stretches across validation and final test; readout physical-amplitude units). Top: mean amplitude RMSE by method. Bottom: precision-weighted combined paired contrasts $\overline{\Delta} = \overline{e_A^2 - e_B^2}$ over the eight eligible final-test files, with joint calibration/evaluation block-bootstrap 95% intervals (2000 replicates); the unweighted mean is shown beside it because file `0019` carries 98.7% of the weight. A positive contrast favours B. From `reviews/2026-09-10-experiment-phase-report.md`, §4.5–4.6.

	zero	cal. mean	PCA	EMPCA	bin read	quad. OF	Wiener	shrink. cal.
mean RMSE	0.654	0.658	0.661	0.655	0.246	0.246	0.243	0.241

Contrast $A - B$	weighted $\overline{\Delta}$	95% interval	unweighted
zero − EMPCA	$+1.6 \times 10^{-5}$	$[-8.9 \times 10^{-5}, +1.2 \times 10^{-4}]$	$+7.8 \times 10^{-3}$
cal. mean − EMPCA	$+4.6 \times 10^{-4}$	$[-3.8 \times 10^{-5}, +9.6 \times 10^{-4}]$	$+8.7 \times 10^{-3}$
PCA − EMPCA	-1.9×10^{-5}	$[-2.2 \times 10^{-4}, +1.8 \times 10^{-4}]$	$+1.5 \times 10^{-2}$
bin read − EMPCA	-1.19×10^{-2}	$[-1.31 \times 10^{-2}, -1.08 \times 10^{-2}]$	-0.40

and a selected-band predeclaration (`reviews/2026-09-11-PREDECLARED_T22.md`) written before any hold-out outcome was scored. The hold-out stretches are *new time regions* of files already examined, starting at sample 37 027 840 — the whole-window stretch end plus two guards, so they do not overlap — which makes this a temporal hold-out and *not* independent-file replication. $B = 1$ is the identity control (a rank-2 projector on two real coordinates is the identity, so band-PCA = band-EMPCA = bin read); it was asserted to a relative 10^{-12} on all six development stretches and measured between -1.2×10^{-18} and $+2.9 \times 10^{-17}$. The declared rule — the smallest admissible candidate within one paired-bootstrap standard deviation of the best — selected $B = 9$, giving $p/n = 18/350 = 0.051$ against 187.

The dimensional hypothesis is confirmed in direction. Over the seven eligible stretches the learned rank-2 reconstruction's mean amplitude RMSE falls from 0.655 to 0.249 while the zero predictor stays at 0.654–0.689 and the bin read at 0.246–0.261: it moves from indistinguishable from predicting zero, and 2.7× worse than a bin read, to *better* than the bin read (Fig. 11). On the frozen final test the equal-file contrast $\overline{\Delta}$(bin read, band-EMPCA) is $+1.58 \times 10^{-3}$ $[+3.5 \times 10^{-4}, +3.2 \times 10^{-3}]$, positive on all four eligible files with each file's own interval excluding zero; validation gave the same direction.

All three registered verdicts are nevertheless ***inconclusive****.* The frozen dependence assessment did not stabilise — the aggregate block-variance curve is still rising at the largest usable block, $V(32)/V(16) = 1.745$, leaving nine complete blocks, the same limiting count as the whole-window stage's 150/16 — and the rule declared before devel-

opment withholds every benefit verdict in that case, not only the limited-recovery one. Had it stabilised, limited recovery ($+0.3927$ $[+0.0045, +0.798]$ against zero and $+0.3935$ $[+0.0046, +0.798]$ against the calibration mean; Holm-adjusted $p = 0.020$) and denoising benefit ($p = 6.0 \times 10^{-4}$) would have been declared and metric contribution would not.

The metric contribution is absent: $\overline{\Delta}(\text{band-PCA}, \text{band-EMPCA}) = -3.8 \times 10^{-5}$ $[-2.3 \times 10^{-4}, +1.3 \times 10^{-4}]$, $p = 0.70$, with the point estimate marginally favouring band-PCA, so the registered direction is *not met*. The explanation is post hoc but structural: the two differ only in the metric, and inside a nine-bin band the noise PSD is nearly flat. The measured in-band dynamic range $\max_k N_k / \min_k N_k$ is 1.16–4.66 across the eight hold-out stretches against 4.77×10^4 over the interior bins of the whole window. The restriction that fixes the estimation problem therefore removes most of the metric's leverage — the paper's two determinants trading off against each other — and whether a band wide enough to carry PSD structure and narrow enough to keep p/n small exists on this instrument is untested. File `0016`, eligible at the whole window, is *chirped* in the new time region (drift 5.54 bins against the declared 0.5) and is excluded by the declared rule, leaving four eligible final-test files; the inverse-variance combination is again dominated by one file ($I^2 = 0.986$), which is why the equal-file mean is primary; and the equal-file t-interval on the point estimates alone includes zero on every contrast while the two-stage bootstrap excludes it on three. 240 of a declared cap of 250 fits were spent. The full report, including the one amendment made on development material and the deterministic re-score that followed, is `reviews/2026-09-11-T22-report.md`.

Calibration of the noise model. Unlike the GWOSC interval, a simple global Gaussian PSD model passes the two predeclared second-moment calibration checks on this detector's noise. On `abra_validation_0000.h5` over 512 windows of 4095 degrees of freedom each, the global model gave χ^2 per degree of freedom 1.0031 and held-out score dispersion 1.0142, both inside the predeclared $[0.9, 1.1]$ band and close to the stationary control's own 1.0002 and 1.0140. Local models at radii of 8, 16, and 32 windows gave dispersions of 1.0500, 1.0535, and 1.0276, so no local refinement was needed, and the amplitude cost of the mismatched model was a standard-deviation inflation of 1.0002. This is a positive result about this file and its approximately 0.2-second sample range, not about SQUID readout in general, and it is a second-moment result: a mean normalized energy and a score dispersion near one do not establish Gaussian tails, independence across frequency or time, or calibrated coverage at a stated level, which non-Gaussian processes can also satisfy; no calibrated-uncertainty or false-alarm claim is made on it. It does not weaken the GWOSC finding: together the two say that whether a stationary Gaussian model is adequate is an empirical property of the instrument rather than of the framework.

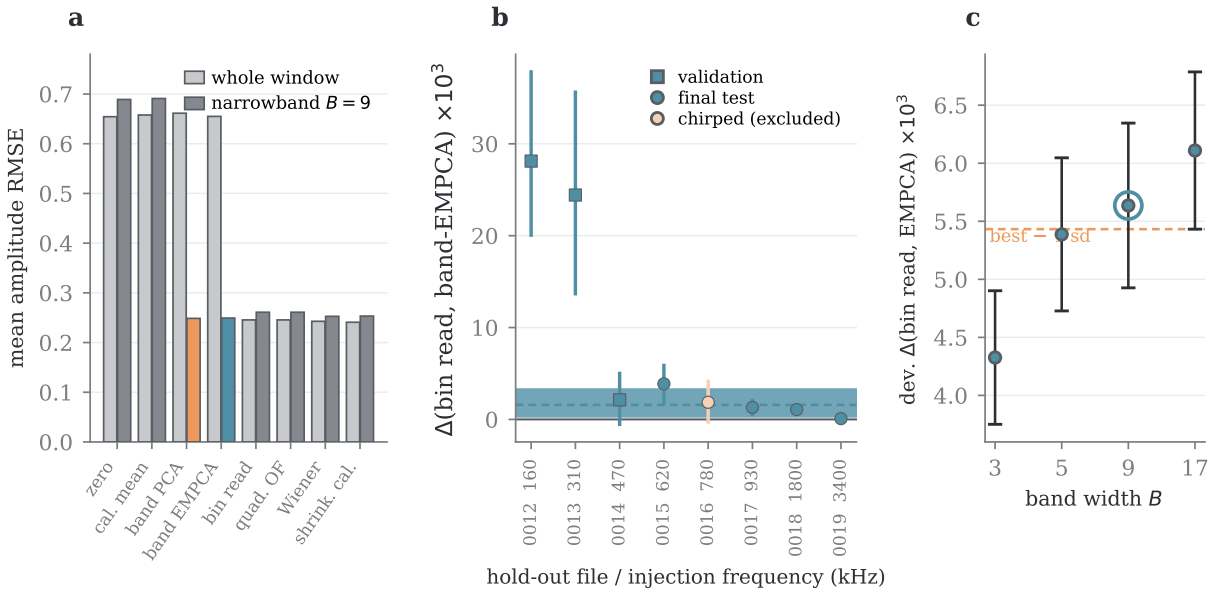


Figure 11: T22 narrowband hold-out at $w = 65\,536$, rank 2, $B = 9$. (a) Mean amplitude RMSE by method: the whole-window stage (light, eleven eligible stretches) against the narrowband stage (dark, seven eligible stretches). (b) Per-file $\overline{\Delta}(\text{bin read}, \text{band-EMPCA})$ with joint calibration/evaluation block-bootstrap intervals; squares are validation, circles the frozen final test, the pale marker the chirped file excluded by the declared rule; the band and dashed line are the equal-file combined estimate and its two-stage bootstrap interval. (c) Development band selection: equal-stretch mean with paired-bootstrap standard deviation per candidate, the "best minus one sd" threshold, and the selected $B = 9$ (circled). Registered inputs: `generated/SOURCE_MAP_part2.md`.

Detector-facing denoising score. Scored through the release's own unmodified benchmark script, the noise-aware linear Wiener estimator achieved a coarse denoising score of -1.8025, below the no-denoising baseline of 1.10 and below every published baseline except the ill-conditioned `savgol` (moving average 0.64; FCNet 6.55; PUNet 3.84; Transformer 4.18 [18]). The predeclared question was how much of the gap between no denoising and a trained deep denoiser a correct noise geometry with a linear model explains, and the honest answer at this operating point is a negative amount. A likelihood-optimal rank-2 estimator for a narrowband injection is not a broadband time-series denoiser, and we make no competitiveness claim on this metric.

Rank prediction and negative controls. The analytic rank-2 prediction, that an unknown-phase sinusoid spans exactly $\mathrm{span}\{\cos, \sin\}$, is confirmed in theory and in controlled synthetic data. Three attempts to confirm it on real TIDMAD validation data did not succeed—including one file selected by a data-quality rule frozen before its outcome was known, on which rank-2 amplitude RMSE 4.752 was statistically indistinguishable from rank-1 4.752, paired CI $[-0.056, 0.098]$. The diagnoses offered are hypotheses rather than established causes, and none implicates the estimator: a persistent two-tone structure in one file's reference channel, a richer manifold than the single-frequency model assumes; weak reference-to-readout coupling in another; and, on the file passing every predeclared quality check, most plausibly insufficient power at 150 evaluation windows. We therefore claim no real-data confirmation of the rank prediction. Of five predeclared negative controls, three degraded the primary metric with 95% CI excluding zero—wrong injection frequency (+0.357), permuted frequency bins (+0.016), and wrong PSD weighting (+0.017), against a nominal amplitude RMSE of 0.0880. Two did not: noise-only training was flat (-0.0025, CI spanning zero), and a random weighted basis improved slightly (-0.0062, CI entirely negative) because it collapses to the zero-recovery level, whose RMSE of 0.0747 is itself below the nominal, leaving that control saturated rather than informative. A norm-matched random basis, scaled to the whitened energy of the fitted basis, was implemented for the specified rerun; a synthetic wiring check found that it also collapses to the zero-recovery level, because a random *direction* recovers nothing however its energy is scaled, so that control is not expected to resolve the saturation. As with the CRESST wrong-PSD exceptions, we report both rather than adjusting the controls.

10 Discussion

The theory and experiments support one organizing statement: under the stated linear Gaussian model, a reconstruction method is the pair $(\Sigma^{-1}, \mathcal{F})$ of a residual metric fixed

by the measured noise and an admissible class fixed by the architecture. Once the metric is fixed, the four classes of this paper — a fixed template (OF), a learned subspace (EM-PCA), its tied encoder–decoder form, and its Kronecker-separable restriction (NFPA) — are not competing likelihoods but different feasible sets for one Σ^{-1}-orthogonal projection, and the separable class is the first at which the architecture rather than the metric decides what can be represented. The equivalences are deterministic identities between optimization problems, exact under matched preprocessing, weighting, normalization and linear model assumptions; the noise model enters only in choosing the weight. Untying the encoder does not change the optimal reconstruction under full-rank data and a spectral gap (Appendix E.2); it changes the parameterization that an optimizer must traverse.

Two results of this version say what happens to that statement when the covariance is estimated and the sample is finite. The metric is only as good as the covariance supplied, and the sensitivity of the learned subspace to covariance error is not a function of the error's size: Proposition 5.2 locates it in the coupling block of the relative error between retained and discarded eigenspaces, divided by the pairwise gaps, and the controlled study of Section 9.2.2 (Fig. 5) found the three regimes the proposition separates — a first-order regime in which the displacement follows the coupling block within a factor of two, an ordering regime in which a diagonal error of any size in the fitted eigenbasis leaves the subspace untouched until the spectrum reorders and then exchanges it, and a conditioning regime, near-singular empirical covariance, that no perturbation statement covers and that shrinkage repairs only in its spectrum. This is why two perturbations of equal relative norm differed by two orders of magnitude in effect, and why the global bound of Remark 5.2, valid wherever its hypothesis held, was uninformative there. The class, in turn, trades approximation for estimation: Proposition 6.3 separates the finite-sample contrast between the separable and the unrestricted class into a population approximation cost, provably positive off the separable case, and two estimation excesses, and the accounting of Section 9.2.3 (Fig. 6) observed both regimes on one configuration, with the crossover bracketed between $N = 800$ and $N = 1600$ on fresh seeds. Neither result alters the identities of Sections 2–7; both say what those identities cost when their inputs are estimated.

A third result of this version says what happened when the whole-window diagnosis was tested directly. Restricting the same rank-2 fit to a nine-bin band at the same TIDMAD window, on new time regions of the same files and under two recorded freezes, moved the learned reconstruction from the level of predicting zero to below a no-denoising bin read — the dimensional diagnosis confirmed in direction. Two things then happened that the design did not anticipate. Its block-dependence assessment did not stabilise, leaving nine complete blocks, the same limiting count as the whole-window stage, so the rule declared before development withheld all three benefit verdicts. And the metric's contribution

disappeared: inside a nine-bin band the noise PSD is nearly flat (in-band dynamic range 1.2–4.7 against 4.8×10^4 across the window), so band-PCA and band-EMPCA nearly coincide. That is the sharpest statement this paper can make about the interaction of its two determinants: on this instrument, the restriction that repairs the estimation problem removes most of the geometry's leverage, and a band that does both is not among the four tested.

The resulting loss-design consequence is practical. Isotropic MSE assumes white Gaussian noise. When the covariance is structured, it emphasizes raw variance rather than inverse-noise-weighted information, so its model ranking can disagree with the ranking implied by the assumed likelihood. The controlled white-noise agreement and colored-noise metric reversal isolate this mechanism without changing rank or data. Covariance-weighted residuals are therefore the appropriate primary comparison when a Gaussian covariance model is the intended statistical description *and* the target is the likelihood or the whitened clean-signal risk, which the residual predicts exactly, in population and up to a trace correction, for any fixed linear reconstruction (Proposition 7.1). For a Euclidean clean-signal or a physical-parameter target the ordering is a hypothesis to be tested against recorded truth (Example 7.2), which is what the TIDMAD stage does. This conclusion does not say that the Gaussian model is always adequate; the GWOSC result shows why model adequacy must itself be tested.

The public CRESST result supplies the positive real-detector component that the synthetic theory alone cannot provide. Every one of the seven channels favored rank-6 EMPCA over a single-template OF on held-out weighted residual, and every channel met the multi-template-OF competitiveness margin. The channel-balanced OF difference was 731.87, with a post-freeze hierarchical 95% interval $[110.69, 1537.48]$. That result mainly tests the signal-model hierarchy: rank-6 EMPCA can represent held-out pulse-shape variation that a fixed template cannot. It should not be attributed wholly to covariance weighting. The same-rank PCA–EMPCA contrast addresses weighting more directly. EMPCA had the lower channel-mean weighted residual in all seven channels, but its fractional advantage ranged from 1.6% to 62.0%. The covariance effect is therefore directionally consistent and heterogeneous, rather than uniformly large; its unnormalized channel-balanced difference was 3.78 with hierarchical 95% interval $[1.22, 6.69]$. The negative controls check the pipeline. The tied-linear-AE row is a closed-form parameterization consistency check evaluated through the same weighted-PCA routine that fits EMPCA (Section 9.3); the near-zero AE/EMPCA angle is therefore not independent evidence for the theorem, and in the default build the independently trained tied-AE evidence is the synthetic test S3 only.

The CRESST claim should consequently remain narrow. The basis and PSD were estimated and evaluated within each channel. The experiment supports a within-channel

linear reconstruction hierarchy for the released pulses and provides descriptive real-data evidence for covariance weighting. It does not demonstrate transfer to an unseen channel, a physical energy-resolution improvement, or trigger sensitivity. The inconclusive trigger grid reinforces the separation between reconstruction and detection.

The GWOSC analysis establishes a complementary boundary. Agreement with GWpy for identical PSD windows shows that the repository PSD calculation is not trivially different from its reference. Agreement between explicit FFT and GWpy convolutions under shared FIR coefficients and normalization shows software identity for that common finite statistic. Neither check implies calibration. Held-out score spreads were too large under the global PSD, and the fixed 64-second local PSD also failed on real H1 and L1 although its stationary synthetic control passed. Thus implementation correctness is necessary but not sufficient: calibrated likelihood inference additionally requires a noise model that describes the evaluation distribution. The predeclared empirical hypothesis that the 64-second local PSD would restore calibration was falsified on this interval. This contradicts that application- level hypothesis, not the conditional linear-Gaussian projection results. The present experiment does not identify whether the remaining mismatch is due to non-stationarity, transients, data-quality structure, or statistic-specific finite-window effects.

The TIDMAD stage was added because CRESST and GWOSC share a structural limitation: neither records the signal that was actually present, so both compare residuals under an assumed covariance rather than distances to a known truth. TIDMAD's injected reference channel removes that circularity. Its contribution is specific and partial. The estimator identities hold on real SQUID noise, and on all three archived validation files the three metrics ordered as predicted, with raw MSE preferring one reconstruction while the weighted residual and the injected truth preferred the other. On the primary file the paired truth-error interval also excluded zero ($+0.0190$, window-level 95% CI $[0.0045, 0.0321]$; block-level $[0.0027, 0.0344]$), which is the paper's only real-data confirmation that the likelihood-aligned ranking and the recorded truth — scored as injected-amplitude error, a physical-parameter risk that Proposition 7.1 does not cover — agree where raw MSE disagrees with both. On the two secondary files the same ordering held but the intervals included zero. The confirmation is an ordering: at the frozen operating point the root-mean-square of the injected amplitude itself, 0.0747, is below both estimators' amplitude RMSE (0.0880 and 0.1209), so neither reconstruction recovers the injection better than a constant. The predeclared held-out test of Section 9.5 then asked whether a longer window would change this, on fourteen untouched files, and it does not: at $w = 65536$ the rank-2 reconstructions are indistinguishable from the zero predictor and lose by a factor of 2.7 to a direct read of the injection bin, at a dimension-to-sample ratio $p/n = 187$ that makes estimation error in the whole-window subspace the leading hypothesis. EMPCA still beats PCA in the predicted direction, and the operating point at which

either would be useful is not reached by rank-2 whole-window reconstruction on this release. The test that decides the hypothesis keeps the long window and reduces the learned dimension: a small multi-bin Fourier band around the injection frequency, frozen from calibration information about drift and leakage, on which PCA and EMPCA receive identical bins, rank, training windows and preprocessing, with a single-bin control (where a rank-2 projector is the identity and any "recovery" is the bin read itself) and the zero, calibration-mean, bin-read and Wiener baselines retained, so that beating zero, beating the bin read and beating matched-band PCA are reported as three separate results. The calibration finding is a second positive result, and it is a positive result rather than a boundary lesson: a simple global Gaussian PSD model passes its predeclared second-moment calibration checks on this detector's noise, where the same class of model failed on GWOSC strain; tails, dependence and coverage were not tested. Read together, the two say that stationarity is an empirical property of the instrument and not a defect of the framework, which is a more useful statement than either result alone.

Where TIDMAD did not deliver should be stated with equal directness. The rank-2 quadrature prediction was not confirmed on any of the three real files attempted, and the noise-aware linear estimator scored below the no-denoising baseline on the release's own detector-facing benchmark. Neither outcome contradicts the theory—the diagnoses point to multi-tone structure, weak channel coupling, limited statistical power, and a mismatch between a narrowband rank-2 estimator and a broadband denoising score—but neither supports it either, and we do not present them as if they did. The held-out test replaced one of those hypotheses with a measurement — coupling is strong at high injection frequency and weak at low, the reverse of the predeclared expectation — and sharpened another: the leading hypothesis is now estimation error at $p/n = 187$ rather than statistical power, and it is a hypothesis with a designed test, not a measured cause. That test keeps the window, the training observations, the injection, the endpoint and the PCA-versus-EMPCA comparison of the held-out test and removes only the whole-window dimension, so that its three questions — whether a narrowband subspace recovers what the whole-window one could not, whether the metric contributes beyond band selection, and whether either beats the direct bin read — are answered separately; and it predeclares an equal-file estimand, because the held-out test's precision-weighted combination was, in effect, one file.

Together these results clarify the level at which the framework should be used. The covariance-weighted objective is the correct likelihood geometry conditional on Σ. Empirical validation must then ask three separate questions: whether the constrained signal model reconstructs held-out signals better within that geometry, whether the estimated covariance actually calibrates residuals or scores on the target data, and whether the preferred reconstruction is in fact closer to the true signal in the named target metric

(Section 7.1). CRESST answers the first positively for the within-channel linear hierarchy. GWOSC answers the second negatively for two frozen PSD models on one interval, and TIDMAD answers it positively, at the level of second-moment checks, for one file of a different instrument. The third question is the one only a release with injected truth can address.

Untested design suggestions. Proposition 3.1 suggests two rules for architectures that tokenize a measurement, and we record them here as suggestions rather than results. First, forming tokens from the whitened trace $\tilde{x} = \Sigma^{-1/2}x$ makes Euclidean dot products of the *inputs* agree with the likelihood metric; it fixes nothing about the attention scores themselves, because learned query and key projections change the dot-product geometry, and inverse-PSD weighting of Fourier features (the eigenvalue weights of the proposition) is not the same operator as whitening by the inverse square root (the $1/\sqrt{J_k}$ scaling). Second, when the measurement has more than one axis, tokenizing along the axis in which Σ^{-1} is diagonal turns the metric into a per-token weight rather than a dense coupling across tokens; on the TIDMAD readout that axis is frequency. Both concern the metric only, supply no locality or deformation stability (Appendix E.5), and neither has been tested in this paper.

What is left to a companion study. Propositions 7.3 and 7.4 were stated here because they follow from the same metric and because they turn two deployment questions into predictions: a frozen model that meets noise whose covariance differs from the training one must respond within the excited span, and one that meets data outside its excited support must respond along the complement, with the pulled-back form $J^T\Sigma^{-1}J$ as the statistic that measures both. Neither deployment prediction is tested in this paper, and the one excitation study it reports is not such a test: T21 (Section 9.2.4) varies the support *at formation time*, in the training draw from which the representation is fitted, and asks which planted directions the fitted subspace recovers; the companion study's question is the post-freeze one, whether a fixed model meeting data outside its excited support can be recognized as doing so. What T21 found bears on that question only as a caution: recovery of a planted physical direction is direction-dependent — on the exploratory reading, set by the population spike strengths relative to d/N and by the direction's loadings on the population eigenvectors, not by a scalar excitation — so "excited" and "detectable" are different properties, and no abstention threshold was derived; that is a hypothesis a companion study can test, not a result it can use. A companion study is designed to take them to frozen public models on real detectors and to report consequence in physical units; until it does, the two propositions are contributions of this paper only as stated identities.

11 Limitations

The equivalence results concern linear subspaces and Gaussian noise with a known or independently estimated positive-definite covariance. They do not cover curved signal manifolds, saturation, pile-up, heavy-tailed contamination, or covariance estimation from the same event being reconstructed. Timing variation can be approximated locally by additional linear directions, but a finite subspace need not represent large shifts or other nonlinear waveform changes. Likewise, a Mahalanobis objective is efficient only when its noise model is adequate; covariance drift or outliers can invalidate its uncertainty interpretation even when the optimization is implemented correctly. Two failures of the assumptions made here — a covariance that differs from the one the model was trained under, and a deployment distribution whose physical strata the training never excited — are deliberately outside this paper (Section 10). The tied-versus-untied ablation and the July excited-support experiment (P1-E11, two binary support choices) belong to the held evidence (Section 1) and are not reported in this version.The excitation study that *is* reported, T21 (Section 9.2.4), is a different experiment: it scales one physical latent continuously in the training draw and measures formation-time recoverability at fixed rank on one generator at one nonseparability, and its spectrum-and-loadings reading is post hoc — checked on the runs that suggested it — and is stated as a hypothesis for an independent test, not as a result.

Noise covariance also has an identifiability limit that is separate from the linear reconstruction results. An eigendecomposition of a covariance or cross-spectral-density matrix produces decorrelated statistical modes, not uniquely labelled physical sources. Nothing in this paper recovers a physical noise decomposition from a measured total spectrum, and no result should be read as attributing residual improvement to a named noise mechanism. Where a component budget is supplied with a dataset, its labels are an input to the analysis rather than an output of it, and a conditional attribution computed under such a budget does not establish that the component is separable in hardware or that removing it would improve a real detector by the same amount.

The CRESST inference is based on seven run/channel clusters. The primary predeclared interval resamples 19,335 pooled events and therefore quantifies event-level uncertainty conditional on these channels; it does not quantify the variation that would arise from sampling a new detector channel. We show all channel effects and an equal-channel point estimate to expose scale heterogeneity. A post-freeze channel-then-event analysis with 10,000 resamples retains the primary effect directions, but its intervals are coarse and potentially unstable because only seven clusters from two runs are available. Pulse and noise streams were each split chronologically, separately; the design does not use a common cross-stream timestamp cutoff. It does not test alternative split fractions,

cross-channel generalization, or independent acquisition campaigns. One negative-control exception—wrong-PSD weighting on `35/58.0`—also shows that pooled control behavior need not hold for every channel.

The CRESST trigger-efficiency calculation was underpowered over the frozen amplitude grid. Six channels were inconclusive at the primary 5% false-positive rate, and none reached 90% efficiency. Reconstruction gains cannot therefore be translated into a threshold, exposure, or detector- sensitivity improvement. Such a claim would require a separately frozen signal-injection and background protocol with an amplitude range that brackets the relevant efficiencies.

The GWOSC evidence covers one 256-second interval centered on GW150914. Windows and chronological blocks from that interval are not independent astrophysical replications. The global and local PSD failures are interval-specific boundary results, not evidence that the framework fails on all gravitational-wave data. Conversely, the passed PSD and shared-FIR checks do not validate event significance. The original PSD-domain GLS statistic and a finite-duration FIR statistic were not shown to be mathematically equivalent. New background models, data-quality rules, or frequency choices must be frozen and tested on untouched intervals rather than tuned until this interval passes.

The TIDMAD evidence is narrower than the number of phases run suggests. Each result comes from named validation files at a single frozen operating point, and no result should be read as a survey of the release; the injected frequency sweeps within each file and the reference-to-readout coupling varies by more than three orders of magnitude across files, so file-to-file transfer is not assumed. The calibration finding covers one file and an approximately 0.2-second sample range and consists of two second-moment checks, not a test of tails, dependence or coverage, and the metric-reversal confirmation covers one file at one frozen operating point: it is a single interval that excludes zero, not a replicated effect, and the two secondary files did not reach significance; the held-out test covers fourteen files once, but its precision-weighted summary is dominated by one file and its dependence assessment was inconclusive, so it establishes a boundary and a mechanism, not an effect size; at the frozen operating point both estimators sit above the zero-predictor amplitude RMSE, so the confirmation is an ordering and the operating point at which the estimator recovers the injection better than a constant is not established. The rank-2 prediction has no real-data confirmation: three attempts failed, and although each has an identified non-implementation cause, a diagnosed non-confirmation is not a confirmation deferred. Metric reversal and the rank prediction are separate claims, and confirming the first on a file does not confirm the second on that file. The denoising score is a genuine negative under its predeclared claim boundary. Two of five negative controls did not degrade, one of them because it had saturated at the zero-recovery level, which means the control set here demonstrates falsifiability but cannot rank control severity; a norm-

matched random-basis control was implemented and has not been run on the release, and a synthetic wiring check indicates it would saturate for the same reason.

12 Code and data availability

The CRESST pulse-shape data are publicly described in Ref. [16] and hosted by the ORIGINS Dark Matter Data Center (`www.origins-cluster.de/odsl/dark-matter-data-center`); the collaboration asks that the description and Ref. [56] be cited by work based on the release, and we do so. The GWOSC strain is the O1 bulk release [17, 57], CC BY 4.0. This research has made use of data or software obtained from the Gravitational Wave Open Science Center (`gwosc.org`), a service of the LIGO Scientific Collaboration, the Virgo Collaboration, and KAGRA. The corrective evidence archive uses schema `paper-revision-evidence/1.0.0`, bundle identifier `paper1-revision-20260726-partial-v1`, and manifest SHA-256 `aa7ea1632983d3c59318fca5c1abd6288f0ebd13281d5a97920692e561e52944`. It records experiment-repository commit `34112bfc313a127840a05e17ce93f23720457f14` and the confirmatory S5 run `S5-30seed-heldout-20260726`. All results reported in this paper come from public datasets; no internal or unreleased detector data, noise budget, or instrument simulation is used or described. The CRESST run is tied to source commit `9c9f696ff6bca6e6b9729cdb90e41ee92d576787`, seed 22, and the archived configuration `p2_cresst_validation_seed22.config.yaml`. The post-freeze CRESST sensitivity archive is indexed under `20260713_channel_robustness_v1`; it records 10,000 hierarchical resamples with seed 20260713, lossless event residuals, analysis code, an environment record, a leakage audit, and checksums. The GWOSC baseline archive has run identifier `20260622T175125Z_b169c1f595a4` and base commit `b169c1f595a46d2701417ff4cbce292330817ad2`; the shared-FIR and local-PSD follow-up runs record source commit `806a849305a9a99d50d545106ba4e63e8de8e7c7` and are archived with their configurations and JSON outputs. The TIDMAD data are the public release `github.com/jessicafry/TIDMAD` under CC-BY-4.0 [18]; gate G0 used the full twenty-file validation set, and the reported phases used validation files `abra_validation_0000.h5`, `abra_validation_0005.h5`, and `abra_validation_0010.h5` together with the release's science files for real-noise windows. The frozen TIDMAD operating point was 4096-sample windows, rank 2, a 70/30 chronological split with a four-window guard gap, and 300 bootstrap resamples with block size 16, at seed 20260727 and source commit `59db37151d45f18e67d2d60142a7140eaf02f47e`; the channel convention, per-file injection frequencies, and valid sample ranges are recorded in the archived data contract. The primary Phase 2 run on `abra_validation_0000.h5` is archived as `data/tidmad/current/TIDMAD_P2_real_`

`seed20260727.json`, experiment identifier `TIDMAD_P2_REAL`, manifest SHA-256 `dbddcccb844734a1087375299abc94f944c6d75323155f8cada22225d100a24b`; the two secondary files are archived alongside it under distinct run identifiers. Raw public strain and pulse arrays are not duplicated in the manuscript archive; source locations and evidence checksums are retained so that the inputs can be retrieved from their public releases and the reported results audited.

13 Conclusion

Under a linear Gaussian measurement model, inverse noise covariance defines the geometry of reconstruction, and the architecture chooses only the admissible class within it: OF is a fixed rank-1 likelihood projection, weighted low-rank reconstruction or EMPCA learns the projection subspace, the tied noise-aware linear autoencoder recovers the same subspace under the stated conditions, and the Kronecker-separable restriction solves the same objective over a product Grassmannian. Isotropic MSE is the corresponding likelihood only for white noise. Two results of this version go beyond that hierarchy. First, the effect of a wrong covariance on the learned representation is local, not a function of the size of the error: to first order the subspace moves by the coupling block of the relative error between retained and discarded eigenspaces divided by the pairwise gaps (Proposition 5.2); a diagonal error leaves it unchanged until it reorders the spectrum; and a near-singular estimate is a third regime that no perturbation statement covers. Perturbations of equal norm differed by two orders of magnitude in effect, as the mechanism predicts and the global bound cannot. Second, for the restricted class the finite-sample contrast with the unrestricted one decomposes exactly into a population approximation cost, provably positive off the separable case, and two estimation excesses (Proposition 6.3); on fresh seeds the restricted learner won at $N = 800$ and lost at $N = 1600$ at both nonzero nonseparabilities, so the regularizing and the approximation-dominated regimes have both been observed and the crossover is bracketed. A third study found that excitation in the training data does not by itself make a physical direction recoverable: recovery followed the population spectrum and the direction's eigenvector loadings, a reading that remains to be tested independently. Controlled simulations support the estimator identities, while public CRESST data support the narrow claim that rank-6 weighted subspace reconstruction improves within-channel held-out residuals relative to fixed-template baselines, with the smaller and heterogeneous same-rank PCA–EMPCA contrast as the direct real-data evidence about covariance weighting. Public GWOSC data show the complementary limit: correct PSD and shared-statistic implementations were insufficient for calibrated inference under the tested global and local PSD models. Public TIDMAD data, which unlike the other two releases record an injected reference signal, supply the estimator identities

on real SQUID noise, a global noise model that passes second-moment calibration checks where GWOSC's did not, and — on one validation file — the paper's only real-data confirmation of metric reversal against recorded truth, an ordering at an operating point where neither reconstruction beats a constant predictor. A predeclared test on fourteen held-out files extends that boundary to a second operating point: it finds no evidence of useful whole-window rank-2 recovery under the declared criterion, with both reconstructions 2.7× worse than a direct read of the injection bin, and its dimension-to-sample ratio of 187 makes estimation error the leading hypothesis and the subject of the designed narrowband test. On that same release the rank-2 prediction was not confirmed and the linear estimator did not compete on the detector-facing denoising score. Cross-channel transfer, detector sensitivity, calibrated GW150914 significance, a real-data rank confirmation, replication of the metric-reversal result beyond one file, an independent test of the spectrum-and-loadings reading, and broader real-noise generalization remain outside the evidence reported here.

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A Detailed Optimal-Filter Derivations and the Cramér–Rao Bound

This appendix expands the summary statements of Section 3.2: unbiasedness, variance, Fisher information, and the energy-resolution formula for the OF estimator.

Convention. The primary convention is a real time-domain trace $x = As_0 + n \in \mathbb{R}^d$ with a real amplitude A and real Gaussian noise $n \sim \mathcal{N}(0, \Sigma)$, Σ circulant under the periodic stationary model of Appendix G, $\Sigma = F^\dagger \operatorname{diag}(J_k) F$ with F the unitary DFT and $J_k = \mathbb{E}|N_k|^2$ the two-sided bin power of the unitary-DFT coefficients $N_k = (Fn)_k$. Writing $X_k = (Fx)_k$ and $S_k = (Fs_0)_k$, the quadratic form is

$$(x - As_0)^T \Sigma^{-1} (x - As_0) = \sum_{k=0}^{d-1} \frac{|X_k - AS_k|^2}{J_k}, \qquad N_\Phi := s_0^T \Sigma^{-1} s_0 = \sum_{k=0}^{d-1} \frac{|S_k|^2}{J_k}, \tag{48}$$

where both sums run over all d bins. On the one-sided rFFT grid the same quantities are obtained with endpoint weights, $\sum_{k=0}^{d-1}(\cdot) = \sum_{k=0}^{\lfloor d/2 \rfloor} w_k(\cdot)$ with $w_k = 1$ for the DC bin and (even d) the Nyquist bin and $w_k = 2$ for the interior bins, because $X_{d-k} = X_k^*$ and $J_{d-k} = J_k$; conversion of J_k to a physical one-sided PSD density is the factor recorded in Appendix G. Every statement below is made in this convention. The proper-complex convention, in which each bin is an independent circular complex observation, is stated separately at the end, because a real trace's full DFT is *not* a vector of d independent proper complex bins.

Unbiasedness. The estimator is $\hat{A} = (s_0^T \Sigma^{-1} x)/N_\Phi = \sum_k \Phi_k^{(0)} X_k$ with $\Phi_k^{(0)} = S_k^*/(J_k N_\Phi)$, the two-sided sum being real by conjugate symmetry. Under $X_k = AS_k + N_k$,

$$\mathbb{E}[\hat{A}] = \sum_k \Phi_k^{(0)} A S_k = A \sum_k \frac{S_k^* S_k}{J_k N_\Phi} = A \cdot \frac{N_\Phi}{N_\Phi} = A. \tag{49}$$

Estimator variance. In detector coordinates, $\hat{A} - A = s_0^T \Sigma^{-1} n / N_\Phi$, so

$$\operatorname{Var}(\hat{A}) = \frac{s_0^T \Sigma^{-1} \Sigma \Sigma^{-1} s_0}{N_\Phi^2} = \frac{1}{N_\Phi} = \Big(\sum_{k=0}^{d-1} \frac{|S_k|^2}{J_k} \Big)^{-1}, \tag{50}$$

which is Eq. (14) of the main text. The same result follows bin by bin from $\operatorname{Var}(\hat{A}) = \sum_k |\Phi_k^{(0)}|^2 J_k$ provided the sum is the two-sided one: the conjugate pairs (N_k, N_{d-k}) are not independent, but their contributions to the real linear functional $\hat{A}$ add to exactly the two-sided formula.

Fisher information and the CRB. The log-likelihood of the real trace is $\log \mathcal{L} = -\frac{1}{2}(x - As_0)^T \Sigma^{-1}(x - As_0) + \text{const}$, so

$$\mathcal{I}(A) = -\mathbb{E}\left[\frac{\partial^2 \log \mathcal{L}}{\partial A^2}\right] = s_0^T \Sigma^{-1} s_0 = \sum_{k=0}^{d-1} \frac{|S_k|^2}{J_k} = N_\Phi, \tag{51}$$

and $\mathrm{Var}(\hat{A}) = 1/\mathcal{I}(A)$: the CRB is attained in the same model in which the variance was computed, so the OF estimator is efficient and hence the minimum-variance unbiased estimator of the real-amplitude Gaussian model.

The proper-complex convention. If instead the observation is a vector of m independent proper (circular) complex bins, $X_k = AS_k + N_k$ with $\mathbb{E}|N_k|^2 = J_k$ and $\mathbb{E}[N_k^2] = 0$, and A is *real*, then the ML estimator is $\hat{A} = \mathrm{Re}\big(\sum_k S_k^* X_k / J_k\big) / \sum_k |S_k|^2/J_k$, with $\mathrm{Var}(\hat{A}) = 1/(2\sum_k |S_k|^2/J_k)$ and $\mathcal{I}(A) = 2\sum_k |S_k|^2/J_k$, again attaining the bound. If A is *complex*, the ML estimator is $\hat{A} = \sum_k S_k^* X_k/J_k \big/ \sum_k |S_k|^2/J_k$ with $\mathbb{E}|\hat{A} - A|^2 = 1/\sum_k |S_k|^2/J_k$, and the Fisher matrix for $(\mathrm{Re}\, A, \mathrm{Im}\, A)$ is $2\sum_k |S_k|^2/J_k \cdot I_2$. These are the statements one obtains by treating the positive-frequency bins of a real trace as independent, and they differ from (50)–(51) by exactly the factor of two that the conjugate-symmetric pairs contribute; the real convention above is the one that describes a real time trace.

Energy resolution. With amplitude calibrated to energy, $A = E/E_0$ for a known calibration energy E_0 (e.g. a K-α X-ray), the baseline energy resolution is

$$\sigma_E = E_0 \cdot \sigma_A = \frac{E_0}{\sqrt{N_\Phi}} = E_0 \left(\sum_k \frac{|S_k|^2}{J_k}\right)^{-1/2}, \tag{52}$$

which is Eq. (15). Resolution improves when signal power is concentrated at frequencies where the noise PSD is small; N_Φ is both the filter normalization and the total signal-to-noise budget.

Bias under template mismatch. When the assumed template s_0 differs from the true signal s,

$$\mathbb{E}[\hat{A}_{\mathrm{OF}}] = A \cdot \frac{s_0^\dagger \Sigma^{-1} s}{s_0^\dagger \Sigma^{-1} s_0} = A \cdot \frac{\|\tilde{s}\|}{\|\tilde{s}_0\|}\, \rho, \qquad \rho = \frac{\tilde{s}_0^\dagger \tilde{s}}{\|\tilde{s}_0\|\, \|\tilde{s}\|}, \tag{53}$$

with $|\rho| = \cos\theta_w$. Two effects are mixed in this factor and should be kept apart. A *normalization* mismatch, $s = \gamma s_0$ with $\gamma > 0$ real, gives $\rho = 1$ and $\mathbb{E}[\hat{A}] = \gamma A$: the amplitude calibration changes, with no loss of SNR and no bias toward zero in particular (for a signed or complex γ, $\rho = \gamma/|\gamma|$ carries the sign or phase, and ρ is a signed correlation, not the cosine of a principal angle). A *direction* mismatch, $\|\tilde{s}\| = \|\tilde{s}_0\|$ and $|\rho| < 1$, scales

the expectation by the signed (real case) or complex ρ: the attenuation magnitude is $|\rho| = \cos\theta_w$, and a negative real ρ reverses the sign of the response. In general the factor can be larger or smaller than one, so "biased toward zero" is not the right description. The variance formula (50) is then no longer the CRB for the true signal; the effective SNR retention is

$$\frac{|s_0^\dagger \Sigma^{-1} s|^2}{(s_0^\dagger \Sigma^{-1} s_0)(s^\dagger \Sigma^{-1} s)} = \cos^2\theta_w, \tag{54}$$

the squared cosine of the whitened angle between s_0 and s (modulus squared in the complex case). In whitened coordinates, with $u = \tilde{s}_0/\|\tilde{s}_0\|$ the unit template direction and $\hat{R}_{\tilde{x}} = \Sigma^{-1/2}\hat{R}_x\Sigma^{-1/2}$, the OF projector retains the fraction $u^\dagger \hat{R}_{\tilde{x}} u / \operatorname{tr}\hat{R}_{\tilde{x}}$ of the *observed* whitened power (signal plus noise), a quantity invariant under rescaling of the template; the corresponding signal-only fraction is $u^\dagger R_t u / \operatorname{tr} R_t$ with R_t the whitened signal second moment of Proposition 7.4, and neither is a signal-to-noise ratio. When the true family is not one-dimensional, the remaining signal power is distributed across additional directions that rank-k EMPCA can retain.

Reference implementation. For completeness, the single-event OF amplitude estimator in the two-sided convention of (48) is:

1. Compute *unitary* DFTs $X_k \leftarrow d^{-1/2}\,\mathrm{FFT}(x)$, $S_k \leftarrow d^{-1/2}\,\mathrm{FFT}(s_0)$, $k = 0, \ldots, d-1$, so that $J_k = \mathbb{E}|N_k|^2$ is the bin power of the same unitary transform (with an unnormalized FFT, J_k must be replaced by dJ_k).

2. Numerator $\mathcal{N} \leftarrow \mathrm{Re}\big(\sum_{k=0}^{d-1} S_k^* X_k / J_k\big)$ (the real part removes rounding residue only; the two-sided sum is real).

3. Denominator $\mathcal{D} \leftarrow \sum_{k=0}^{d-1} |S_k|^2/J_k = N_\Phi$.

4. $\hat{A} \leftarrow \mathcal{N}/\mathcal{D}$; residual $\chi^2 \leftarrow \sum_{k=0}^{d-1} |X_k - \hat{A}S_k|^2/J_k$; predicted variance $1/N_\Phi$.

All operations are $O(d)$ after the one-time $O(d \log d)$ FFT. The amplitude ratio $\mathcal{N}/\mathcal{D}$ is invariant under a common rescaling of J_k and therefore cannot reveal a normalization mismatch; χ^2, N_Φ and the predicted variance are not, and they must equal their time-domain values $(x - \hat{A}s_0)^T\Sigma^{-1}(x - \hat{A}s_0)$, $s_0^T\Sigma^{-1}s_0$ and $1/(s_0^T\Sigma^{-1}s_0)$ under one stated convention (the regression check `scripts/check_revision_claims.py` compares all four quantities between the time-domain, two-sided and endpoint-weighted rFFT forms). An implementation on the rFFT grid replaces each sum by its endpoint-weighted one-sided form, and an implementation that stores J_k as a physical one-sided PSD density carries the conversion factor of Appendix G in numerator, denominator and residual alike. The normalization actually implemented in the experiment repository is checked against the

bound (50) by the controlled test S1 of Section 9. For time-shift OF, step 1 is augmented by a grid search over candidate shifts with $S_k \to S_k e^{-2\pi i k \tau_j / N}$, selecting the shift that minimizes χ^2 (Appendix B.1).

B Time-Shift and Joint-Channel Optimal Filtering

This appendix gives the full derivations for the two OF extensions summarized in Section 3.3, and proves the degeneracy propositions referenced there.

B.1 Time-shift optimal filtering

Include an unknown arrival-time shift t_0 via

$$S_k(t_0) = e^{-i\omega_k t_0} S_k, \qquad X_k = A\, S_k(t_0) + N_k, \tag{55}$$

with weighted objective

$$\chi^2(A, t_0) = \sum_k \frac{|X_k - A e^{-i\omega_k t_0} S_k|^2}{J_k}. \tag{56}$$

For fixed t_0, the stationarity condition $\partial\chi^2/\partial A^* = 0$ yields

$$\hat{A}(t_0) = \frac{\sum_k e^{i\omega_k t_0} S_k^* X_k / J_k}{\sum_k |S_k|^2 / J_k} = \sum_k e^{i\omega_k t_0} \Phi_k^{(0)} X_k, \tag{57}$$

so $\hat{A}(t_0)$ is the inverse DFT of the filtered spectrum [5]. To profile out the amplitude, define

$$\chi_0^2 = \sum_k \frac{|X_k|^2}{J_k}, \qquad B(t_0) = \sum_k \frac{e^{i\omega_k t_0} S_k^* X_k}{J_k}. \tag{58}$$

Then

$$\chi^2(A, t_0) = \chi_0^2 - A^* B(t_0) - A B^*(t_0) + |A|^2 N_\Phi, \tag{59}$$

and at $A = \hat{A}(t_0) = B(t_0)/N_\Phi$,

$$\chi^2(t_0) = \chi_0^2 - \frac{|B(t_0)|^2}{N_\Phi} = \chi_0^2 - |\hat{A}(t_0)|^2 N_\Phi. \tag{60}$$

Hence the optimal delay is

$$\hat{t}_0 = \arg\min_{t_0} \chi^2(t_0) = \arg\max_{t_0} |\hat{A}(t_0)|, \tag{61}$$

the usual "maximize filtered amplitude in time" criterion for real-valued traces with Hermitian spectra.

Relation to learned subspaces. The shifted templates $\mathcal{M} = \{e^{-i\omega t_0} s_0 : t_0 \in \mathbb{R}\}$ form a one-parameter signal manifold, and time-shift OF performs ML estimation over this manifold. For small shifts, the Taylor expansion

$$e^{-i\omega t_0} s_0 = s_0 - i\omega t_0\, s_0 - \tfrac{1}{2}(\omega t_0)^2 s_0 + \cdots \tag{62}$$

shows that the shift family is approximately contained in $\mathrm{span}\{s_0, \omega s_0, \omega^2 s_0, \dots\}$ — in time-domain notation, $\mathrm{span}\{s_0, \partial_t s_0, \partial_t^2 s_0, \dots\}$.

Remark B.1 (Shifted OF and learned subspaces)**.** *OF with time shift is the ML estimator obtained by restricting the signal model to the nonlinear one-parameter manifold $\mathcal{M}$. Its relation to a learned linear subspace is one of approximation, not degeneracy: the linear span of $\mathcal{M}$ is generically all of $\mathbb{C}^d$, so there is no rank at which a subspace* equals *the shift family, and a rank-k EMPCA subspace can only contain the truncated Taylor family* (62) *to some order. When the dominant variation in the data is timing jitter that is small on the scale of the template's bandwidth, the leading learned directions are expected to approximate* $\mathrm{span}\{s_0, \partial_t s_0, \dots\}$ *and rank-k EMPCA then acts as a linear-subspace approximation to shifted OF; this is a heuristic expectation about the data, and no precise hypothesis under which it holds is claimed here.*

Basic OF, shifted OF, and rank-k EMPCA are thus related in the informal sense that each enlarges the admissible family of the previous one, with the last relation an approximation rather than a containment. The former numerical illustration has been omitted from this revision because its seed-level provenance was not recoverable from the archived evidence; the analytical statement above does not depend on it.

B.2 Joint-channel correlated optimal filtering

Joint observation model. For C channels with d samples per channel, stack the channel traces:

$$x = \begin{bmatrix} x^{(1)} \\ \vdots \\ x^{(C)} \end{bmatrix} \in \mathbb{C}^{Cd}, \qquad s = \begin{bmatrix} s^{(1)} \\ \vdots \\ s^{(C)} \end{bmatrix} \in \mathbb{C}^{Cd}, \tag{63}$$

with noise $n \sim \mathcal{N}_{\mathbb{C}}(0, \Sigma_{\text{joint}})$ and block-structured covariance

$$\Sigma_{\text{joint}} = \begin{bmatrix} \Sigma^{(11)} & \Sigma^{(12)} & \dots & \Sigma^{(1C)} \\ \Sigma^{(21)} & \Sigma^{(22)} & \dots & \Sigma^{(2C)} \\ \vdots & & \ddots & \vdots \\ \Sigma^{(C1)} & \dots & \dots & \Sigma^{(CC)} \end{bmatrix}, \tag{64}$$

where $\Sigma^{(cc)}$ is the intra-channel covariance and $\Sigma^{(cc')}$ the cross-channel correlation; for stationary noise, $\Sigma^{(cc)}$ is circulant and $\Sigma^{(cc')}$ is the cross-spectral density matrix.

Estimator and degeneracy. The Gaussian ML objective is unchanged in form, $\chi^2(\hat{s}) = (x - \hat{s})^\dagger \Sigma_{\text{joint}}^{-1}(x - \hat{s})$. The rank-1 constraint $\hat{s} = As$ gives the joint-channel correlated optimal filter

$$\hat{A}_{\text{JOF}} = \frac{s^\dagger \Sigma_{\text{joint}}^{-1} x}{s^\dagger \Sigma_{\text{joint}}^{-1} s}, \tag{65}$$

which is algebraically the rank-1 weighted subspace estimator (7) with $U = s$.

Proposition B.1 (Degeneracy to joint-channel correlated OF)**.** *Weighted EMPCA with the full inverse covariance $\Sigma_{\text{joint}}^{-1}$ is the multi-channel subspace generalization of joint-channel correlated optimal filtering. In the rank-1 case, when the EMPCA basis spans the joint template direction, weighted EMPCA reduces exactly to joint-channel correlated OF.*

EMPCA does not change the underlying likelihood; it replaces a fixed joint template by a learned joint signal subspace, with noise correlations handled identically through the quadratic form.

Whitening and decoupling. The joint whitening map is any W with $W\Sigma_{\text{joint}}W^\dagger = I$, e.g. the Hermitian $\Sigma_{\text{joint}}^{-1/2}$ from an eigendecomposition or L^{-1} from a Cholesky factor (Appendix C.8). For stationary noise with per-frequency cross-spectral matrices $\Sigma_f \in \mathbb{C}^{C\times C}$, whitening decomposes into independent $C \times C$ factors per frequency bin, reducing to C scalar divisions when channels are uncorrelated. If $\Sigma^{(cc')} = 0$ for $c \neq c'$, the joint objective is the sum of the C per-channel quadratic forms. What this implies depends on the signal model. For joint-channel OF with a fixed joint template the amplitude is shared, and the estimator is the Fisher-information-weighted mean of the per-channel OF estimates. For a learned rank-k subspace of $\mathbb{C}^{Cd}$ the objective does *not* decouple into independent per-channel subspace problems: the optimal joint subspace is the top-k eigenspace of the whitened joint second moment, whose cross-channel blocks are set by the signal, and a latent that excites two channels couples them however independent the noise is (with $\Sigma_{\text{joint}} = I$ and a signal second moment $\left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ across two single-sample channels, the rank-one joint direction is $(1, 1)/\sqrt{2}$ and uses both). Independent single-channel EMPCA is the further restriction of the class to block-diagonal bases with per-channel coefficients, which is an assumption about the signal, not a consequence of independent noise.

C EMPCA Updates, Convergence, and Conditioning

This appendix collects the algorithmic content supporting Section 4: the EM updates and their derivation, the frequency-domain variant, the coordinate-descent identity with the AE objective, the convergence proof, and the practical machinery (initialization, model-order selection, whitening numerics, and conditioning).

C.1 Classical PCA and its noise assumption

For a mean-subtracted data matrix, classical PCA seeks a rank-k model $X \approx \Gamma^\dagger U^\dagger$ (rows of X are events) minimizing the Euclidean reconstruction error $\|X - \Gamma^\dagger U^\dagger\|_F^2$ subject to $U^\dagger U = I_k$; the solution is the span of the top-k eigenvectors of the sample covariance. PCA is the ML subspace estimator only under isotropic noise $\Sigma = \sigma^2 I$; any colored or heteroskedastic noise breaks this optimality. EMPCA [8] restores ML optimality by carrying the weight Σ^{-1} (or per-observation weights) through the estimation.

C.2 EM updates

Write the joint objective in the variables (U, Γ), with $\Gamma = [c_1, \ldots, c_N]$ the coefficient matrix,

$$\mathcal{L}(U, \Gamma) = \frac{1}{N} \sum_{i=1}^{N} (x_i - U c_i)^\dagger \Sigma^{-1} (x_i - U c_i), \tag{66}$$

which is also the tied-AE loss of Section 5 with the encoder constraint imposed at convergence.

E-step (coefficients). For fixed U, $\mathcal{L}$ is strictly convex and quadratic in each c_i; the unique minimizer solves the weighted normal equations

$$(U^\dagger \Sigma^{-1} U)\, c_i = U^\dagger \Sigma^{-1} x_i, \qquad \hat{c}_i = (U^\dagger \Sigma^{-1} U)^{-1} U^\dagger \Sigma^{-1} x_i, \tag{67}$$

which reduces to $\hat{c}_i = U^\dagger \Sigma^{-1} x_i$ under the Σ^{-1}-orthonormality constraint — exactly the tied-AE encoder.

M-step (basis). For fixed Γ, with aggregates $G = \Gamma\Gamma^\dagger/N$ and $H = \Gamma X/N$ (so that $H^\dagger = N^{-1} \sum_i x_i c_i^\dagger$), the unconstrained minimizer (unique when $G \succ 0$) is

$$U_{\text{unc}} = H^\dagger G^{-1} = \Big(\frac{1}{N} \sum_i x_i c_i^\dagger\Big) \Big(\frac{1}{N} \sum_i c_i c_i^\dagger\Big)^{-1}, \tag{68}$$

followed by re-orthonormalization in the Σ^{-1} inner product *with the compensating change of coefficients*: with the thin QR factorization $\Sigma^{-1/2} U_{\text{unc}} = QR$, set

$$U_{\text{new}} = \Sigma^{1/2} Q, \qquad \Gamma_{\text{new}} = R\,\Gamma, \qquad \text{so that} \qquad U_{\text{new}} \Gamma_{\text{new}} = U_{\text{unc}} \Gamma. \tag{69}$$

The normalization is then a change of gauge that leaves the product, and hence the objective, unchanged; the next E-step recomputes the coefficients from U_{new} in any case. Normalizing U with Γ held fixed is *not* a descent step: for the scalar example $x = 2$, $c = 1$, $U_{\text{unc}} = 2$ the loss is zero before normalization and one after replacing U by 1 with c

fixed. The weight Σ^{-1} does not appear in (68): the stationarity condition of the weighted objective in U is $\Sigma^{-1}\sum_i (x_i - Uc_i)c_i^\dagger = 0$, and Σ^{-1} is invertible (an update that carries an extra Σ^{-1} in H is monotone but converges to a different subspace). Substituting the E-step coefficients $c_i = U^\dagger \Sigma^{-1} x_i$ gives $U_{\mathrm{unc}} = \hat{R}_x \Sigma^{-1} U\, G^{-1}$, so in whitened coordinates $W = \Sigma^{-1/2} U$ the update is $\mathrm{span}(W_{\mathrm{new}}) = \mathrm{span}(\hat{R}_{\tilde{x}} W)$ with $\hat{R}_{\tilde{x}} = \Sigma^{-1/2} \hat{R}_x \Sigma^{-1/2}$: EMPCA with exact block steps is orthogonal (subspace) iteration on the whitened second moment [40], which is what its convergence theory below rests on.

Coordinate-descent view. Each step is the exact block minimizer of (66) over its block, the M-step being taken over unconstrained U and followed by the gauge change (69), so

$$\mathcal{L}(U^{(r+1)}, \Gamma^{(r+1)}) \le \mathcal{L}(U^{(r)}, \Gamma^{(r+1)}) \le \mathcal{L}(U^{(r)}, \Gamma^{(r)}), \tag{70}$$

where $(U^{(r+1)}, \Gamma^{(r+1)})$ denotes the normalized pair $(U_{\mathrm{new}}, \Gamma_{\mathrm{new}})$ of (69), whose objective equals that of $(U_{\mathrm{unc}}, \Gamma^{(r+1)})$; i.e. the EMPCA E- and M-steps are block-coordinate descent on the noise-aware AE loss. The objective is convex in each block separately, but the feasible set of the constrained basis (an orthonormality constraint) is not convex, so the problem is not biconvex in the usual sense; the descent argument uses only the exact block minimizations. This is the algorithmic face of the Bridge Theorem. Equivalently one may drop the coefficients after the M-step and describe the algorithm as the *profiled* subspace iteration $W \mapsto \mathrm{orth}(\hat{R}_{\tilde{x}} W)$ on the whitened second moment, which is what the convergence theory below analyses.

Three algorithms, one optimum. Three procedures compute or approximate the minimizer of (66), and they are not interchangeable. (a) A direct eigendecomposition of $\hat{R}_{\tilde{x}}$, or a thin SVD of the whitened data matrix, returns the top-k eigenspace in one step; this is how the archived weighted-PCA fits of Section 9 are computed, and it needs no convergence theory. (b) The common-covariance EMPCA iteration above with exact block steps is orthogonal iteration on $\hat{R}_{\tilde{x}}$, and the convergence statements of this appendix are statements about it. (c) Heterogeneous-weight EMPCA [8], with per-event or per-element weights Σ_i^{-1} or ω_{kj} for which no single whitening exists, is a genuinely alternating scheme: it retains the monotone descent (70) but is not a subspace iteration on a fixed matrix, and the identification with (a)–(b) does not apply to it.

Deflation variant. Equivalent to Bailey's one-component deflation, components may be solved sequentially, updating $X \leftarrow X - u_m c_m$ before solving the next component.

Reference iteration. Rank-k EMPCA with weighting Σ^{-1}:

1. Initialize $U^{(0)}$ (random orthonormal, SVD warm-start, or template-seeded; see below).

2. E-step: solve (67) for all events ($O(k^2 dN)$).

3. M-step: form (68); re-orthonormalize via QR of $\Sigma^{-1/2} U_{\text{unc}}$ and transport the coefficients, $\Gamma \leftarrow R\Gamma$, as in (69) ($O(kdN) + O(k^2 d)$); or discard Γ and let the next E-step recompute it (profiled iteration).

4. Evaluate $\mathcal{L}^{(r+1)}$; stop when the relative decrease is below tolerance.

For $k = 1$ the iteration is the power method on $\hat{R}_{\tilde{x}}$. From a template seed it starts near the fixed point whenever the leading sample eigenvector is close to the whitened template, but the seed is not preserved: one step already rotates it toward the sample eigenvector (for $\hat{R}_{\tilde{x}} = \text{diag}(2, 1)$ the seed $(1, 1)/\sqrt{2}$ moves to $(2, 1)/\sqrt{5}$, by 18.4°), so the number of iterations to a given tolerance depends on the sample spectrum and no finite-sample guarantee of convergence in $O(1)$ iterations follows from the seed.

C.3 Frequency-domain variant and dual interpretation

Yu's microcalorimeter setting [9] motivates a frequency-domain variant robust to random time shifts. In Fourier space, $X_k^{(j)} = A_j e^{-i\omega_k t_{0,j}} S_k + N_k^{(j)}$, and direct PCA is confounded because the effective coefficient depends on the frequency bin. Using phase differences $\Delta\phi_k = \phi_k - \phi_{k-1}$, a time shift contributes $(\omega_k - \omega_{k-1}) t_0$, which is constant for uniform frequency spacing; the transformed coefficients then behave as a single complex factor per event, restoring a linear subspace model.

With per-element weights ω_{kj} (e.g. inverse PSD; the symbol W is reserved for the whitened basis), the weighted EMPCA objective is

$$\chi^2_{\text{EMPCA}} = \sum_k \sum_j \omega_{kj} \left| X_{kj} - [U\Gamma]_{kj} \right|^2 , \tag{71}$$

with weighted E-step normal equations $(U^\dagger \Omega_j U) c_j = U^\dagger \Omega_j X_{:,j}$ per event, where $\Omega_j = \text{diag}_k(\omega_{kj})$, and weighted M-step row solves $(\Gamma \Omega_k \Gamma^\dagger) u_k = \Gamma \Omega_k X^\dagger_{k,:}$ per frequency, where $\Omega_k = \text{diag}_j(\omega_{kj})$. For stationary noise the covariance is diagonal in the Fourier basis, Σ^{-1} acts element-wise, and weighted inner products reduce to Hadamard products: the Hadamard-product implementation is the exact dual PCA formulation under a diagonal noise covariance, not an approximation.

C.4 Amplitude estimation details

Gauge-invariant amplitude estimators. Because the basis is identified only up to $\mathcal{U}(k)$ rotation, physical amplitudes use gauge-invariant functionals: the whitened reconstruction norm $\|\hat{s}\|_{\Sigma^{-1}} = \|c\|_2$ (under $U^\dagger \Sigma^{-1} U = I_k$), calibrated against a reference line; or the projection onto a calibration template, Eq. (21).

Ridge-regularized coefficient solves (fixed basis). Two Gram matrices must be kept apart. The *basis* Gram matrix $U^\dagger \Sigma^{-1} U$ is ill-conditioned when basis vectors are nearly collinear in whitened space; it is I_k for a Σ^{-1}-orthonormal basis whatever the rank of the data. The *coefficient* Gram matrix $\Gamma\Gamma^\dagger/N$ has rank at most $\operatorname{rank} X$, so it is singular whenever k exceeds the data rank, and no coefficient regularization changes that. For a *fixed* basis with a nearly collinear basis Gram matrix, Tikhonov regularization

$$\hat{c}_\mu = (U^\dagger \Sigma^{-1} U + \mu I_k)^{-1} U^\dagger \Sigma^{-1} x \tag{72}$$

controls the bias–variance tradeoff; the ridge parameter μ (distinct from the eigenvalues λ_j) may be chosen by generalized cross-validation [59], though for $N \gg k$ a small numerical floor suffices after proper re-orthonormalization.

C.5 Convergence proof

Theorem C.1 (Monotone convergence of EMPCA)**.** *Consider the common-covariance iteration of Appendix C.2 with exact E- and M-steps and the gauge transport* (69)*, and write $W^{(r)} = \Sigma^{-1/2} U^{(r)}$ for the whitened basis. Assume* $\operatorname{rank} \hat{R}_{\tilde{x}} \geq k$ *and that the initial basis satisfies* $\operatorname{rank}(\hat{R}_{\tilde{x}} W^{(0)}) = k$*, i.e.* $\operatorname{range} W^{(0)} \cap \ker \hat{R}_{\tilde{x}} = \{0\}$*, which holds for almost every $W^{(0)}$ (Haar measure on the Stiefel manifold). Then the coefficient Gram matrix $G^{(r)} = \Gamma^{(r)} \Gamma^{(r)\dagger}/N$ is positive definite at every iteration, so the M-step* (68) *is defined and unique throughout, and: (i) $\mathcal{L}^{(r+1)} \leq \mathcal{L}^{(r)}$ for all r; (ii) the sequence $\{(U^{(r)}, \Gamma^{(r)})\}$ has at least one limit point, and G is positive definite at every limit point; (iii) every limit point is a fixed point of the alternating-minimization map, and hence satisfies the first-order stationarity conditions of* (19) *in the projector variable $UU^\dagger \Sigma^{-1}$. When* $\operatorname{rank} \hat{R}_{\tilde{x}} < k$ *the coefficient Gram matrix is singular from the first iteration, the M-step is undefined, and the iteration as written must stop with a recorded rank failure or be restarted with $k \leq \operatorname{rank} \hat{R}_{\tilde{x}}$; a ridge on the coefficient solve does not remove the singularity (Appendix C.4), and no regularized alternating algorithm is defined or analysed here.*

Proof. With the E-step coefficients $\Gamma = W^\dagger \tilde{X}^\dagger$ the Gram matrix is $G = W^\dagger \hat{R}_{\tilde{x}} W$, which is positive definite exactly when $\operatorname{range} W \cap \ker \hat{R}_{\tilde{x}} = \{0\}$; this is the assumption at $r = 0$. For $r \geq 1$, $\operatorname{range} W^{(r)} = \hat{R}_{\tilde{x}} \operatorname{range} W^{(r-1)} \subseteq \operatorname{range} \hat{R}_{\tilde{x}}$ (Appendix C.2), on which $\hat{R}_{\tilde{x}}$ is

bounded below by its smallest nonzero eigenvalue $\lambda^+_{\min}$, so $G^{(r)} \succeq \lambda^+_{\min} I_k$ for all $r \geq 1$; the set of bases with range inside range $\hat{R}_{\tilde{x}}$ is closed, so the same bound holds at every limit point. This is the Gram condition used below, and it is not an extra hypothesis. The E-step solution (67) is the unique global minimizer of a strictly convex quadratic, so $\mathcal{L}(U^{(r)}, \Gamma^{(r+1)}) \leq \mathcal{L}(U^{(r)}, \Gamma^{(r)})$. The M-step minimizes the quadratic over unconstrained U, and the re-orthonormalization with coefficient transport (69) leaves the product $U\Gamma$, hence the objective, unchanged. Chaining gives monotone decrease (70); since $\mathcal{L} \geq 0$, the sequence converges. The iterates live in a compact set (orthonormality constraint plus bounded coefficients), so a limit point exists. For (iii): on the set where $G \succ 0$ the M-step (68) is a continuous function of Γ, the thin QR re-orthonormalization is continuous up to the $\mathcal{U}(k)$ gauge, and the E-step is continuous in U, so the update map $F : (U, \Gamma) \mapsto (U^{(r+1)}, \Gamma^{(r+1)})$ is continuous there up to gauge. Along a subsequence converging to a limit point (U^*, Γ^*) with $G^* \succ 0$, $\mathcal{L}(F(U^*, \Gamma^*)) = \lim \mathcal{L}^{(r+1)} = \mathcal{L}(U^*, \Gamma^*)$; because the E-step minimizer is unique, equality in the E-step forces Γ^* to be the E-step of U^*, and because the M-step minimizer is unique when $G^* \succ 0$, equality in the M-step forces U^* to be the M-step of Γ^* up to gauge. Thus $F(U^*, \Gamma^*) = (U^*, \Gamma^*)$ up to gauge, and a fixed point of both exact block minimizations satisfies the first-order conditions of (19). The initial condition is a genericity condition on $W^{(0)}$: it fails on the null set of initial subspaces that meet $\ker \hat{R}_{\tilde{x}}$, and it cannot hold at all when k exceeds the data rank. □

Remark C.1 (Stationary points and almost-everywhere convergence)**.** *The objective is non-convex jointly in* (U, Γ) *for* $k \geq 2$*, and its stationary points are* not *all equivalent. In the projector variable the problem is the Rayleigh–Ritz maximization of* $\mathrm{Tr}(W^\dagger \hat{R}_{\tilde{x}} W)$ *over* $W^\dagger W = I_k$ *(Appendix D.3), whose stationary points are all* k*-dimensional invariant subspaces of* $\hat{R}_{\tilde{x}}$ *— any choice of* k *eigenvectors when the eigenvalues are distinct — and only the top-*k *one is the global minimizer of* $\mathcal{L}$*. Rotating a retained eigenvector* v_i *toward an unretained* v_j*,* $w(\theta) = \cos\theta\, v_i + \sin\theta\, v_j$*, changes the loss by* $\mathcal{L}(\theta) - \mathcal{L}(0) = (\lambda_i - \lambda_j)\sin^2\theta$*. Hence, for distinct eigenvalues and* $0 < k < d$*, an invariant subspace is a local minimum of* $\mathcal{L}$ *if and only if every retained eigenvalue exceeds every unretained one (the top-*k *subspace), a local* maximum *if and only if every retained eigenvalue is below every unretained one (the bottom-*k *subspace; for* $k = 1$ *and* $\hat{R}_{\tilde{x}} = \mathrm{diag}(2, 1)$ *the loss along* $w = (\sin\theta, \cos\theta)$ *is* $2 - \sin^2\theta$*, maximal at the bottom eigenvector), and a saddle otherwise [43]. With repeated eigenvalues the critical sets are manifolds and the classification is by the same inequalities applied to the multiset;* $k = 0$ *and* $k = d$ *are trivial. This classification does not by itself say where the alternating algorithm goes. What does is the identification of the exact-block-step iteration with orthogonal iteration on* $\hat{R}_{\tilde{x}}$ *(Appendix C.2): when* $\lambda_k > \lambda_{k+1}$*, the iterates converge to the top-*k *invariant subspace for every initial* W *whose projection onto that subspace is nonsingular — almost every* W *— at the linear rate* λ_{k+1}/λ_k *[40, Thm. 8.2.2], and the coefficient-Gram condition*

of Theorem C.1 holds along the way. The bottom-k and mixed invariant subspaces are reached only from the null set of initial subspaces orthogonal to some top eigenvector. Multiple random restarts (Appendix C.6) remain the check used in this paper, and the guarantee does not *transfer to the alternating scheme of the separable class (Section 6.3), which is not a subspace iteration on a fixed matrix.*

Remark C.2 (Rate)**.** *When $\lambda_{k+1} < \lambda_k$, the iteration converges at the linear rate λ_{k+1}/λ_k of orthogonal iteration [40], consistent with the local rate that block-coordinate-descent theory gives for the quadratic objective [39].*

C.6 Initialization

Random: draw $U^{(0)}$ Haar-uniformly on the Stiefel manifold; simple, effective when $N \gg kd$; multiple restarts check convergence to the same subspace (Remark C.1). *SVD warm-start:* top-k right singular vectors of the unweighted data matrix; a good warm start when noise is not far from isotropic. *Template-seeded:* set $u_1 \propto \tilde{s}/\|\tilde{s}\|$ from a calibration template, extending to rank k with random whitened-orthogonal directions. This starts the iteration inside the containment condition of Section 4.3 but does not preserve it: the iteration moves the seed toward the sample eigenvectors (Appendix C.2), so containment after learning holds only when it is imposed as a constraint — fixing u_1 and learning the remaining $k-1$ directions in its whitened orthogonal complement — and is otherwise an empirical property of the fit.

C.7 Model-order selection

The rank is a modelling choice with a stated target, and the target has to be one that a held-out residual can estimate. The held-out self-reconstruction residual is *not* such a criterion: for nested fitted subspaces the whitened residual of every trace, held out or not, is nonincreasing in k and reaches zero at $k = d$, so "the k at which the out-of-sample χ^2 stops decreasing" rewards retaining noisy dimensions and is not a denoising-rank rule. Three criteria with stated assumptions are used instead.

Corrected held-out risk. For a projector fitted on training data and evaluated on held-out traces with a correct Σ, Proposition 7.1 gives the whitened clean-signal risk as $\mathcal{R}_{\text{clean},\Sigma}(k) = \mathcal{R}_{\text{obs}}(k) + 2k - d$, so the rank that minimizes the population clean-signal risk within the projector class is estimated by

$$\hat{k} = \arg\min_k \big[\hat{\mathcal{R}}_{\text{obs}}(k) + 2k\big], \tag{73}$$

the held-out mean whitened residual per trace plus the degrees-of-freedom penalty $2k$. Its population interpretation is transparent in the ideal nested case: if the fitted subspaces

are the top-k eigenspaces of the whitened signal second moment R_t with eigenvalues $\mu_1 \geq \mu_2 \geq \cdots$, then $\mathcal{R}_{\text{clean},\Sigma}(k) = \sum_{j>k} \mu_j + k$ and the increment from k to $k+1$ is $1 - \mu_{k+1}$, so the population optimum retains exactly the signal eigenvalues above one (ties at $\mu_{k+1} = 1$ are indifferent). Pure noise ($\mu \equiv 0$) has population optimum $k = 0$, which a finite validation sample estimates but does not guarantee; and "total signal energy above d" is only the condition for the identity map to beat the zero map, not a characterization of the optimal rank (with $\mu = (100, 0)$ and $d = 2$ the risks for $k = 0, 1, 2$ are $100, 1, 2$). The rule is a risk *estimate*: it is valid for fixed projectors evaluated on data they were not fitted to, requires the whitened noise second moment to be I (a wrong Σ biases the penalty), and compares projectors only; the zero and mean-only reconstructions ($k = 0$ with and without a fitted mean) are its baselines, and a shrinkage estimator uses the trace correction of Proposition 7.1. Rank selection and covariance accuracy are therefore coupled (Remark 5.2).

Noise-aware spectral gap. In whitened coordinates the noise floor has expected eigenvalue 1, and for N traces in d dimensions the sample noise eigenvalues spread up to the Marchenko–Pastur edge $(1 + \sqrt{d/N})^2$ (Section 4.3); a rank chosen as the number of whitened sample eigenvalues above that edge is a criterion with an explicit assumption (isotropic whitened noise, one-spike asymptotics used as a scale), which should be stated when it is used.

Probabilistic marginal likelihood. The model with event-specific fitted coefficients has a parameter count that grows with N, so the standard AIC/BIC penalties do not apply to it without a marginal likelihood that integrates the coefficients out. If that route is taken, the subspace itself contributes $2k(d-k)$ real parameters (the dimension of the complex Grassmannian $\mathrm{Gr}(k, d)$, or $k(d-k)$ for real data) — not $2kd - k^2$, which is the dimension of the Stiefel manifold and still contains the k^2 gauge parameters — and the Laplace-approximated evidence of probabilistic PCA [60] is the reference criterion in isotropic whitened noise.

A physical upper bound on k (the number of independent pulse-shape variation modes) usefully narrows the search. Whichever criterion is used, the selection is made on development data and the selected rank is then evaluated once on a separate untouched split; the archived experiments of Section 9 fix the rank in the frozen configuration rather than selecting it.

C.8 Whitening numerics and conditioning

Computing $\Sigma^{-1/2}$. For circulant Σ (stationary noise), whitening is element-wise division by $\sqrt{J_k}$ in the frequency domain. Otherwise: eigendecomposition $\Sigma = V \Lambda V^\dagger$ gives $\Sigma^{-1/2} = V \Lambda^{-1/2} V^\dagger$ (numerically stable, $O(d^3)$); for column observations and the Cholesky

factorization $\Sigma = LL^\dagger$ the whitening map is $W = L^{-1}$, which satisfies $W\Sigma W^\dagger = I$ (the transposed choice $L^{-\dagger}$ does not: for $\Sigma = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ it gives $L^{-\dagger}\Sigma L^{-1} = \left(\begin{smallmatrix} 1 & -1 \\ -1 & 2 \end{smallmatrix}\right)$). W is triangular, not Hermitian, and differs from $\Sigma^{-1/2}$ by a unitary factor, $W = Q\Sigma^{-1/2}$ with $Q = L^{-1}\Sigma^{1/2}$; every Σ^{-1} inner product, projector, residual and principal angle is the same under either map, so the choice is numerical (Cholesky is faster and preferred for well-conditioned dense Σ), but a nonsymmetric whitening map should be written W rather than $\Sigma^{-1/2}$; the archived fits use the eigendecomposition route.

Ill-conditioned noise PSDs. Bins with $J_k \approx 0$ amplify noise after whitening. Remedies: (i) an additive floor $J_k \to \max(J_k, \delta \max_k J_k)$ with $\delta \in [10^{-4}, 10^{-2}]$; (ii) frequency masking to a band $[f_{\min}, f_{\max}]$ where the signal has appreciable power — lossless when the signal has negligible power outside the band, and standard in cryogenic detector OF; (iii) Tikhonov regularization $\Sigma^{-1} \to (\Sigma + \varepsilon I)^{-1}$ when the noise floor itself is uncertain.

Rank deficiency and ill-conditioned bases. If the coefficient Gram matrix $\Gamma\Gamma^\dagger/N$ is singular (data rank below k, or an initialization meeting $\ker \hat{R}_{\tilde{x}}$), the iteration of Appendix C.2 stops with a recorded rank failure and is restarted with a smaller k; this is the policy assumed by Theorem C.1. A large condition number of the basis Gram matrix $U^\dagger\Sigma^{-1}U$ between re-orthonormalizations is a separate numerical symptom of near-collinear basis vectors, cured by re-orthonormalizing every M-step (standard); the coefficient ridge (72) is a fixed-basis shrinkage device and is not part of the alternating algorithm.

Computational notes. Sliding-window inference uses fixed eigenvectors with fast coefficient solves, producing amplitude, χ^2, and likelihood traces efficiently. The E-step at $O(k^2 dN)$ dominates at large k but is negligible relative to FFT cost at the small ranks ($k \le 6$) used in this paper; batched GPU linear algebra (e.g. batched GEMM and fused weighting/projection kernels) accelerates batch workloads.

D Proofs of the Theorem Statements

D.1 Extended proof of Theorem 4.1

We restate and prove the rank-1 OF–EMPCA equivalence in full detail. We work in whitened coordinates $\tilde{x}_i = \Sigma^{-1/2}x_i$, $\tilde{s} = \Sigma^{-1/2}s$, $\tilde{n}_i = \Sigma^{-1/2}n_i$, so that

$$\tilde{x}_i = a_i\tilde{s} + \tilde{n}_i, \qquad \tilde{n}_i \sim \mathcal{N}(0, I). \tag{74}$$

Step 1: population second moment is a spiked identity. By assumption (ii) of the theorem, $\mathbb{E}[\tilde{n}_i\tilde{n}_i^\dagger] = I$ and the signal–noise cross moment vanishes, $\mathbb{E}[a_i\tilde{n}_i^\dagger] = 0$; by assumption (i), $s \neq 0$ and $\nu := \mathbb{E}|a_i|^2 > 0$. Then

$$R := \mathbb{E}[\tilde{x}_i\tilde{x}_i^\dagger] = \nu\,\tilde{s}\tilde{s}^\dagger + I. \tag{75}$$

The eigenvalues of R are $\lambda_1 = 1+\nu\|\tilde{s}\|^2$ (simple, eigenvector $\tilde{s}/\|\tilde{s}\|$) and $\lambda_j = 1$ for $j \geq 2$. All three assumptions are used: without $s \neq 0$ or $\nu > 0$ there is no spike, and a nonzero cross moment adds a term $\mathbb{E}[a_i\tilde{n}_i^\dagger]\tilde{s}^\dagger$ plus its adjoint that moves the leading eigenvector off $\tilde{s}$.

Step 2: EMPCA recovers the leading eigenvector. Rank-1 EMPCA maximizes the explained population second moment (the data are not centered; R is the uncentered second moment), i.e. finds the leading eigenvector of R:

$$u_1 = \frac{\tilde{s}}{\|\tilde{s}\|}. \tag{76}$$

The eigenvalue gap $\lambda_1 - \lambda_2 = \nu\|\tilde{s}\|^2 > 0$ guarantees uniqueness whenever $\nu > 0$ and $s \neq 0$.

Step 3: the EMPCA projection coefficient. The raw projection coefficient is $u_1^\dagger\tilde{x}_i = \tilde{s}^\dagger\tilde{x}_i/\|\tilde{s}\|$. Rescaling so that a unit-amplitude signal gives $\hat{a}_i = 1$,

$$\hat{a}_i^{(\mathrm{EMPCA})} = \frac{\tilde{s}^\dagger\tilde{x}_i}{\tilde{s}^\dagger\tilde{s}} = \frac{s^\dagger\Sigma^{-1}x_i}{s^\dagger\Sigma^{-1}s}. \tag{77}$$

Step 4: comparison to the OF estimator. Minimizing $\chi^2(a) = (x_i - as)^\dagger\Sigma^{-1}(x_i - as)$ over a gives

$$\hat{a}_i^{(\mathrm{OF})} = \frac{s^\dagger\Sigma^{-1}x_i}{s^\dagger\Sigma^{-1}s}, \tag{78}$$

identical to (77). Subspace equivalence therefore implies amplitude equivalence, which implies estimator equivalence.

Step 5: efficiency. Both estimators are unbiased with common variance

$$\mathrm{Var}(\hat{a}_i) = \frac{1}{s^\dagger\Sigma^{-1}s}, \tag{79}$$

equal to the CRB for model (74), so the shared estimator is efficient. □

D.2 Proof of Theorem 4.2

For rank-1 EMPCA with basis u,

$$\hat{c} = \frac{u^\dagger \Sigma^{-1} x}{u^\dagger \Sigma^{-1} u}. \tag{80}$$

Substituting $u = \gamma s_0$,

$$\hat{c} = \frac{\gamma^* s_0^\dagger \Sigma^{-1} x}{|\gamma|^2 s_0^\dagger \Sigma^{-1} s_0} = \frac{1}{\gamma} \frac{s_0^\dagger \Sigma^{-1} x}{s_0^\dagger \Sigma^{-1} s_0} = \frac{\hat{A}_{\mathrm{OF}}}{\gamma}, \tag{81}$$

hence

$$\hat{s}_{\mathrm{EMPCA}} = \hat{c}\, u = \frac{\hat{A}_{\mathrm{OF}}}{\gamma}(\gamma s_0) = \hat{A}_{\mathrm{OF}}\, s_0 = \hat{s}_{\mathrm{OF}}, \tag{82}$$

and the weighted residual $(x-\hat{s})^\dagger \Sigma^{-1}(x-\hat{s})$ is identical for both estimators. The remaining freedom $u \mapsto \alpha u$, $c \mapsto c/\alpha$ is the rank-1 gauge transformation: it changes the coefficient convention but not the reconstructed signal. □

D.3 Extended proof of the Bridge Theorem (Theorem 5.1)

Step A: reduction to whitened form. Set $\tilde{x}_i = \Sigma^{-1/2} x_i$ and $W = \Sigma^{-1/2} P$; the constraint $P^\dagger \Sigma^{-1} P = I_k$ becomes $W^\dagger W = I_k$, and since $P P^\dagger \Sigma^{-1} x_i = \Sigma^{1/2} W W^\dagger \tilde{x}_i$, the loss (28) becomes

$$\mathcal{L}(W) = \frac{1}{N} \|\tilde{X}(I - W W^\dagger)\|_F^2. \tag{83}$$

Step B: expansion and Rayleigh–Ritz. Expanding,

$$\mathcal{L}(W) = \frac{1}{N}\|\tilde{X}\|_F^2 - \mathrm{Tr}\big(W^\dagger \hat{R}_{\tilde{x}} W\big), \qquad \hat{R}_{\tilde{x}} = \frac{1}{N}\tilde{X}^\dagger \tilde{X} = \Sigma^{-1/2} \hat{R}_x \Sigma^{-1/2}, \tag{84}$$

so minimizing $\mathcal{L}$ over $W^\dagger W = I_k$ is the Rayleigh–Ritz problem maximizing $\mathrm{Tr}(W^\dagger \hat{R}_{\tilde{x}} W)$, solved by the top-$k$ eigenvectors $W^* = [v_1, \ldots, v_k]$ of $\hat{R}_{\tilde{x}}$.

Step B′: the set of minimizers. Rayleigh–Ritz gives $\mathrm{Tr}(W^\dagger \hat{R}_{\tilde{x}} W) \le \sum_{j \le k} \lambda_j$ for every $W^\dagger W = I_k$ (Ky Fan's maximum principle), with equality if and only if $\mathrm{range}(W)$ is an invariant subspace of $\hat{R}_{\tilde{x}}$ spanned by eigenvectors of $\lambda_1, \ldots, \lambda_k$. If $\lambda_k > \lambda_{k+1}$ that subspace is unique. If $\lambda_k = \lambda_{k+1}$, any k-dimensional subspace that contains the eigenspaces of all eigenvalues strictly above λ_k and is completed inside the eigenspace of λ_k attains the bound; when $\mathrm{rank}\, \hat{R}_{\tilde{x}} < k$ this includes every completion inside the whitened null space $\ker \hat{R}_{\tilde{x}} = \Sigma^{1/2} \ker \hat{R}_x$, since the corresponding λ_j vanish; in detector coordinates ($P = \Sigma^{1/2} W$) these completions are the directions of $\Sigma \ker \hat{R}_x$, the null space of $\hat{R}_x \Sigma^{-1}$. This is statement 1–2 of the theorem; the feasible range $1 \le k \le d$ is the range in which $W^\dagger W = I_k$ has solutions.

Step C: translation back and connection to EMPCA. With $P^* = \Sigma^{1/2} W^*$: left-multiplying $\Sigma^{-1/2} \hat{R}_x \Sigma^{-1/2} v_j = \lambda_j v_j$ by $\Sigma^{1/2}$ gives $\hat{R}_x \Sigma^{-1} (\Sigma^{1/2} v_j) = \lambda_j (\Sigma^{1/2} v_j)$, the generalized eigenproblem whose top-k eigenspace is the EMPCA optimum (Appendix C); every k-dimensional invariant subspace is a fixed point of the EMPCA iteration, and the top-k one is the global minimizer (Appendix C.5). Hence, under the gap condition, $\mathrm{range}(P^*) = \mathrm{range}(\hat{U})$ up to a unitary rotation within the subspace.

Step D: minimum value. At the optimum, $\mathrm{Tr}(W^{*\dagger} \hat{R}_{\tilde{x}} W^*) = \sum_{j=1}^{k} \lambda_j$, so

$$\mathcal{L}^* = \frac{1}{N} \|\tilde{X}\|_F^2 - \sum_{j=1}^{k} \lambda_j = \mathrm{Tr}(\Sigma^{-1} \hat{R}_x) - \sum_{j=1}^{k} \lambda_j, \tag{85}$$

the unexplained variance in whitened coordinates. □

Remark on uniqueness. Under the gap condition W^* is unique only up to right multiplication by $Q \in \mathcal{U}(k)$; without it the family of Step B′ applies. Physical estimators (reconstructed signals, likelihoods, χ^2 values) depend only on the projector $WW^\dagger$, which is invariant under such rotations. The population version of the theorem replaces $\hat{R}_x$ by $\mathbb{E}[xx^\dagger]$ and connects to the asymptotic regime of Theorem 4.1.

E Symmetry, Gauge Freedom, and Geometric Interpretation

This appendix develops the structural background summarized in Section 5.4: the symmetry-axiomatic derivation of the noise-aware architecture, the Grassmannian geometry of the subspace family, and the finite-sample stability statement.

E.1 Symmetry-axiomatic derivation of the noise-aware architecture

Definition E.1 (Noise isometry group)**.**

$$G_{\mathrm{noise}} = \{Q \in GL(\mathbb{C}^d) : Q^\dagger \Sigma^{-1} Q = \Sigma^{-1}\} \tag{86}$$

is the group of linear isometries of the noise inner product $\langle u, v \rangle_{\Sigma^{-1}} = u^\dagger \Sigma^{-1} v$. In whitened coordinates, every $Q \in G_{\mathrm{noise}}$ becomes a unitary map $\tilde{Q} = \Sigma^{-1/2} Q \Sigma^{1/2} \in \mathcal{U}(d)$.

Definition E.2 (Latent gauge group)**.** *For a rank-k decoder basis $P \in \mathbb{C}^{d\times k}$, the latent gauge group is $G_{\text{gauge}} = \mathcal{U}(k)$ acting by*

$$P \mapsto PQ, \qquad z \mapsto Q^\dagger z, \qquad Q \in \mathcal{U}(k), \tag{87}$$

which changes the coordinate basis of the learned subspace but leaves the reconstruction Pz invariant. The physical object is the rank-k signal subspace and its projector, not a particular basis. This is a global *gauge symmetry — one unitary acting on the whole latent space — not a local gauge field over a base manifold in the geometric-deep-learning sense [20].*

Theorem E.1 (Symmetry-axiomatic uniqueness)**.** *Consider linear rank-k reconstructions $\hat{x} = DEx$ with $D(z) = Pz$, $P^\dagger\Sigma^{-1}P = I_k$, and quadratic residual loss $\mathcal{L}_M(x, \hat{x}) = (x - \hat{x})^\dagger M(x - \hat{x})$, $M \succ 0$. Suppose:*

1. noise-metric invariance*: $\mathcal{L}_M(Qx, Q\hat{x}) = \mathcal{L}_M(x, \hat{x})$ for all $Q \in G_{\text{noise}}$;*
2. gauge invariance*: $P \mapsto PQ$, $z \mapsto Q^\dagger z$ leaves $\hat{x}$ and the loss unchanged for all $Q \in \mathcal{U}(k)$;*
3. subspace unbiasedness*: $DEp = p$ for every $p \in \text{range}(P)$;*
4. metric-orthogonal projection*: among rank-k linear maps with range* range(P)*, DE minimizes the noise-metric residual norm.*

Then $M = \kappa\Sigma^{-1}$ for some $\kappa > 0$ (the loss is the Gaussian χ^2 up to scale); the encoder is the Σ^{-1}-adjoint of the decoder,

$$E = D^\sharp = P^\dagger\Sigma^{-1}, \qquad DE = PP^\dagger\Sigma^{-1}; \tag{88}$$

and minimizing the resulting loss over all such rank-k subspaces yields the rank-k EMPCA subspace.

Proof. **Loss uniqueness.** Noise-metric invariance gives $Q^\dagger MQ = M$ for all $Q \in G_{\text{noise}}$. In whitened coordinates, with $\tilde{M} = \Sigma^{1/2}M\Sigma^{1/2}$ and $\tilde{Q} = \Sigma^{-1/2}Q\Sigma^{1/2}$, this reads $\tilde{Q}^\dagger\tilde{M}\tilde{Q} = \tilde{M}$ for every $\tilde{Q} \in \mathcal{U}(d)$. The only matrices commuting with all of $\mathcal{U}(d)$ are scalar multiples of the identity, so $\tilde{M} = \kappa I$ with $\kappa > 0$, i.e. $M = \kappa\Sigma^{-1}$. Equivalently, information geometry selects the same loss: for the Gaussian family with fixed covariance Σ, the Fisher information metric on the mean parameter is Σ^{-1}, and the corresponding squared geodesic distance is the Mahalanobis χ^2 distance [29].

Adjointness and gauge invariance. For $D(z) = Pz$, the adjoint with respect to $\langle\cdot,\cdot\rangle_{\Sigma^{-1}}$ in data space and the Euclidean inner product in latent space is the unique

$D^\sharp$ with $\langle Dz, x\rangle_{\Sigma^{-1}} = \langle z, D^\sharp x\rangle_2$; since $\langle Pz, x\rangle_{\Sigma^{-1}} = z^\dagger P^\dagger \Sigma^{-1} x$, we get $D^\sharp x = P^\dagger \Sigma^{-1} x$. The metric-orthogonal rank-k projection onto range(P) is then $DD^\sharp = PP^\dagger \Sigma^{-1}$ using $P^\dagger \Sigma^{-1} P = I_k$. Under $P \mapsto PQ$ the encoder transforms as $Q^\dagger P^\dagger \Sigma^{-1} x$ and the reconstruction $PQQ^\dagger P^\dagger \Sigma^{-1} x = PP^\dagger \Sigma^{-1} x$ is unchanged, proving both the tied form and the gauge invariance.

EMPCA minimizer. Substituting $E = P^\dagger \Sigma^{-1}$ and $M = \kappa \Sigma^{-1}$ gives the loss (26) up to scale; expanding gives Eq. (27), and minimizing over $P^\dagger \Sigma^{-1} P = I_k$ is the Rayleigh–Ritz problem for the top-k whitened eigenspace — the rank-k EMPCA subspace. □

Corollary E.2 (Bridge Theorem from symmetry)**.** *The noise-aware tied linear AE of Eq.* (28) *is the unique linear rank-k architecture–loss pair satisfying the requirements of Theorem E.1, up to latent gauge rotations and an overall positive scaling of the loss; its minimizer therefore coincides with the rank-k EMPCA subspace, recovering Theorem 5.1.*

Remark E.1 (Tied weights as the adjoint condition)**.** *The tied-weight constraint is not a regularization heuristic; it states that encoder and decoder are mutual adjoints under the correct inner products. What is lost without it is made precise next: not the optimal reconstruction, but the guarantee that intermediate reconstruction maps are Σ^{-1}-orthogonal projections at all.*

E.2 Untied linear encoder–decoder pairs

Let $E \in \mathbb{C}^{k\times d}$ and $D \in \mathbb{C}^{d\times k}$ be unconstrained, and consider the same weighted objective

$$\mathcal{L}(D, E) = \frac{1}{N} \sum_{i=1}^{N} (x_i - DEx_i)^\dagger \Sigma^{-1} (x_i - DEx_i). \tag{89}$$

In whitened coordinates $\tilde{D} = \Sigma^{-1/2} D$, $\tilde{E} = E\Sigma^{1/2}$, $\tilde{R} = \Sigma^{-1/2} \hat{R}_x \Sigma^{-1/2}$, the objective depends on the pair only through the product $B = \tilde{D}\tilde{E}$, a matrix of rank at most k:

$$\mathcal{L}(D, E) = \operatorname{tr}\big((I - B)\tilde{R}(I - B)^\dagger\big) = \big\|(I - B)\tilde{R}^{1/2}\big\|_F^2. \tag{90}$$

Proposition E.3 (Untied optimum)**.** *Let $\tilde{R} = V\Lambda V^\dagger$ with $\lambda_1 \geq \cdots \geq \lambda_d$. (i) The minimum of* (90) *over all B of rank at most k equals $\sum_{j>k} \lambda_j$, the tied minimum of Theorem 5.1, and is attained at $B = V_k V_k^\dagger$, the orthogonal projector onto a top-k eigenspace. (ii) If $\tilde{R} \succ 0$ (the whitened data have full rank, which requires $N \geq d$) and $\lambda_k > \lambda_{k+1}$, this minimizer is unique: every optimal untied pair has $DE = P^* P^{*\dagger} \Sigma^{-1}$ with P^* the tied optimum, so the untied optimum is the Σ^{-1}-orthogonal projection and tied and untied classes have the same optimal reconstruction map. (iii) If $\tilde{R}$ is singular, B is determined only on* range $\tilde{R}$ *and may act arbitrarily on its null space; if $\lambda_k = \lambda_{k+1}$ the eigenspace is not unique, as in Theorem 5.1. (iv) The parameterization is redundant: $(\tilde{D}, \tilde{E}) \mapsto (\tilde{D}S, S^{-1}\tilde{E})$ for*

any invertible $S \in GL(k)$ leaves B and the objective unchanged, a k^2-dimensional gauge that the tie $E = D^\sharp$ reduces to the $\mathcal{U}(k)$ of Definition E.2.

Proof. When $\tilde{R} \succ 0$, $\{B\tilde{R}^{1/2} : \operatorname{rank} B \le k\}$ is the set of all matrices of rank at most k, so (i)–(ii) are the Eckart–Young theorem applied to $\tilde{R}^{1/2} = V\Lambda^{1/2}V^\dagger$: the best rank-$k$ approximation is $V_k\Lambda_k^{1/2}V_k^\dagger$, unique under the gap condition, and $B = V_k\Lambda_k^{1/2}V_k^\dagger\tilde{R}^{-1/2} = V_kV_k^\dagger$. Unwhitening gives $DE = \Sigma^{1/2}V_kV_k^\dagger\Sigma^{-1/2} = P^*P^{*\dagger}\Sigma^{-1}$ with $P^* = \Sigma^{1/2}V_k$. When $\tilde{R}$ is singular, $B\tilde{R}^{1/2}$ ranges only over matrices whose row space lies in $\operatorname{range}\tilde{R}$, which determines B on that range and not on its complement, giving (iii). Statement (iv) is immediate from $B = \tilde{D}\tilde{E}$. □

The proposition fixes what may and may not be said about untying. It may *not* be said that an untied linear autoencoder trained under the Σ^{-1} loss has a different optimal reconstruction from the tied one: under full-rank data and a spectral gap the optima coincide. What is true is that away from the optimum the product DE of an untied pair is in general not idempotent and not Σ^{-1}-self-adjoint — it is a non-self-adjoint reconstruction map, not a projection — and that the optimizer must traverse the $GL(k)$ gauge of statement (iv), whose flat directions an anisotropic Σ can make badly conditioned for a first-order method. Whether a first-order optimizer traverses that gauge in practice is an empirical question; the gauge-conditioning explanation is a plausible mechanism, not a proved one.

Remark E.2 (Connection to geometric deep learning)**.** *In geometric-deep-learning terms [20], G_{noise} is the symmetry group of the observation-space metric induced by the noise model, and $G_{\text{gauge}} = \mathcal{U}(k)$ is the global gauge freedom of the latent basis. The blueprint usually derives architectures from symmetries of the input domain; here the symmetry is information-geometric, imposed by the detector covariance. The Bridge Theorem is thus not an algebraic coincidence but the linear reconstruction method selected by the noise metric and the latent gauge freedom.*

E.3 Grassmannian geometry of the subspace family

When $\tilde{U}^\dagger\tilde{U} = I_k$, the EMPCA objective depends on $\tilde{U}$ only through the orthogonal projector $\Pi_{\tilde{U}} = \tilde{U}\tilde{U}^\dagger$: EMPCA optimizes not over bases but over the space of k-dimensional subspaces of $\mathbb{C}^d$, the complex Grassmannian

$$\operatorname{Gr}(k,d) = \mathcal{U}(d)/(\mathcal{U}(k)\times\mathcal{U}(d-k)), \tag{91}$$

equivalently the space of rank-k orthogonal projectors in whitened coordinates. The $\mathcal{U}(k)$ gauge freedom is precisely the freedom to choose a basis within a fixed point of $\operatorname{Gr}(k,d)$. For $k = 1$, $\operatorname{Gr}(1,d) = \mathbb{C}P^{d-1}$ and basic OF corresponds to the fixed point represented by

the whitened template direction $[\tilde{s}_0]$. This places EMPCA within the standard geometry of Grassmannian subspace optimization [42, 43].

E.4 Product gauge of the separable class

For the NFPA class of Section 6 the reconstruction depends on the factors only through the projectors $U_c U_c^T$ and $U_t U_t^T$, so the gauge group is the product $O(k_c) \times O(k_t)$ acting by $U_c \mapsto U_c Q_c$, $U_t \mapsto U_t Q_t$. Under the embedding $\mathrm{range}(U_t) \otimes \mathrm{range}(U_c)$ this is the subgroup $\{Q_t \otimes Q_c\}$ of the $O(k_c k_t)$ gauge of the unrestricted rank-$k_c k_t$ class: separability removes not only most of the subspaces but also most of the basis freedom within a subspace. The practical rule is unchanged — interpret projectors, subspaces, reconstructions, and principal angles, never raw factor coordinates — and it now applies axis by axis. In particular a comparison of a learned factor with a planted one must be made between the projectors $U_c U_c^T$ and $V_c V_c^T$ (or between subspaces), which is gauge-invariant, and not between the matrices U_c and V_c, which is not.

Proof of Proposition 6.3. The identity (37) telescopes. $a \geq 0$ because $\mathcal{F} \subset \mathrm{Gr}(k, Cd)$ and $P_U^\star$ minimizes R_K over all rank-k orthogonal projectors (Ky Fan's maximum principle: $\mathrm{tr}(PK) \leq \sum_{i \leq k} \mu_i$); $x_{\mathcal{F}}, x_U \geq 0$ by the definition of the two optima. For the first inequality of (38) write $w_i = v_i^T P_F v_i \in [0, 1]$, so that $\sum_i w_i = \mathrm{tr}\, P_F = k$ and $\mathrm{tr}(P_F K) = \sum_i \mu_i w_i$. Then

$$\begin{aligned} \mathrm{tr}(P_F K) &\leq \sum_{i \leq k} \mu_i w_i + \mu_{k+1} \sum_{i > k} w_i = \sum_{i \leq k} \mu_i - \sum_{i \leq k} (\mu_i - \mu_{k+1})(1 - w_i) \\ &\leq \mathrm{tr}(P_U^\star K) - \Delta_k \sum_{i \leq k} (1 - w_i), \end{aligned}$$

and $\sum_{i \leq k}(1 - w_i) = k - \mathrm{tr}(P_F P_U^\star)$. For the second, $P_F = P_t \otimes P_c \preceq I \otimes P_c$ and $P_F \preceq P_t \otimes I$ in the Loewner order, and $P_U^\star \succeq 0$, so $\mathrm{tr}(P_F P_U^\star) \leq \mathrm{tr}\big((I \otimes P_c) P_U^\star\big) = \mathrm{tr}(P_c M_c)$ with $M_c = \mathrm{tr}_t\, P_U^\star$ the partial trace over the temporal factor, a positive semidefinite $C \times C$ matrix of trace k; over rank-k_c projectors $\mathrm{tr}(P_c M_c)$ is at most the sum of the k_c largest eigenvalues of M_c (Ky Fan again), and likewise $\mathrm{tr}(P_F P_U^\star) \leq \sum_{i \leq k_t} \mu_i(M_t)$ with $M_t = \mathrm{tr}_c\, P_U^\star$; the cap k is $\mathrm{tr}\, P_F$. Finally, if $\Delta_k > 0$ and $a = 0$ then some $P_F \in \mathcal{F}$ attains $R_K(P_U^\star)$, so $\mathrm{tr}(P_F P_U^\star) = k$ by the first inequality, and two rank-k orthogonal projectors with $\mathrm{tr}(P_F P_U^\star) = k$ are equal; conversely $P_U^\star \in \mathcal{F}$ gives $a = 0$ directly. The bound is computed on whatever K is supplied; in Section 9.2.3 that is an estimate of the generator's clean-signal covariance from 2×10^5 events, so the numbers there carry that estimate's error and floating-point error, and nothing more. □

E.5 Metric symmetry versus domain symmetry

It is essential not to conflate the metric symmetry G_{noise} with a domain symmetry. G_{noise} preserves the residual norm; it acts on observation space and knows nothing about the detector layout. It is not time translation, channel relabeling, detector rotation, or a graph automorphism — transformations that act on the measurement domain and may (approximately) commute with the physical signal-generating process, and from which the geometric-deep-learning blueprint derives its architectures [20]. Prewhitening therefore fixes the residual metric but does not by itself impose convolutional locality, graph equivariance, or deformation stability; those inductive biases must be encoded by the architecture, and their validity is a property of the detector, not of the noise model. The same distinction separates the two symmetry groups of Theorem E.1: G_{noise} is information-geometric and selects the loss; G_{gauge} is a global redundancy of the parameterization and selects nothing. Neither is a domain symmetry, and the separable prior of Section 6 is the first place in the hierarchy where a domain assumption — that channel and time structure factor — enters at all.

E.6 Representation geometry: proofs of Propositions 7.3 and 7.4

Both proofs are for real observations, real latents and real Gaussian noise, as the propositions are stated.

Proof of Proposition 7.3. The log-likelihood of x given ϑ is $\ell(\vartheta) = -\frac{1}{2}(x - g(\vartheta))^T \Sigma^{-1}(x - g(\vartheta)) + \text{const}$, with score $\nabla_\vartheta \ell = J_g(\vartheta)^T \Sigma^{-1}(x - g(\vartheta))$. Since $\mathbb{E}[(x - g(\vartheta))(x - g(\vartheta))^T] = \Sigma$, the Fisher information is $\mathbb{E}[\nabla_\vartheta \ell \, \nabla_\vartheta \ell^T] = J_g^T \Sigma^{-1} \Sigma \Sigma^{-1} J_g = J_g^T \Sigma^{-1} J_g$, which is positive semidefinite, and positive definite exactly when $J_g v = 0$ has only the trivial solution. For the representation statement, compose $h \mapsto y = g_{\text{out}}(h) \in \mathbb{R}^d$ with the observation model: a displacement δh moves the predicted observation by $J_h \delta h$ to first order, whose squared whitened length is $\delta h^T J_h^T \Sigma^{-1} J_h \delta h$; the form is degenerate on $\ker J_h$. The construction requires y to live in observation space, because it is Σ that supplies the inner product. The identifiability statement is the first-order one: if $J_g(\vartheta) v = 0$ then $g(\vartheta + \epsilon v) = g(\vartheta) + O(\epsilon^2)$, so no estimator distinguishes ϑ from $\vartheta + \epsilon v$ at rate ϵ; this does not exclude that g is injective near ϑ. Throughout, g is assumed continuously differentiable at the point of linearization. □

Proof of Proposition 7.4. Under (iii) the reconstruction acts on $u = t + w$ with $t \in T_S$, $\mathbb{E}[t] = 0$ by (iv), $\mathbb{E}[tw^T] = 0$ by (ii), and $\mathbb{E}[ww^T] = I$ by (i), so $\mathbb{E}[uu^T] = R_t + I$. For a rank-$k$ orthogonal projector Π, $J(\Pi) = \operatorname{tr}\big((I - \Pi)(R_t + I)(I - \Pi)\big) = \operatorname{tr}\big((I - \Pi)R_t\big) + \operatorname{tr}(I - \Pi) = \operatorname{tr} R_t - \operatorname{tr}(\Pi R_t) + (d - k)$, using $(I - \Pi)^2 = I - \Pi$ and the cyclicity of the trace; this is

statement 1. If $Q \in O(d)$ fixes T_S pointwise then, because $\operatorname{range} R_t \subseteq T_S$, $QR_tQ^T = R_t$ and $\operatorname{tr}(Q\Pi Q^T R_t) = \operatorname{tr}(\Pi\, Q^T R_t Q) = \operatorname{tr}(\Pi R_t)$, so $J(Q\Pi Q^T) = J(\Pi)$ exactly. For statement 2, $\operatorname{tr}(\Pi R_t) \le \sum_{j\le k} \mu_j$ with μ_j the eigenvalues of R_t (Ky Fan), with equality if and only if $\operatorname{range}\Pi$ contains a top-k eigenspace of R_t; when $k \ge r = \operatorname{rank} R_t$ the top-k eigenspaces are exactly the k-dimensional subspaces containing $\operatorname{range} R_t$, whose remaining $k - r$ dimensions are arbitrary in $(\operatorname{range} R_t)^\perp$, and when $k < r$ the maximizers are the top-k eigenspaces of R_t with the usual gap condition. For statement 3, with $B = \Sigma^{-1/2} A_{\cdot S}$ and $K \succeq 0$, $\operatorname{range}(BKB^T) = \operatorname{range}(BK^{1/2}) = B(\operatorname{range} K)$, so $\operatorname{range} R_t = \operatorname{range} B = T_S$ if and only if $B(\operatorname{range} K) = B(\mathbb{R}^{|S|})$, i.e. $\operatorname{range} K + \ker B = \mathbb{R}^{|S|}$; taking orthogonal complements, $\ker K \cap (\ker B)^\perp = \{0\}$, which is positive definiteness of K on $\operatorname{range}(B^T)$. If $K \succ 0$ the condition holds for every B; if B is injective, $\ker B = \{0\}$ and the condition reduces to $K \succ 0$. The condition is not symmetric in the two factors: $B = [1\ 1]$ with $K = \operatorname{diag}(1, 0)$ gives $BKB^T = 1$ and $\operatorname{range} R_t = \operatorname{range} B$ although K is singular and B is not injective. □

The proof makes visible why the finite-sample situation is different. Replacing $R_t + I$ by the sample second moment $\hat{R}_u = N^{-1} \sum_i u_i u_i^T$ replaces the isotropic term I by a sample noise covariance and R_t by a finite-sample estimate. For Gaussian w and $d/N \to \gamma$, the nonzero eigenvalues of the sample noise covariance follow the Marchenko–Pastur law on $[(1-\sqrt{\gamma})^2, (1+\sqrt{\gamma})^2]$, and when $N < d$ there are in addition $d - N$ eigenvalues exactly at zero [23]. The population objective's indifference to the complement then becomes a small but nonzero preference set by the noise realization. For a single population direction of excess variance θ added to isotropic noise, the leading sample eigenvalue separates from the bulk, and the leading sample eigenvector acquires a nonvanishing asymptotic overlap with that direction, if and only if $\theta > \sqrt{\gamma}$; below the transition the overlap vanishes, and above it equals $\big((1-\gamma/\theta^2)/(1+\gamma/\theta)\big)^{1/2}$, not one [24, 25]. These are statements about an isolated spike, isotropic noise and unstructured PCA; for several planted directions and for the structured NFPA estimator they are used in the controlled evidence (the controlled separable-class evidence, held from this version (Section 1)) only as diagnostics, not as a threshold.

E.7 Finite-sample stability: extended Davis–Kahan discussion

Two perturbations act on the whitened operator whose top-k eigenspace the estimators of this paper return, and both are handled by the Davis–Kahan theorem in the same coordinates. Throughout, $f(M)$ maps a Hermitian operator to its top-k eigenspace, a point of $\mathrm{Gr}(k, d)$, $\sin\Theta$ is the sine of the largest principal angle (equivalently the operator-norm distance between the two orthogonal projectors, which is a metric on $\mathrm{Gr}(k, d)$), and

the form of the theorem used is that of Yu, Wang and Samworth [27]: for Hermitian M and $\hat{M}$ with $\Delta_\lambda = \lambda_k(M) - \lambda_{k+1}(M) > 0$, $\sin\Theta(f(\hat{M}), f(M)) \le 2\|\hat{M} - M\|_{\mathrm{op}}/\Delta_\lambda$.

Sampling error at known covariance. With the true Σ, $A = \Sigma^{-1/2} R_x \Sigma^{-1/2}$ and $\hat{A} = \Sigma^{-1/2}\hat{R}_x\Sigma^{-1/2} = A + \Sigma^{-1/2}\Delta R\,\Sigma^{-1/2}$, so

$$\sin\Theta\big(f(\hat{A}), f(A)\big) \le \frac{2\,\|\Sigma^{-1/2}\Delta R\,\Sigma^{-1/2}\|_{\mathrm{op}}}{\Delta_\lambda}, \tag{92}$$

which is (24): numerator and gap are both whitened. The numerator is a covariance-estimation error and is controlled under the tail and sample-size assumptions stated in Section 4.4; an unqualified $O(\sqrt{d/N})$ is not claimed.

Covariance error (proof of the bound in Remark 5.2). Let $\hat{\Sigma} = \Sigma + \Delta\Sigma$ with $E = \Sigma^{-1/2}\Delta\Sigma\,\Sigma^{-1/2}$, $\varepsilon = \|E\|_{\mathrm{op}} \le \frac{1}{2}$, so that $\hat{\Sigma} = \Sigma^{1/2}(I+E)\Sigma^{1/2}$ and $I + E \succ 0$. A basis $\hat{P}$ fitted with $\hat{\Sigma}$ satisfies the generalized eigenproblem $R_x\hat{\Sigma}^{-1}\hat{p} = \lambda\hat{p}$ (Theorem 5.1, item 1, with $\hat{\Sigma}$ in place of Σ). Substituting $\hat{p} = \Sigma^{1/2}u$ gives, in the true whitened coordinates, $A(I+E)^{-1}u = \lambda u$, and $A(I+E)^{-1} = (I+E)^{1/2}\tilde{A}\,(I+E)^{-1/2}$ with the Hermitian operator $\tilde{A} = (I+E)^{-1/2}A\,(I+E)^{-1/2}$. Hence $\hat{\mathcal{S}} = \mathrm{range}(\Sigma^{-1/2}\hat{P}) = (I+E)^{1/2} f(\tilde{A})$, while $\mathcal{S} = f(A)$. Two steps bound the distance. (i) With $G = (I+E)^{-1/2} - I$, $\|G\| \le (1-\varepsilon)^{-1/2} - 1 \le \varepsilon$ for $\varepsilon \le \frac{1}{2}$, and $\tilde{A} - A = GA + AG + GAG$, so $\|\tilde{A} - A\| \le (2\varepsilon + \varepsilon^2)\lambda_1(A) \le \frac{5}{2}\varepsilon\lambda_1(A)$ and $\sin\Theta(f(\tilde{A}), f(A)) \le 5\varepsilon\lambda_1(A)/\Delta_\lambda$. (ii) With $H = (I+E)^{1/2} - I$, $\|H\| \le \varepsilon/(1+\sqrt{1-\varepsilon})$, and for any k-dimensional subspace Z and unit $z \in Z$ the vector $(I+H)z$ lies within $\|H\|$ of Z and has norm at least $1 - \|H\|$, so $\sin\Theta((I+H)Z, Z) \le \|H\|/(1-\|H\|) \le \varepsilon$ for $\varepsilon \le \frac{1}{2}$. The triangle inequality for $\sin\Theta$ then gives $\sin\Theta(\hat{\mathcal{S}}, \mathcal{S}) \le 5\varepsilon\lambda_1/\Delta_\lambda + \varepsilon \le 6\varepsilon\lambda_1/\Delta_\lambda$, which is (30). The bound is invariant under a common scaling of Σ, $\Delta\Sigma$ and R_x, as it must be; the regression check `scripts/check_revision_claims.py` verifies it on random instances and reproduces the scaling counterexample that rules out an absolute bound in $\|\Delta\Sigma\|$. If the fitted bases are compared in detector coordinates, the comparison must be made in the Σ^{-1} metric, which is what the whitened statement does; a Euclidean comparison in detector coordinates additionally depends on the changing unwhitening map $\hat{\Sigma}^{1/2}$ and is not the quantity bounded here. When both perturbations act at once — a sample second moment $\hat{A} = A + \Delta A$ whitened with an estimated covariance — the operator whose eigenspace is fitted is, in true whitened coordinates and up to the same transport, $(I+E)^{-1/2}(A+\Delta A)(I+E)^{-1/2} = \tilde{A} + (I+E)^{-1/2}\Delta A(I+E)^{-1/2}$; the sample perturbation is conjugated, not added unchanged, and $\|(I+E)^{-1/2}\Delta A(I+E)^{-1/2}\| \le \|\Delta A\|/(1-\varepsilon)$. Hence, under the same hypotheses, $\sin\Theta(\hat{\mathcal{S}}, \mathcal{S}) \le 2\big[(2\varepsilon+\varepsilon^2)\lambda_1(A) + \|\Delta A\|/(1-\varepsilon)\big]/\Delta_\lambda + \varepsilon$, by the same two steps.

First-order displacement (proof of Proposition 5.2). In the true whitened coordinates the basis fitted with $\hat{\Sigma}$ spans the top-k invariant subspace of $A(I+E)^{-1}$ (previous paragraph). For $\varepsilon < 1$, $(I+E)^{-1} = I - E + O(\varepsilon^2)$, so $A(I+E)^{-1} = A + M + O(\varepsilon^2)$ with the non-symmetric perturbation $M = -AE$ of the symmetric operator A, $\|M\| \leq \varepsilon\lambda_1(A)$. First-order perturbation of a simple invariant subspace [55, Ch. V] applies: with $Q_{\mathcal{R}} = [q_1, \ldots, q_k]$ and $Q_{\mathcal{R}^\perp}$ its complement, the top-k invariant subspace of $A + M$ is spanned by the columns of $Q_{\mathcal{R}} + Q_{\mathcal{R}^\perp}X$ with $X_{ji} = M_{ji}/(\lambda_i - \lambda_j) + O(\|M\|^2/\delta^2)$ for $i \in \mathcal{R}$, $j \notin \mathcal{R}$, where $M_{ji} = q_j^T M q_i = -\lambda_j E_{ji}$ and $\delta = \min|\lambda_i - \lambda_j|$ over such pairs. The orthogonal projector onto that subspace is $P + Q_{\mathcal{R}^\perp}XQ_{\mathcal{R}}^T + Q_{\mathcal{R}}X^TQ_{\mathcal{R}^\perp}^T + O(\|X\|^2)$, which is (31). The blocks $M_{\mathcal{R}\mathcal{R}}$ and $M_{\mathcal{R}^\perp\mathcal{R}^\perp}$ enter X only at second order — they change the eigenbasis inside $\mathcal{R}$ and inside $\mathcal{R}^\perp$, not the two subspaces — which is item 1. Item 2 reads off the Frobenius norm of the two off-diagonal blocks and uses $\sin\Theta(\hat{\mathcal{S}}, \mathcal{S}) = \|(I-P)\hat{P}\|_{\mathrm{op}} = \|Q_{\mathcal{R}^\perp}XQ_{\mathcal{R}}^T\|_{\mathrm{op}} + O(\|X\|^2)$. For item 3, if $E = \sum_i e_i q_i q_i^T$ then $A(I+E)^{-1} = \sum_i \lambda_i(1+e_i)^{-1}q_iq_i^T$ exactly; its eigenvectors are the q_i, and its top-k eigenvectors are those of $\mathcal{R}$ if and only if the stated inequality holds. For item 4, replace A by $A + D$ before expanding: $(A+D)(I+E)^{-1} = A + (D - AE) + O(\varepsilon^2, \varepsilon\|D\|)$, so $M_{ji} = D_{ji} - \lambda_j E_{ji}$ in the eigenbasis of A; expanding about $\hat{A} = A + D$ in its own eigenbasis instead, $M = -\hat{A}E$ and D does not appear. The remainder statement is the second-order term of the same expansion, $\|X\| \leq \|M\|/\delta + O(\|M\|^2/\delta^2)$, which with $\|M\| \leq \varepsilon\lambda_1(A)$ is $O(\varepsilon^2\lambda_1^2/\delta^2)$. Because $\delta \leq \lambda_k \leq \lambda_1$, this dominates the $O(\varepsilon^2\lambda_1/\delta)$ contributed by truncating $(I+E)^{-1} = I - E + O(\varepsilon^2)$, and it is the order the proposition states: the expansion is informative when $\varepsilon\lambda_1/\delta \ll 1$ rather than merely when $\varepsilon \ll 1$. (An earlier statement of the proposition gave the remainder as $O(\varepsilon^2\lambda_1/\delta)$; that is violated as soon as λ_1/δ exceeds one, as `scripts/check_revision_claims.py` exhibits on a coupling-block perturbation.) Note that the transported subspace $(I+E)^{1/2}f(\tilde{A})$ of the previous paragraph and the invariant subspace of $A(I+E)^{-1}$ used here are the same object; the proposition expands it directly rather than through the symmetrized $\tilde{A}$, which is why the within-block entries of E drop out exactly at first order.

Reading. Structurally the bound plays the role of a deformation-stability statement in geometric deep learning [20]: small perturbations of the estimated object produce small changes in the geometric estimator when the stability constant — here the conditioning ratio λ_1/Δ_λ of the retained whitened spectrum — is controlled. The relevant perturbation is a *signal-space* deformation, the covariance estimate changing on the fixed observation domain, as opposed to a *domain* deformation as in graph or manifold settings. Experiment S7 (Section 9) is the empirical counterpart; the CRESST tied-AE row is a closed-form parameterization check (Section 9.3) and carries no information about stability. For the separable class the analogous statement would bound the angle between the NFPA and

EMPCA subspaces by the best Tucker-2 approximation error to the EMPCA projector [31] divided by the spectral gap; Section 6.4 states this as an open target.

F Supplementary Evidence Status

Earlier drafts included quantitative panels for principal-component interpretation, Cramér–Rao checks, resolution scaling, template mismatch, time-shift recovery, rank saturation, timing-bias reduction, residual calibration, and iterative convergence. The current evidence bundle does not map those displayed values to exact configurations, seeds, machine-readable metrics, and a code revision. In particular, it does not independently substantiate the repeated residual values 1029.2, 1028.9, and 1028.6, the reported scaling slope, the stated timing-bias reduction, or the claimed Kolmogorov–Smirnov transition. Those quantitative claims and their legacy figures are therefore omitted from this revision rather than treated as verified results. The controlled S1–S9 records reported in Section 9 remain the synthetic evidence for the claims made in the main text.

A conditional noise-budget attribution analysis, which would have compared the discrete optimal-filter information computed from a baseline total PSD against the information obtained after removing or rescaling individual additive components, was also prepared for an earlier draft. It is not reported here. Such quantities are conditional attributions under a declared additive model: they do not show that a component can be physically removed, that its label is uniquely identifiable from a total spectrum, or that hardware performance would improve by the corresponding amount. Reporting one would additionally require the sampling frequency, record length, template normalization, endpoint factors, frequency support, and acquisition transfer function to be frozen and independently verified, which the available preparation record does not establish. The claim is therefore withdrawn from this revision rather than reported conditionally.

G Noise Sampling and Spectral Conventions

This appendix fixes the discrete spectral conventions used by the synthetic experiments of Section 9 and referenced by the optimal-filter derivations of Appendix A. It concerns sampling and normalization only; no detector, instrument, or measured noise budget is described here.

G.1 Generating colored noise from a one-sided PSD

The synthetic noise used throughout is a real Gaussian time-domain trace drawn from a specified one-sided PSD density $P(f)$. For trace length N, sampling frequency f_s,

spacing $\Delta f = f_s/N$, and rFFT bins $f_k = k\Delta f$, we adopt the NumPy convention of an unnormalised forward transform with a factor $1/N$ in `irfft`. For each interior positive-frequency bin we draw

$$z_k = \frac{a_k + ib_k}{\sqrt{2}}, \qquad a_k, b_k \overset{\text{iid}}{\sim} \mathcal{N}(0,1), \qquad \tilde{\eta}_k = \sqrt{\frac{f_s N P_k}{2}}\, z_k, \tag{93}$$

where $P_k = P(f_k)$, so that $\mathbb{E}|\tilde{\eta}_k|^2 = f_s N P_k/2$. The DC coefficient and, for even N, the Nyquist coefficient have no independent imaginary part and are drawn as real Gaussians,

$$\tilde{\eta}_k = \sqrt{f_s N P_k}\, a_k, \qquad \mathbb{E}|\tilde{\eta}_k|^2 = f_s N P_k. \tag{94}$$

For odd N, the highest positive-frequency bin remains an interior complex bin. With this convention, Parseval's identity gives the discrete one-sided relation

$$\mathbb{E}\left[\frac{1}{N}\sum_{n=0}^{N-1}\eta_n^2\right] = \Delta f \sum_{k=0}^{\lfloor N/2\rfloor} P_k, \tag{95}$$

which is the identity used to convert between DFT-normalized quantities and physical one-sided PSD units in the optimal-filter expressions. Conventions in the literature differ by a factor of two between one-sided and two-sided PSDs, and by a further factor of two between real-valued and complex Gaussian models; the two-factor combination is held fixed across all methods within each comparison so that no ordering reported in this paper depends on it. The stochastic DC component is set to zero and any deterministic mean is represented separately. Built-in spectral shapes and custom tabular PSDs use the same density-to-bin-power conversion.

G.2 Sampling contract and its verification

Drawing independent coefficients on a finite DFT grid defines a periodic stationary process whose covariance is circulant, not Toeplitz. When the requested covariance is a genuinely non-periodic Toeplitz matrix, a separate exact finite-Toeplitz sampler is used instead; these are distinct sampling contracts and are not interchangeable at finite N. The distinction matters for the residual-calibration test S8, where the whitened residual is compared against its nominal χ^2 distribution.

Figure 12 is a second-order spectral check only: agreement of the ensemble mean spectrum with the target density verifies Eqs. (93)–(95) but does not by itself establish Gaussianity. Fourier-bin normality, periodogram variability, endpoint scaling, and finite-grid covariance agreement are tested separately in the generator validation suite and are not reproduced here.

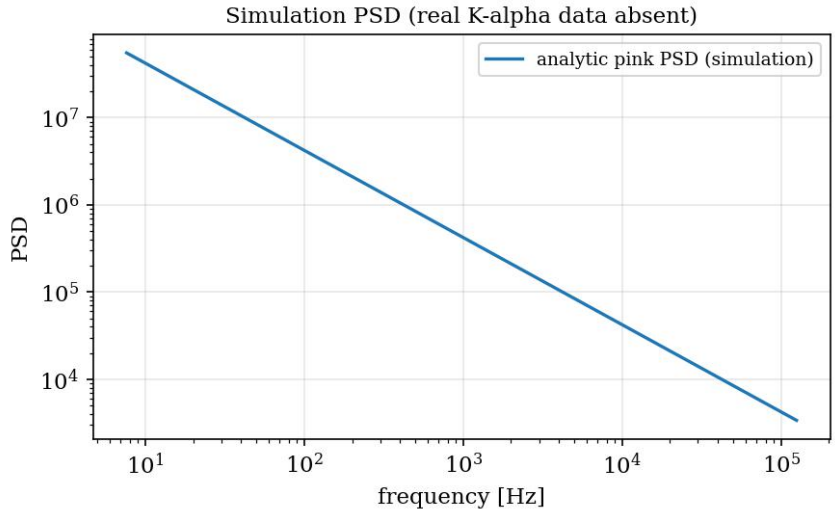


Figure 12: Second-order PSD audit for the one-sided convention of Eq. (93) used by the synthetic OF and EMPCA experiments. Agreement of the ensemble mean spectrum with the target does not by itself establish Gaussianity; Fourier-bin normality, periodogram variability, endpoint scaling, and finite-grid covariance are tested separately.

H List of Symbols

Notation used throughout the paper. Complex notation with † (conjugate transpose) is used by default; all statements specialize to real data with $\dagger \to T$. Vectors are columns; whitened quantities carry tildes.

Category	**Symbol**	**Meaning**
Dimensions	d	Trace length (samples or frequency bins per event).
	N	Number of events in the dataset.
	k	Rank of the learned signal subspace (also frequency-bin index; clear from context).
	C	Number of detector channels.
Indices	i	Event index, $i = 1, \ldots, N$.
	k	Frequency-bin index of the DFT.
	c	Channel index, $c = 1, \ldots, C$.
	r	EM iteration index.
Data	$x,\ x_i$	Observed trace (column vector in $\mathbb{C}^d$).

Category	Symbol	Meaning
	X_k	Frequency-domain coefficient of the observed trace at bin k.
	X	Data matrix, rows $x_i^\dagger$ ($N \times d$).
	$\tilde{x},\ \tilde{X}$	Whitened trace / data matrix, $\tilde{x} = \Sigma^{-1/2}x$.
Signal model	$s_0,\ S_k$	Fixed signal template (time / frequency domain).
	$\hat{s}$	Reconstructed signal.
	$A,\ a_i$	Signal amplitude (pulse height) of an event.
	t_0	Signal arrival time (time offset).
	U	Learned rank-k basis, $U \in \mathbb{C}^{d\times k}$, $U^\dagger\Sigma^{-1}U = I_k$.
	$c,\ \Gamma$	Latent coefficient vector / matrix $\Gamma = [c_1, \ldots, c_N]$ for the rank-k model.
	P	Decoder basis of the noise-aware linear AE, $P^\dagger\Sigma^{-1}P = I_k$.
	z	Latent code of the autoencoder, $z = P^\dagger\Sigma^{-1}x$.
	$E,\ D$	Encoder / decoder of a linear autoencoder; $E = D^\sharp$ in the tied form.
	W	Whitened decoder basis $W = \Sigma^{-1/2}P$, $W^\dagger W = I_k$ (Appendix D.3).
	$\omega_{kj},\ \Omega$	Per-element weights of the frequency-domain EMPCA variant (Appendix C.3).
Noise	$n,\ N_k$	Additive noise (time / frequency domain).
	Σ	Noise covariance matrix ($\succ 0$); diagonal in frequency space for stationary noise.
	Σ^{-1}	Inverse covariance; the observation-space metric.
	$\Sigma^{-1/2}$	Whitening transform.
	$J(f),\ J_k$	Noise power spectral density (PSD).
Optimal filtering	χ^2	Mahalanobis / weighted least-squares objective, Eq. (4).
	$\Phi_k,\ \Phi_k^{(0)}$	Unnormalized / normalized OF kernels.

Category	Symbol	Meaning
	N_Φ	Filter normalization $\sum_k \lvert S_k\rvert^2/J_k$; total SNR budget.
	$\sigma_A,\ \sigma_E$	Amplitude / energy resolution.
	$B(t_0)$	Delayed matched-filter response (time-shift OF).
Subspace estimation	$\hat{R}_x$	Sample covariance $N^{-1}\sum_i x_i x_i^\dagger$.
	λ_j	Eigenvalues of $\Sigma^{-1}\hat{R}_x$ (descending).
	Δ_λ	Spectral gap between signal and noise eigenvalues.
	$\rho_w,\ \theta_w$	Weighted (whitened-space) subspace cosine / angle.
	$\Theta(\cdot,\cdot)$	Vector of principal angles between two subspaces; $\sin\Theta$ denotes the corresponding sine (Remark 5.2).
	$\hat{\Sigma}$	Covariance estimated from calibration data, used in place of Σ.
	$\Delta\Sigma$	Covariance estimation error, $\Delta\Sigma = \hat{\Sigma} - \Sigma$.
	ΔR	Sampling error of the covariance, $\hat{R}_x = R_x + \Delta R$ (Appendix E.7).
	μ	Ridge parameter of the regularized coefficient solve, Eq. (72).
	$P_{\tilde{U}}$	Orthogonal projector onto the whitened subspace.
	$\mathrm{Gr}(k,d)$	Grassmannian of k-dimensional subspaces.
Separable class	$Z,\ \tilde{Z}$	Two-axis (channel $\times$ time) event matrix in $\mathbb{R}^{C\times d}$ and its per-axis whitened form.
	$\Sigma_c,\ \Sigma_t$	Channel and temporal factors of a separable covariance, $\Sigma = \Sigma_t \otimes \Sigma_c$.
	$U_c,\ U_t$	Orthonormal channel / temporal factor bases of the NFPA, $U_c \in \mathbb{R}^{C\times k_c}$, $U_t \in \mathbb{R}^{d\times k_t}$.
	$k_c,\ k_t$	Channel / temporal factor ranks; total rank $k_c k_t$.

Category	Symbol	Meaning
	Γ_i	NFPA code matrix $U_c^T \tilde{Z}_i U_t \in \mathbb{R}^{k_c \times k_t}$.
	α	Nonseparability of the planted truth in the controlled evidence ($\alpha = 0$: exactly separable).
	$\otimes$	Kronecker product; $\mathrm{vec}(AZB) = (B^T \otimes A)\mathrm{vec}(Z)$.
	$O(k_c) \times O(k_t)$	Product gauge of the separable class (Appendix E.4).
Symmetry and loss	G_{noise}	Noise isometry group preserving Σ^{-1} (Definition E.1).
	$\mathcal{U}(k)$	Latent gauge group of basis rotations (Definition E.2).
	M	Weight of a quadratic residual loss $\mathcal{L}_M$; $M = \kappa\Sigma^{-1}$ under noise-isometry invariance.
	κ	Positive scale of the invariant loss (Proposition 5.3).
	$\mathcal{F}$	Admissible reconstruction class; a method is the pair $(\Sigma^{-1}, \mathcal{F})$.
	B	Whitened reconstruction map $\tilde{D}\tilde{E}$ of an untied pair (Appendix E.2).
Representation geometry	ϑ	Physical latent variables of a forward model $x = g(\vartheta) + n$, $\vartheta \in \mathbb{R}^m$.
	m	Number of physical latent variables.
	$g,\ J_g$	Forward model and its Jacobian $\partial g/\partial\vartheta \in \mathbb{R}^{d\times m}$.
	$I(\vartheta)$	Fisher information $J_g^T\Sigma^{-1}J_g$, the pulled-back noise metric (Proposition 7.3).
	$h,\ g_{\text{out}},\ J_h$	A representation, its output map to observation space, and the Jacobian $\partial y/\partial h$.
	$S,\ T_S$	Excited latent coordinates and the excited tangent span range$(\Sigma^{-1/2}A_{\cdot S})$ in whitened coordinates.

Category	Symbol	Meaning
	R_t	Second moment BKB^T of the whitened signal displacement over the training distribution, $B = \Sigma^{-1/2} A_{\cdot S}$, $K = \mathbb{E}[\delta_S \delta_S^T]$.
	Π	Rank-k orthogonal projector in whitened coordinates.

I Configuration Inventory

The operating point of every stage, and every run-level override, is listed in Table 4. The complete mechanically extracted inventory of the frozen configuration files (363 rows from 25 files, experiment P1-E4b) is not reproduced here: it is a separately built supplementary document, `supplement_configuration.pdf`, generated from the registered file `transfer_paper/data/paper1/P1-E4b/hyperparameters.csv` by `figures_src/make_paper1_evidence.py` and archived with the paper evidence. The inventory records what the configuration files say; which configuration a run used is recorded by the `config_hash` in that run's manifest.